Agricultural Equipment Group
Strategic Plan Review

July 1979

Corporate Planning Department

AGRICULTURAL EQUIPMENT GROUP

1980-1984 STRATEGIC PLAN REVIEW

JULY 31, 1979

AGRICULTURAL EQUIPMENT GROUP
STRATEGIC PLAN REVIEW
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IH SBU STRATEGIC COMMENTARY (P4):

STRATEGIC GUIDELINES

- Become global sales leader in agricultural equipment and achieve first market share position for major products in key markets.
- Improve financial and operating ratios of AEG to equal or exceed those of leading competitors.
- Lead in innovative designs and new products which will meet the emerging needs of customers.
- Distribution will be made through the leading dealer and distributor organizations, which IH will support and improve. They will be financially sound, aggressive in the marketplace, and show continued profitability for further growth.
- Set the industry standard in the reduction of manufacturing costs. This will require the introduction of innovative manufacturing methodologies, improved utilization of assets, and the utilization of emerging technologies as they affect product and manufacturing costs.
- Provide a professional and working environment that will attract and retain the highest level of qualified individuals, consistent with the needs of the worldwide AEG organization.

MARKET AND COMPETITION - AEG is competing for an equipment market that is estimated to approach $16 billion (excluding parts) worldwide in 1979. Over the next five years this market is forecasted to average 10% annual growth, of which about 3% is real growth. Plans call for increasing IH's share of this market from 14% in 1978, and an estimated 16% in 1979, to over 20% by 1984. This aggressive growth and attendant improvement in the quantity and quality of earnings is consistent with the Strategic Guidelines established for AEG.

John Deere is the leading agricultural equipment competitor on a global basis.

IH PRODUCTS AND PRODUCTION - The Plan reflects the results of programs aimed at positioning AEG as the leader in market share, product offerings, technological innovation, quality, and cost effectiveness. Market share improvements are the result of major new product introductions supported by aggressive marketing programs. "New product" for 1980-84 will account for 9.6% of Group sales (compared to 5.1% during 1977-79). Engineering expenditures to support the product programs will remain at 2.2% of sales, with 18% devoted to "new product."

Capital expenditures over the Plan period average $216 MM per year (compares to Deere's announced plans to average $400 million per year).

PRODUCTIVITY - Net productivity accounts for 65% of operating income generated over the planning period. Cumulative gross productivity results are forecasted to be favorable to target through 1982, but fall $52 million and $130 million short of target in 1983 and 1984 respectively.

OPPORTUNITIES AND VULNERABILITIES - Over the Plan period, identified vulnerabilities of $185 MM in PBT compare with identified opportunities of $184 MM in PBT. The Group's objective is to achieve the opportunities and develop necessary programs to avoid the vulnerabilities.

July 31, 1979	B. H. Warren	Group	AEG
Date	Manager	SBU	Group

STRATEGIC PLAN HIGHLIGHTS (1980-1984)

		HISTORY		PROJECTION						GROWTH	
		1977	1978	1979	1980	1981	1982	1983	1984	77-79	79-84
A.NET SALES	$MM	2529.79	2474.23	3073.47	3600.16	3965.37	4667.13	5390.21	6019.32	10.22	14.39
B. BREAKEVEN SALES %	%		80.25	71.07	68.95	66.00	61.41	56.57	54.72		
C. PLN.PROD/TOTAL COST	%		2.11	2.56	2.10	2.62	1.98	1.35	1.19		
D.PROFIT BEFORE TAX	$MM	162.44	90.26	203.16	250.02	315.02	449.99	625.01	750.03	11.83	29.85
E.ROS	%	6.42	3.65	6.61	6.94	7.94	9.64	11.60	12.46		
F.TOTAL AVG ASSETS	$MM	1091.59	1205.16	1245.13	1452.87	1723.21	1973.56	2188.45	2455.73		
G. ASSET(AVG) TURNOVER	X	2.32	2.05	2.47	2.48	2.30	2.36	2.46	2.45		
H. WORKING CAPITAL TURNOVER (AVG)	X	5.62	5.30	6.91	7.52	7.89	8.38	8.43	8.14		
I.ROA (AVG ASSETS)	%	14.88	7.49	16.32	17.21	18.28	22.80	28.56	30.54		
J.ESTIMATED CAPITAL EXPENDITURES	$MM	61.59	77.98	108.23	275.55	254.40	185.75	197.58	167.76		
K.M&AE EXPENSE	$MM	266.71	296.08	347.23	412.00	468.82	571.48	671.20	744.11	14.10	16.47
L. MAE/SALES	%	10.54	11.97	11.30	11.44	11.82	12.24	12.45	12.36		
M.R&D EXPENSE	$MM	54.90	57.90	67.30	80.40	90.40	101.40	116.20	133.10	10.72	14.61
N. R&D/SALES	%	2.17	2.34	2.19	2.23	2.28	2.17	2.16	2.21		
O.CHANGE IN WORKING CAPITAL	$MM		17.10	(22.56)	33.76	24.43	54.23	82.12	100.00		
P.CASH FLOW (OPERATIONAL)	$MM	0	51.80	138.86	(10.13)	107.95	245.26	326.13	415.95		
Q.CONTINGENCY	$MM	0	0	0	(76.68)	(126.30)	(220.69)	(297.76)	(337.37)		

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COMPARISON OF 1980 STRATEGIC PLAN WITH 1979 STRATEGIC PLAN

RESULTS FOR THE YEARS 1980-1983
1980 PLAN BETTER (WORSE) THAN 1979 PLAN

	1980			1981			1982			1983		
	'80 Plan	B(W) '79 Plan		'80 Plan	B(W) '79 Plan		'80 Plan	B(W) '79 Plan		'80 Plan	B(W) '79 Plan	
Sales	$3,600.2	$ 594.5	19.8%	$3,965.4	$ 633.7	19.0%	$4,667.1	$ 961.4	25.9%	$5,390.2	$1,338.7	33.0%
PBT	$ 250.0	$ (82.4)	(24.8)%	$ 315.0	$(111.3)	(26.1)%	$ 450.0	$(102.9)	(18.6)%	$ 625.0	$ (11.7)	(1.8)%
Total Ending Assets	$1,560.2	$(148.5)	(10.5)%	$1,822.7	$(267.8)	(17.2)%	$2,016.4	$(349.6)	(21.0)%	$2,229.3	$ (444.1)	(24.9)%
Return on Sales	6.9%	(4.2)pts.		7.9%	(4.9)pts.		9.6%	(5.3)pts.		11.6%	(4.1)pts.	
Asset Turnover (Ending)	2.31X	.18X		2.18X	.04X		2.31X	.09X		2.42X	.15X	

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SALES, INCOME, AND ASSET RECONCILIATION
1980 STRATEGIC PLAN FOR 1980 - 1979 STRATEGIC PLAN FOR 1980
($ Millions)

	Sales	Pre-Tax Income	Ending Net Assets
1980 Strategic Plan for 1980	$3,600.2	$ 250.0	$1,560.2
1979 Strategic Plan for 1980	3,005.7	332.4	1,411.7
Change - Better/(Worse)	$ 594.5	$ (82.4)	$ (148.5)
Causal Factor (Change)			
Volume			
Parts	$ 8.7	$ 2.4	$ (1.9)
Industry Volume	205.0	56.2	(44.0)
IH Market Share	110.5	30.3	(23.7)
Wholesale Inventory	43.7	12.0	$ (9.4)
Mix	30.7	8.4	(6.6)
Intergroup Sales	(24.2)	(6.6)	-
Sub-Total Volume	$ 374.4	$ 102.7	$ (85.6)
Economics			
Selling Price Increase			
Economic Increase	$ 113.6	$ 113.6	$ -
Sales Allowance Increase	(41.0)	(41.0)	-
Net Selling Price Increase	$ 72.6	$ 72.6	$ -
Cost Increase	-	(115.0)	(24.4)
Sub-Total Economics	$ 72.6	$ (42.4)	$ (24.4)
Performance			
Cost Improvement Programs	$ -	$ (7.4)	$ -
Engineering Expense	-	(2.0)	-
M&A Expense	-	(96.7)	-
Interest Cost	-	(10.6)	-
Sub-Total Performance	$ -	$(116.7)	$ -
Other			
Exchange Rate	$ 147.5	$ 7.6	$ (68.4)
Cost Improvement Prgs. Expense	-	(42.2)	-
Capital Expenditures	-	-	(22.1)
Discounted Receivables	-	-	52.0
Work Interrupt/Projection Delays	-	8.6	-
Sub-Total Other	$ 147.5	$ (26.0)	$ (38.5)
Total Change	$ 594.5	$ (82.4)	$ (148.5)

Comments:

The forecasted 1980 contingency of $45.1 MM was lumped as a "one-line" item in the October 1978, P1 submission. The causal factor explanation of change presented has been adjusted to account for the spreading of that contingency by expense category - consistent with this year's contingency handling - as follows:

	$ MM
M&A Expense	$19.5
Engineering Expense	17.0
Work Interruptions/Production Delays	8.6
Total	$45.1

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TOTAL PLAN SUMMARY (A)

THE PLAN: GROWTH AND PROFITABILITY - Strengthen dealer/distributor organization, expend capital to reduce manufacturing costs and introduce new products.

$ MARKET GROWTH	
Smooth growth 1980-84.	
Real	3.1%
Inflation	6.7
Total	9.8%

NET SALES GROWTH	
Fluctuating-highs of 17+% in '80 &'82; lows of 10-11% in '81 & '84.	
Real	7.5%
Pricing	6.9%
Total	14.4%

COMPETITIVE PRESSURE
High-John Deere is dominant competitor worldwide.

$ MARKET SHARE
Smooth increase from 16.0% (1979) to 20.3% (1984); NAO leads with increase from 19.1% (1979) to 24.2% (1984).

PROFITABILITY
ROA increases from 16.3% (1979) to 30.5% (1984).
ROS increases from 6.6% (1979) to 12.5% (1984).

CUMULATIVE CASH FLOW (B)
$1,085 million (1980-84), with $742 MM occurring in 1983-84, and negative flow in 1980.

PRICING
Average 6.9% annual increase is greater than market inflation rate, but equal to cost inflation rates.

COST INCREASE
Average variable and fixed cost inflation rates 6.9%.

PRODUCTIVITY %-OPERATING INCOME
Average 65.2% (1980-84).

M & A
Increases from 11.3% (1979) to 12.4% (1984) of sales.

R & D
Level 1980-84 at 2.2% of sales.

CAPITAL EXPENDITURES
$1,081 million total (1980-84), with $530 MM occurring 1980-81.

(A) After contingencies.

(B) Cash flow has been adjusted for contingencies at Group level only. Individual SBU/Segment Pl cash flows have not been adjusted.

PROFILE (P-1)

		HISTORY		PROJECTION						GROWTH	
		1977	1978	1979	1980	1981	1982	1983	1984	77-79	79-84
A.AVAIL RETAIL MARKET	$MM	13723.3	14202.7	15768.8	17067.6	18805.4	20554.3	22741.0	25153.8	7.2	9.8
B.IH RETAIL MARKET SHARE	%	14.26	14.09	15.98	17.01	17.51	18.62	19.48	20.26		
C.LARGEST COMP SHARE	%	17.70	18.30	17.90	18.40	18.60	18.60	19.10	19.50		
D.COMPETITIVE STRENGTH		M	M	M	M	M	H	H	H		
E.NET SALES	$MM	2529.8	2474.2	3073.5	3600.2	3965.4	4667.1	5390.2	6019.3	10.2	14.4
F.CONTRIBUTION MARGIN	%	29.00	28.74	28.62	27.96	29.07	29.75	30.69	31.17		
G.TOTAL FIXED COST	$MM	520.9	570.6	625.2	694.2	760.8	852.7	935.9	1026.8	9.6	10.4
H.OPERATING INCOME	$MM	284.0	145.3	263.7	312.6	392.3	536.1	718.5	850.1	(3.5)	26.4
I.PROFIT BEFORE TAX	$MM	162.4	90.3	203.2	250.0	315.0	450.0	625.0	750.0	11.8	29.9
J.TOTAL AVG ASSETS	$MM	1091.6	1205.2	1245.1	1452.9	1723.2	1973.6	2188.5	2455.7		
K.TOTAL ENDING ASSETS	$MM	1092.2	1226.1	1260.7	1560.2	1822.7	2016.4	2229.3	2501.4	7.4	14.7
L.CASH FLOW (OPERATIONAL)	$MM	0	51.8	138.9	(10.1)	107.9	245.3	326.1	415.9		
M.MARKET GROWTH	%		3.49	11.03	8.24	10.18	9.30	10.64	10.61		
N.SALES GROWTH	%		(2.20)	24.22	17.14	10.14	17.70	15.49	11.67		
O.PBT GROWTH	%		(44.44)	125.09	23.06	26.00	42.84	38.89	20.00		
P.ANNUNAL MKT INFLATION	INDX		1.084	1.073	1.068	1.067	1.067	1.067	1.067		
Q.ANNUAL PR INFL	INDX		1.089	1.080	1.069	1.066	1.072	1.070	1.067		
R.ANNUAL VC INFL	INDX		1.085	1.076	1.071	1.068	1.070	1.071	1.067		
S.ANNUAL FC INFL	INDX		1.085	1.074	1.071	1.067	1.069	1.069	1.069		
T.ROA (AVG ASSETS)	%	14.9	7.5	16.3	17.2	18.3	22.8	28.6	30.5		
U.ROS	%	6.4	3.6	6.6	6.9	7.9	9.6	11.6	12.5		
V.ASSET(AVG) TURNOVER	X	2.32	2.05	2.47	2.48	2.30	2.36	2.46	2.45		
W.NET PRDTVTY/OP.INC	%	0	34.7	47.2	62.3	74.1	69.8	61.0	58.8		
X.R&D/SALES	%	2.2	2.3	2.2	2.2	2.3	2.2	2.2	2.2		
Y.MAE/SALES	%	10.5	12.0	11.3	11.4	11.8	12.2	12.5	12.4		
Z.ESTIMATED CAPITAL EXPENDITURES	$MM	61.6	78.0	108.2	275.6	254.4	185.8	197.6	167.8		
AA. PRODUCT CAPITAL EXPENDITURES	%TOT	54.5	36.9	43.1	57.9	50.0	51.2	57.3	57.6		
BA. COST-REDUCTION CAP. EXPEND.	%TOT	6.3	16.0	24.7	16.0	17.1	8.6	6.1	7.6		
CA. OTHER CAPITAL EXPENDITURES	%	39.1	47.1	32.2	26.1	32.9	40.2	36.6	34.8		
DA.R&D EXPENSE	$MM	54.9	57.9	67.3	80.4	90.4	101.4	116.2	133.1	10.7	14.6
EA. NEW PRODUCT R&D	%R&D	21.7	39.9	27.7	21.4	18.2	17.1	17.1	15.3		
FA.M&AE EXPENSE	$MM	266.7	296.1	347.2	412.0	468.8	571.5	671.2	744.1	14.1	16.5
GA.NET PLANNED PRODUCTIVITY (CUM)	$MM	0	50.4	124.4	194.9	290.8	374.1	438.1	500.3		
HA.CONTINGENCY	$MM	0	0	0	(76.7)	(126.3)	(220.7)	(297.8)	(337.4)		
IA.PARTS REVENUE	$MM	367.5	426.2	499.5	556.8	642.6	742.8	845.4	962.5	16.6	14.0
JA.PARTS VARIABLE MARGIN	%	44.41	41.86	43.31	42.07	42.29	42.72	42.60	43.55		

COMPOUND ANNUAL GROWTH RATES

AGRICULTURAL EQUIPMENT GROUP
CAUSAL FACTOR ANALYSIS

SALES

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR SALES	$MM		3073.47	3600.16	3965.37	4667.13	5390.21
B. AVAIL MARKET EFFECT	$MM		34.44	93.74	80.68	139.62	162.03
C. SELLING PRICE EFFECT	$MM		230.91	246.84	313.19	354.77	375.51
D. MARKET SHARE EFFECT	$MM		165.99	87.62	214.07	182.18	183.76
E. WHL/DLR INV EFFECT	$MM		97.28	(65.81)	110.79	58.17	(94.57)
F. REMAINDER SALES	$MM		(1.94)	2.83	(16.97)	(11.65)	2.37
G.NET SALES	$MM	3073.47	3600.16	3965.37	4667.13	5390.21	6019.32

PROFIT BEFORE TAX

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR	$MM		203.16	250.02	315.02	449.99	625.01
B. AVAIL MARKET EFFECT	$MM		9.86	26.21	23.45	41.54	49.73
C. COST INFL EFFECT	$MM		(216.97)	(226.4)	(269.4)	(307.19)	(326.82)
D. SELLING PRICE EFFECT	$MM		230.91	246.84	313.19	354.77	375.51
E. MARKET SHARE EFFECT	$MM		47.5	24.5	62.23	54.2	56.39
F. WHOLESALE/DEALER INV EFFECT	$MM		27.84	(18.4)	32.2	17.3	(29.02)
G. VARIABLE COST CHG	$MM		(17.56)	44.33	23.89	48.4	29.32
H. FIXED COST CHG	$MM		(22.88)	(18.55)	(36.76)	(22.94)	(24.53)
I. REMAINDER	$MM		(11.85)	(13.52)	(13.84)	(11.05)	(5.57)
J.PROFIT BEFORE TAX	$MM	203.16	250.02	315.02	449.99	625.01	750.03

Sales Commentary

Annual sales in 1984 are expected to be about $2946 MM greater than 1979 sales, a 96% increase. Selling price effects are responsible for about 50% of the increase, market share gains for about 28%, and market growth for 17%. Largest sectors, Europe and N. America, are essentially replacement; thus market growth is due to trend to larger, more sophisticated models, and cyclical recovery in some markets. Five global segments, 115-200 H.P. Tractors, grain harvesters, 40-115 H.P. tractors, planters, and outdoor power products, are categorized for growth.

Income Commentary

Annual income in 1984 is expected to be about $547 MM greater than 1979 income, a 270% increase. Market share gains account for about 45% of the increase, retention of net inflation benefits account for about 32%, market growth for about 28%, and variable cost improvements about 23%. Extensive new product development and product improvement programs, combined with aggressive marketing programs, are expected to bring about greater market penetration. Pricing strategy involves the retention of productivity gains and the pass-through of cost increases to the extent permitted by competitive pressures. Variable cost improvements are expected from the favorable impact of productivity programs, manufacturing efficiencies associated with new products, and reductions in the sales finance, warranty, and sales allowance expenses associated with the new, more competitive product offerings, partially offset by the higher S&A costs of marketing programs, the start-up costs of the High Horsepower tractor program, and the outside sourcing of under 40 H.P. tractors after 1982.

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AGRICULTURAL EQUIPMENT GROUP
PRODUCTIVITY PROGRAMS
($ Millions)

	Through 1979	1980	1981	1982	1983	1984
TARGET/2%	$135	$223	$315	$447	$595	$753
GROUP PLAN	156	252	361	463	543	623
PLAN O/(U) TARGET/2%	21	29	46	16	(52)	(130)

NET PLANNED PRODUCTIVITY PROGRAMS

SBU Identified:

1. LUIP	$ 79.9	$128.2	$166.2	$183.4	$196.0	$209.4
2. Quality	1.3	2.5	2.9	3.2	3.2	4.1
3. Purchasing	24.5	34.3	48.3	63.5	79.8	91.0
4. In-Sourcing	1.8	2.2	3.6	15.5	17.0	18.6
5. Staff Expense	18.9	22.2	26.5	33.0	35.3	37.7
6. Capital Programs	7.1	6.5	34.8	57.2	90.6	128.2
7. Components Pass-Through	-	30.0	32.6	35.6	38.2	43.5
8. Value Analysis	10.8	14.1	17.5	33.6	39.6	43.7
9. Misc. Manufacturing Programs	11.9	12.2	28.3	38.4	43.5	46.5
TOTAL GROSS PRODUCTIVITY	$156.2	$252.2	$360.7	$463.4	$543.2	$622.7
Cost of Programs	($31.8)	(57.3)	(69.9)	(89.3)	(105.1)	(122.4)
TOTAL NET PRODUCTIVITY	$124.4	$194.9	$290.8	$374.1	$438.1	$500.3

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1980-84 STRATEGIC PLAN -
BEFORE AND AFTER CONTINGENCY RECONCILIATION

	1980			1981			1982			1983			1984		
	Before	Cont.	After	Before	Cont.	After	Before	Cont.	After	Before	Cont.	After	Before	Cont.	After
• Net Sales ($MM)															
40-115 H.P. Tractor	1061.4	20.5	1040.9	1111.0	32.5	1078.5	1238.0	46.0	1192.0	1386.6	52.0	1334.6	1664.0	60.0	1604.0
115-200 H.P. Tractor	838.6	20.5	818.1	934.2	32.5	901.7	1133.8	46.0	1087.8	1336.9	52.0	1284.9	1459.4	60.0	1399.4
Grain Harvesting	641.2	-	641.2	765.8	8.0	757.8	869.9	26.0	843.9	1099.9	55.0	1044.9	1199.4	60.0	1139.4
Total AEG	3641.2	41.0	3600.2	4038.4	73.0	3965.4	4785.1	118.0	4667.1	5549.1	159.0	5390.2	6199.3	180.0	6019.3
• M&A Expense ($MM)															
40-115 H.P. Tractor	77.4	12.0	89.4	82.3	14.0	96.3	86.4	26.0	112.4	93.3	36.5	129.8	100.7	45.5	146.2
115-200 H.P. Tractor	94.1	12.0	106.1	104.4	14.0	118.4	120.0	26.0	146.0	139.0	36.5	175.5	154.6	45.5	200.1
Grain Harvesting	71.4	6.0	77.4	80.5	14.0	94.5	89.1	19.0	108.1	105.6	31.0	136.6	116.3	33.0	149.3
Planters	18.9	3.0	21.9	23.0	8.0	31.0	29.9	24.0	53.9	35.4	25.0	60.4	38.3	23.0	61.3
Hay, Forage, M.H.	30.8	3.0	33.8	33.9	3.0	36.9	36.9	8.0	44.9	40.0	10.0	50.0	43.6	11.0	54.6
Total AEG	376.0	36.0	412.0	415.8	53.0	468.8	468.5	103.0	571.5	532.2	139.0	671.2	586.1	158.0	744.1
• Profit Before Tax															
40-115 H.P. Tractor	67.3	32.5	34.8	79.9	46.5	33.4	111.7	72.0	39.7	139.8	88.5	51.3	222.1	105.5	116.6
115-200 H.P. Tractor	94.5	32.5	62.0	102.4	46.5	55.9	160.2	72.0	88.2	232.1	88.5	143.6	264.9	105.5	159.4
Grain Harvesting	110.8	6.0	104.8	153.3	22.0	131.3	187.0	45.0	142.0	279.0	86.0	193.0	315.3	93.0	222.3
Planters	(5.5)	3.0	(8.5)	25.5	8.0	17.5	63.5	24.0	39.5	84.1	25.0	59.1	81.7	23.0	58.7
Hay, Forage, M.H.	17.8	3.0	14.8	23.7	3.0	20.7	31.6	8.0	23.6	43.7	10.0	33.7	48.3	11.0	37.3
Total AEG*	250.0*	77.0	250.0	315.0*	126.0	315.0	450.0*	221.0	450.0	625.0*	298.0	625.0	750.0*	338.0	750.0
• M&A as a % of Sales															
40-115 H.P. Tractor	7.3		8.6	7.4		8.9	7.0		9.4	6.7		9.7	6.0		9.1
115-200 H.P. Tractor	11.2		13.0	11.2		13.1	10.6		13.4	10.4		13.7	10.6		14.3
Grain Harvesting	11.1		12.1	10.5		12.5	10.2		12.8	9.6		13.1	9.7		13.1
Planters	16.4		19.1	12.2		16.4	10.3		18.5	10.6		18.1	11.3		18.1
Hay, Forage, M.H.	12.1		13.3	12.3		13.4	12.1		14.7	11.6		14.5	11.6		14.5
Total AEG	10.3		11.4	10.3		11.8	9.8		12.2	9.6		12.5	9.5		12.4
• ROA															
40-115 H.P. Tractor	14.7		7.6	15.8		6.6	20.4		7.2	22.4		8.2	27.7		14.6
115-200 H.P. Tractor	33.8		22.2	26.2		14.3	34.9		19.2	46.7		28.9	50.8		30.5
Grain Harvesting	47.3		44.7	54.3		46.5	58.6		44.5	77.4		53.5	82.3		58.0
Planters	(7.7)		(11.9)	28.0		19.3	52.6		32.7	64.4		45.3	62.1		44.6
Hay, Forage, M.H.	15.6		13.0	19.0		16.6	22.5		16.8	28.1		21.7	28.8		22.2
Total AEG*	17.2*		17.2	18.3*		18.3	22.8*		22.8	28.6*		28.6	30.5*		30.5
• ROS															
40-115 H.P. Tractor	6.3		3.3	7.2		3.1	9.0		3.3	10.1		3.8	13.3		7.3
115-200 H.P. Tractor	11.3		7.6	11.0		6.2	14.1		8.1	17.4		11.2	18.2		11.4
Grain Harvester	17.3		16.3	20.0		17.3	21.5		16.8	25.4		18.5	26.3		19.5
Planters	(4.8)		(7.4)	13.5		9.3	21.8		13.5	25.2		17.7	24.1		17.3
Hay, Forage, M.H.	7.0		5.8	8.6		7.5	10.3		7.7	12.7		9.8	12.8		9.9
Total AEG*	6.9*		6.9	7.8*		7.9	9.4*		9.6	11.3*		11.6	12.1*		12.5
• MEMO:															
Contingency by Program Type ($MM)															
Market Penetration Programs		26			34			69			93			107	
New Product Introductions		10			19			34			46			51	
Sales Allowances		41			73			118			159			180	
Total Contingency		77			126			221			298			338	
Contingency as % Net Sales		2.1%			3.2%			4.7%			5.5%			5.6%	

*Group level AEG Plan is "after contingency" only.

AGRICULTURAL EQUIPMENT GROUP

GLOBAL SBU/SEGMENT PROFIT BEFORE TAX
FROM EQUIPMENT AND PARTS
($ Millions)

SBU/SEGMENT	1979 Total	1979 Equipment	1979 Parts	1980 Total	1980 Equipment	1980 Parts	1981 Total	1981 Equipment	1981 Parts	1982 Total	1982 Equipment	1982 Parts	1983 Total	1983 Equipment	1983 Parts	1984 Total	1984 Equipment	1984 Parts
Under 40 H.P. Tractors	(6.9)	(8.1)	1.2	(6.2)	(7.1)	0.9	2.6	1.6	1.0	10.8	9.8	1.0	7.4	6.5	0.9	9.3	8.1	1.2
40-115 H.P. Tractors	51.3	15.2	36.1	34.8	1.5	33.3	33.4	(5.2)	38.6	39.7	(5.2)	44.9	51.3	1.8	49.5	116.6	48.2	68.4
115-200 H.P. Tractors	58.5	35.4	23.1	62.0	34.9	27.1	55.9	24.2	31.7	88.2	52.5	35.7	143.6	101.3	42.3	159.4	107.8	51.6
200 + H.P. Tractors	(3.1)	(6.6)	3.5	3.0	(0.9)	3.9	5.4	0.3	5.1	9.1	2.8	6.3	22.5	14.0	8.5	25.4	13.6	11.8
Grain Harvesting	94.1	76.4	17.7	104.8	85.8	19.0	131.3	108.2	23.1	142.0	114.9	27.1	193.0	158.4	34.6	222.3	182.2	40.1
Cotton Harvesting	(3.1)	(6.0)	2.9	2.6	(1.3)	3.9	1.2	(2.4)	3.6	20.4	13.7	6.7	22.5	15.1	7.4	21.5	13.5	8.0
Planters	(3.7)	(8.9)	5.2	(8.5)	(13.7)	5.2	17.5	8.1	9.4	39.5	25.7	13.8	59.1	42.8	16.3	58.7	41.1	17.6
Tillage	15.4	5.0	10.4	25.2	13.7	11.5	35.2	22.0	13.2	56.1	41.4	14.7	68.2	51.9	16.3	73.9	54.6	19.3
Hay, Forage, M. H.	6.8	(1.3)	8.1	14.8	5.5	9.3	20.7	9.8	10.9	23.6	11.0	12.6	33.7	19.0	14.7	37.3	20.2	17.1
Outdoor Power	3.8	2.5	1.3	16.3	14.7	1.6	11.6	9.9	1.7	17.4	15.4	2.0	17.5	15.4	2.1	15.6	13.0	2.6
Total AEG*	213.1	103.6	109.5	248.8	133.1	115.7	314.8	176.5	138.3	446.8	282.0	164.8	618.8	426.2	192.6	740.0	502.3	237.7
OPERATION																		
NAO	157.0	86.0	71.0	203.6	122.6	81.0	251.2	155.8	95.4	375.8	262.0	113.8	530.1	394.1	136.0	584.0	423.1	160.9
EAME	56.3	18.8	37.5	37.0	2.9	34.1	50.0	8.4	41.6	56.4	7.4	49.0	73.9	19.4	54.5	140.1	66.0	74.1
PFELA	(1.0)	(2.0)	1.0	9.1	8.6	.5	14.1	12.8	1.3	17.6	15.9	1.7	20.8	18.8	2.0	25.5	22.7	2.8
Total AEG*	212.3	102.8	109.5	249.7	134.1	115.6	315.3	177.0	138.3	449.8	285.3	164.5	624.8	433.3	192.5	749.6	511.8	237.8

* Differs from Group P-1 because of rounding and inclusions/eliminations.

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AGRICULTURAL EQUIPMENT GROUP

1980 RESOURCE PLAN BY GLOBAL SBU/SEGMENT
($ Millions)

SBU/SEGMENT	CATEGORY	AVERAGE NET ASSETS	TOTAL CAPITAL EXPENDITURES	PRODUCT CAPITAL EXPENDITURES	COST REDUCTION CAPITAL EXPENDITURES	OTHER CAPITAL EXPENDITURES	TOTAL R&D	TOTAL M&A
115-200 H.P. Tractor	Growth (Aggressive)	279.2	121.9	96.6	19.4	5.9	23.7	106.1
Grain Harvesting	Growth (Aggressive)	234.2	32.0	19.1	2.9	10.0	14.0	77.4
40-115 H.P. Tractors	Growth	456.8	55.6	22.1	4.0	29.5	17.3	89.4
Planters	Growth	71.2	10.8	7.3	1.4	2.1	5.8	21.9
Outdoor Power	Growth	27.6	1.8	1.1	0	0.7	2.1	15.8
[redacted]								
Cotton Harvesting	Profit	33.5	9.9	2.8	4.0	3.1	3.5	7.4
Tillage	Profit	116.3	25.3	3.0	7.4	14.9	4.7	34.1
Hay, Forage, M.H.	Profit	114.2	9.9	2.4	3.1	4.4	6.0	33.8
Sub-total	Profit	264.0	45.1	8.2	14.5	22.4	14.2	75.3
< 40 H.P. Tractors	Explore	21.6	0.3	0	0	0.3	.2	5.6
> 200 H.P. Tractors	Explore	30.4	5.9	5.2	0.6	0.1	2.9	10.8
Sub-total	Explore	52.0	6.2	5.2	0.6	0.4	3.1	16.4
Total*		1385.0	273.4	159.6	42.8	71.0	80.2	402.3

* Differs from Group P1 because of rounding and inclusions/eliminations; rows may not add because of rounding.

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AGRICULTURAL EQUIPMENT GROUP
1980 MAJOR CAPITAL PROGRAMS OVER $5.0MM
($ MM)

Program (SBU/Segment)	APPROPRIATION Year	APPROPRIATION Amount	EXPENDITURES 1979	EXPENDITURES 1980	EXPENDITURES 1981	EXPENDITURES 1982	EXPENDITURES 1983
*TR-4 Tractor(115-200 HP) Prg. I&II	1978	$122.6	$ 7.5	$ 93.4	$ 21.7	$	$
*Auto Storage & Dist. System - Farmall (115-200 HP)	1979	25.0	0.6	13.8	10.6		
*New C-85 Control Center(40-115 HP)	1979	28.3	4.0	20.4	3.9		
Disc Blade Mfg. (Tillage)	1979	10.3		5.6	4.7		
New Air Planters Models(Planters)	1979	10.8	0.7	7.2	2.2	0.6	0.1
*Combine Capacity-NAO(Grain Harv.)	1979	12.3	1.2	8.2	2.9		
TR-4 Tractor Program III(115-200 HP)	1980	31.8		6.9	24.9		
Sub-Total(Over $10MM)		$241.1	$14.0	$155.5	$70.9	$ .6	$ .1
Misc. Programs($5-10MM) defined in Segment exhibits		54.4	7.5	20.2	18.6	8.1	-
Total Major Programs(Over $5.0MM)		$295.5	$21.5	$175.7	$89.5	$8.7	$ .1

*Project appropriation approved.

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AGRICULTURAL EQUIPMENT GROUP

1980-84 RESOURCE PLAN BY GLOBAL SBU/SEGMENT
($ Millions)

SBU/SEGMENT	CATEGORY	1984 AVERAGE NET ASSETS	TOTAL CAPITAL EXPENDITURES	PRODUCT CAPITAL EXPENDITURES	COST REDUCTION CAPITAL EXPENDITURES	OTHER CAPITAL EXPENDITURES	CUMULATIVE TOTAL R&D	CUMULATIVE TOTAL M&A
115-200 H.P. Tractors	Growth (Aggressive)	521.8	340.5	252.0	55.1	33.4	126.0	746.1
Grain Harvesting	Growth (Aggressive)	383.3	160.6	95.1	6.1	59.4	89.8	565.9
40-115 H.P. Tractors	Growth	800.5	330.7	139.6	16.1	175.0	128.9	574.1
Planters	Growth	131.6	37.6	20.1	5.7	11.8	44.5	228.5
Outdoor Power	Growth	46.7	21.1	10.5	1.4	9.2	12.5	105.5
[redacted]								
Cotton Harvesting	Profit	55.2	31.1	14.2	6.8	10.1	22.2	48.8
Tillage	Profit	183.2	78.5	21.5	18.4	38.6	34.6	205.2
Hay, Forage, M.H.	Profit	167.8	49.6	22.2	8.8	18.6	38.9	220.2
Sub-total	**Profit**	**393.9**	**159.2**	**57.9**	**34.0**	**67.3**	**95.7**	**474.2**
<40 H.P. Tractors	Explore	16.1	2.3	.8	.2	1.3	1.0	30.7
>200 H.P. Tractors	Explore	61.7	17.3	15.4	1.1	0.8	22.1	76.1
Sub-total	**Explore**	**77.8**	**19.6**	**16.2**	**1.3**	**2.1**	**23.1**	**106.8**
Total*		2355.6	1069.3	591.4	119.7	358.2	520.5	2801.1

*Differs from Group P1 because of rounding and inclusions/eliminations; rows may not add because of rounding.

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AGRICULTURAL EQUIPMENT GROUP
1980-84 MAJOR CAPITAL PROGRAMS OVER $5.0 MM
($ MM)

Program (SBU/Segment)	APPROPRIATION Year	APPROPRIATION Amount	EXPENDITURES -1979	1980	1981	1982	1983	1984	1985+
*TR-4 Tractor (115-200 HP) Prg. I&II	1978	$122.6	$ 7.5	$ 93.4	$ 21.7	$	$	$	$
*Auto Storage & Dist. Systm.-Farmall (115-200 HP)	1979	25.0	0.6	13.8	10.6				
*New C-85 Control Center (40-115 HP)	1979	28.3	4.0	20.4	3.9				
Disc Blade Mfg. (Tillage)	1979	10.3		5.6	4.7				
New Air Planters Models (Planters)	1979	10.8	0.7	7.2	2.2	0.6	0.1		
*Combine Capacity-NAO (Grain Harv.)	1979	12.3	1.2	8.2	2.9				
TR-4 Tractor Program III (115-200 HP)	1980	31.8		6.9	24.9				
New 4 W D Tractors (Over 200 HP)	1981	10.0			5.0	5.0			
3988 Tractor (115-200 HP)	1981	23.3			10.0	13.3			
Increase Farmall Capacity (115-200 HP)	1981	10.0			10.0				
Increase WW Tractor Cap.-II (40-115 HP)	1981	23.5				10.0	13.5		
TR-1 Tractor Program (40-115 HP)	1981	10.0			10.0				
TR-2 Tractor Program (40-115 HP)	1981	58.4				11.1	44.4	2.9	
Vari-Range Transmission (TR-4) (115-200 HP)	1982	36.0				4.0	22.0	10.0	
TR-3 Tractor Program (115-200 HP)	1983	9.7(A)					2.3	5.0	2.4
Doncaster Foundry Expansion -II (40-115 HP)	1983	11.0					1.2	4.9	4.9
TR-3 Tractor Program (40-115 HP)	1983	32.2(A)					9.0	22.1	1.1
European Axial Flow Combines (Grain Harv.)	1984	10.7						1.5	9.2
2100 Series Combines (Grain Harv.)	1984	41.5							41.5
Sub-Total (over $10MM)		$517.4	$14.0	$155.5	$105.9	$44.0	$92.5	$46.4	$59.1
Misc. Programs($5-10 MM) defined in segment exhibits		115.9	7.5	20.2	23.8	29.1	18.1	9.0	8.2
Total Major Programs(Over $5.0 MM)		$633.3	$21.5	$175.7	$129.7	$73.1	$110.6	$55.4	$67.3

(A) Segment Portion of Total TR-3 Program ($41.9 MM).

* Project appropriation approved.

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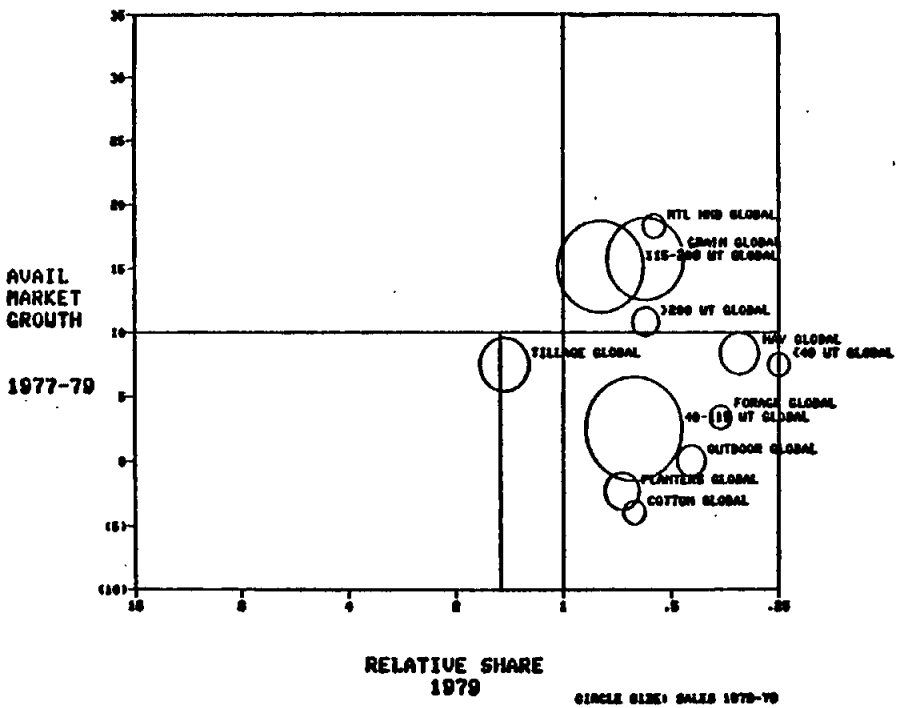
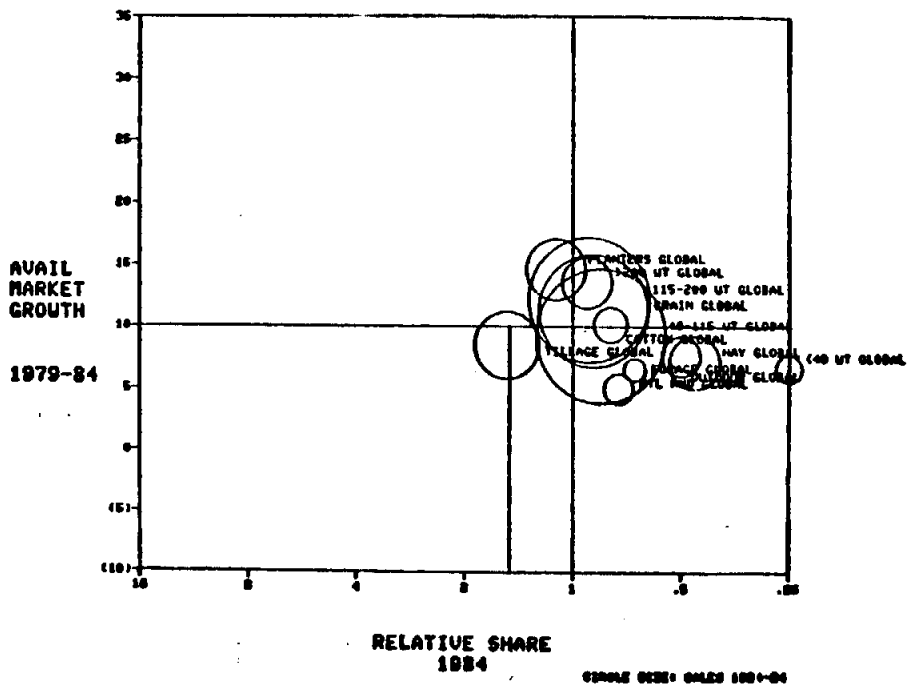
AGRICULTURAL EQUIPMENT GROUP

HISTORIC STRATEGIC POSITIONING



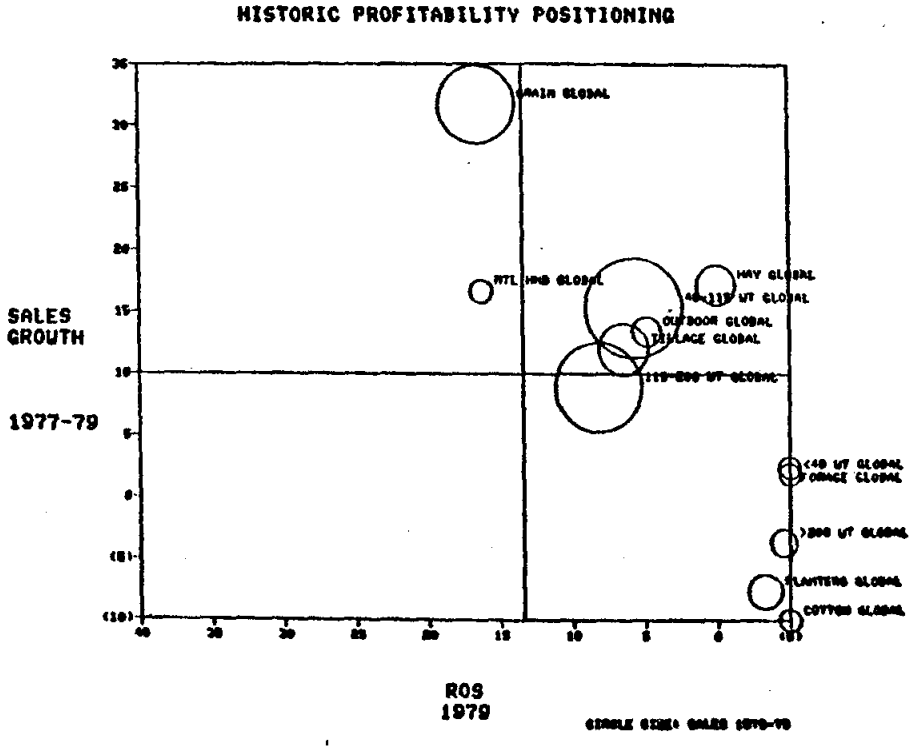
PROJECTED STRATEGIC POSITIONING

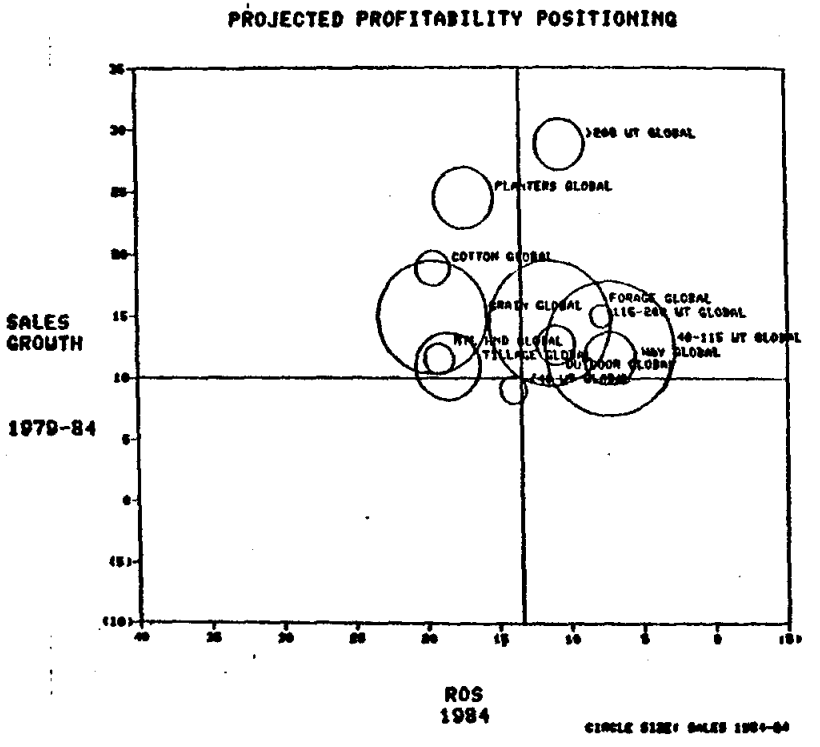


COMMENTS:

- The Historic Strategic Positioning chart shows a dominant share position for Tillage equipment, and close to parity with the weighted average of leading competitors for 115-200 H.P. Tractors. The remainder of the businesses are in the range of .7X to .25X relative share of leading competitors.
- Historical growth rates have been low for most AEG Global Segments.
- The primary change in the portfolio in 1984 is the uniformity of expected market growth rates and increased relative share for almost all businesses.
- In both periods there is a total lack of high real growth markets.

AGRICULTURAL EQUIPMENT GROUP





COMMENTS:

- In 1979 negative ROS is expected for five of AEG's 12 global businesses.
- Grain Harvesting and Material Handling experienced high sales growth and high profitability from 1977 to 1979.
- The projected profitability positioning in 1984 shows significant improvement in most businesses' ROS except for 40-115 H.P. Tractors.
- The dramatic shift is in the rapid growth and simultaneous profitability improvement of businesses with a negative ROS in 1979.
- If the planned profitability improvement does not come with the growth rates projected, AEG will be a serious cash drain on the total company.

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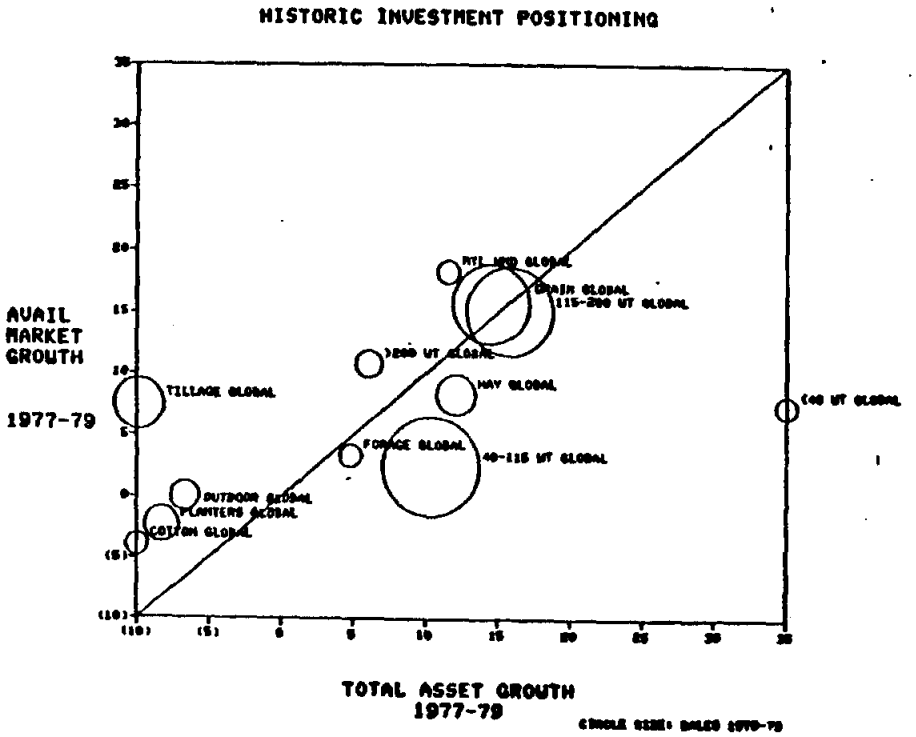

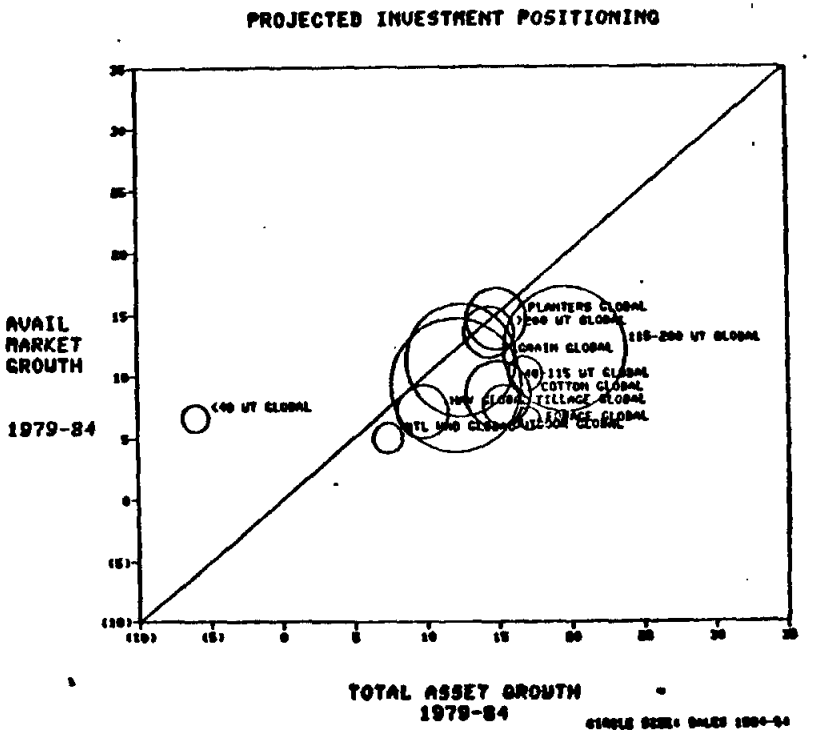

COMMENTS:

- The projected investment positioning chart shows a dramatic shift in the selectivity of AEG investments. From 1977 to 1979 assets grew faster than market growth rates in only two Segments. During 1979-84 assets are planned to grow faster than market growth rates in all but one Segment (OEM sourced).

- Much of the increase in asset commitments to the AEG businesses is in the form of fixed capital. Consequently, high priority should be placed on close monitoring of planned share gain achievement and profitability improvement.

AGRICULTURAL EQUIPMENT GROUP

KEY ISSUES

The AEG plan is an aggressive drive for worldwide leadership in agricultural equipment. Accomplishment of the goals set forth will require large amounts of capital, and increased spending levels in marketing, distribution development, and R&D.

There is little question but that several AEG SBU's are core IH businesses, and that aggressive growth plans are warranted. The key issues at the Group level have to do with how well the Group has differentiated strategies by SBU, targeting of specific strategies and resources on product/market segments with the highest payoff/risk ratio, and monitoring and control of progress in achieving objectives.

Specific questions relating to the above issues are:

- * Has a priority system been established for product/market segments on a worldwide basis? What research has been done on market potential and the competitiveness of AEG to provide a basis for these priorities?(1979-A16)
- What priority is being placed on Comecon Countries, China, and LDC's? What programs are being developed to support these priorities?
- * What progress is being made in implementing segmentation of the NAO tractor market to identify differentials in potential, competitive performance, and specific marketing programs? (1979-A2 & A13)
- What steps are being taken to improve the marketing research data base for marketing decision making, e.g., competitive analysis, customer attitudes, changing farming practices such as low-tillage, etc.?
- What specific programs exist (other than adding 500 salesmen) to upgrade the dealer distributor organization - in service support, in capitalization, in finance, etc.?

* • Research and Development expenditures are expected to grow at the same rate as sales--about 15% per year--but will still equal only 1/2 of Deere's expected expenditures through 1984. Should we spend more? Should spending be focused on a few specific areas or "across the board"? (1979-A17).

• How will the new organization structure aid in decision making, communication, and control? Will the SBU management concept be implemented as part of that new organization structure? Are accounting and reporting systems being established by SBU?

* • What upgrading is required, in both quantity and quality of people, to analyze, implement, and control the marketing programs and increased investments in capital, R&D and M&AE? (1979-A15).

• Are the internal control systems (financial and project monitoring) adequate to control the increased number of projects and dollar investments? What fallback/contingency plans exist with regard to projected cash flow, and especially capital investments, if it becomes apparent that we will not achieve our sales, working capital, or profit goals?

* 1979 Key Issue.

AGRICULTURAL EQUIPMENT GROUP
STRATEGIC ALTERNATIVES

[G] = Group Strategic Objective

(A) = Alternative for consideration

SBU	EXPLORE (Aggressive)	EXPLORE (Wait & See)	GROWTH (Aggressive)	GROWTH	PROFITABILITY	CASH	CASH (WITHDRAW)
Under 40 H.P. Tractors	[G]						
40-115 H.P. Tractors			(A)	[G]			
115-200 H.P. Tractors			[G]				
Over 200 H.P. Tractors	[G]						
Grain Harvesting			[G]				
Cotton Harvesting	(A)				[G]		
Planters				[G]			
Tillage					[G]		
Hay, Forage & Material Handling					[G]	(A)	
Outdoor Power	(A)			[G]			

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UNDER 40 H.P. TRACTORS

GLOBAL SEGMENT

IH SBU STRATEGIC COMMENTARY (P4):

HISTORY - On a worldwide basis, there are a wide variety of tractors in the Under 40 H.P. (DIN) category, with most lines offering many scaled down features that are typically found on higher horsepower tractor models. End user applications generally fit one of the categories: Agricultural - small farming; Commercial - utility and maintenance; or Residential - lawn and gardening.

MARKET AND COMPETITION - Over the past twenty years, traditional worldwide manufacturers of agricultural tractors have gradually turned their attention away from the low horsepower tractor business. This left a market void to be filled; first by specialty producers such as Bolens, Wheel Horse, etc., and then by the Japanese manufacturers in the 1960s. The large Japanese domestic market for these tractors permitted rapid realization of scale economies, and the concurrent benefit of becoming very cost competitive worldwide. Hence, the world market today is dominated by strong, cost efficient and aggressive full-line Japanese manufaturers, led by Kubota, who also intends to aggressively introduce tractors in the 40-115 H.P. category in the U.S. Most U.S. agricultural equipment manufacturers have signed OEM purchase agreements to private label the Japanese manufactured tractors in the U.S.

	Industry Unit Sales			Memo: 1979 IH Market
Market	1979	1984	% Change	Share ($)
NAO	40,000	48,500	21.3%	15.5%
EAME	35,300	32,900	(6.8)	6.0
PFELA	196,800	198,700	0.9	0.2
Total	272,100	280,100	2.9%	3.2%

1977 Global Market Share (Units)	
Kubota	27.1%
Yanmar	14.3
Iseki	7.9
Shibaura	7.1
Hinomoto/Toyosha (Mitsubishi)	6.1
Satoh (Mitsubishi)	6.1

A global market study is being undertaken to determine the marketplace requirements in key market areas.

IH PRODUCTS AND PRODUCTION - The current IH product line is incomplete, inadequate, and obsolete. Weaknesses include: incomplete horsepower range offering; inadequate features offering (no 4-wheel drive, diesel engine, power steering, etc); high product costs; and weak distribution channel representation (most IH dealers also sell competitive brands). Some favorable residual product reputation remains from prior years.

Louisville plant manufacture of the Cub, 184 and 140 tractors will be discontinued in 1979.

STRATEGIES-Explore (Aggressive)-Provide a full line of compact tractors with modern features to reverse the decline in IH market share, to offer IH dealers more sales potential, and to counteract the erosion of the IH dealer organization by Japanese manufacturers. Sourcing alternatives being investigated are: Development of four models from 20-40 H.P. by KIMCO; Purchase of four models from Satoh/Mitsubishi with IH brand identification (P-1 assumes this will occur in 1982, though 1980 would be more desirable).

ALTERNATIVES - Development of a full line of compact tractors for IH manufacture.
- Acquisition of a joint venture with an existing tractor manufacturer.

RISKS - A major risk for this segment is that no agreement can be reached on a source for a full line of Under 40 H.P. tractors.

BUSINESS IDENTIFICATION FOR STRATEGIC PLANNING:

STRATEGIC BUSINESS UNIT (SBU) DIRECTORY (P3)

A. SALES $MM IN 1978 1,317

B. NUMBER OF P3 PAGES FOR THIS SBU 1

PRODUCTS/MANUFACTURING/ASSEMBLY

(1) PRODUCTS	1978 % OF SBU SALES	(2) MANUFACTURING/ASSEMBLY PLANTS			1978 % OF SBU SALES
TRACTORS		Farmall (USA)	33.8	Bradford (Britain)	2.7
Under 40 h.p. (DIN)	3.2	Neuss (Germany)	19.0	Mooka (Japan)	1.3
40-115 h.p. (DIN)	53.2	Doncaster (Britain)	15.1	Geelong(Australia)	1.2
115-200 h.p. (DIN)	39.3	St. Dizier (France)	13.3	Saltillo (Mexico)	1.1
Over 200 h.p. (DIN)	4.3	Fargo (USA)	3.9	Outside Purchase	5.1
		Louisville (USA)	3.5		
TOTAL	100%	TOTAL			100%

DISTRIBUTION/REGIONS

(3) DISTRIBUTOR/DEALERS, OEMS, DIRECT USERS	1978 % OF SBU SALES	(4) GEOGRAPHIC MARKETS	1978 % OF SBU SALES
Distributor/Dealer	97	North America	54.1
OEM	2	Europe/Africa/Middle East	40.6
Direct User	1	Pacific/Far East/Latin America	5.3
TOTAL	100%	TOTAL	100%

END USE MARKETS/COMPETITORS

(5) END USE MARKETS	1978 % OF SBU SALES	(6) COMPETITORS	1978 % MKT. SHARE
		Deere	17
		Massey Ferguson	16
		IH	15
		Ford	11
		Others (Allis-Chalmers, Case, Deutz, Fendt, Fiat, Renault, et al.)	41
TOTAL	100%	TOTAL	100%

July 31, 1979
DATE

Tractors
SBU

AE
GROUP

AGRICULTURAL EQUIPMENT

GROUP

GLOBAL SEGMENT

PLAN SUMMARY

UNDER 40 H.P. TRACTORS

THE PLAN: EXPLORE (AGGRESSIVE) - Major problem surrounding inadequate product line offering.

$ MARKET GROWTH	
Fluctuates due to changes in EAME rate between (2.7)% and 18.9%.	
Real	1.0%
Inflation	5.7
Total	6.7%

NET SALES GROWTH	
Low in 1981; peak in 1982 with new product.	
Real	2.0%
Pricing	7.1
Total	9.1%

COMPETITIVE PRESSURE
High - especially from Kubota and other Japanese manufacturers.

$ MARKET SHARE
Fluctuating-5.5% in 1979, dips to 5.1% in 1981, then recovers to 6.0% by 1984.

PROFITABILITY
Turnaround - ROA from negative (32.3)% (1979) to 57.5% (1984).
ROS from negative (16.0)% (1979) to 13.9% (1984).

CUMULATIVE CASH FLOW
$23 million (1980-84) with $17 MM in 1982-84. EAME is negative each year due to low contribution margin.

PRICING
Average 7.1% annual increase exceeds market inflation rate, but is slightly less than average cost inflation rates.

COST INCREASE
Average variable and fixed cost inflation rates of 7.2% and 7.0% respectively.

PRODUCTIVITY %-OPERATING INCOME
Average is greater than 100% (1980-84) because of minimal operating income levels.

M & A
Decreases from 13.5% (1979) to 11.3% (1984) of sales.

R & D
Minimal levels of R&D(essentially all maintenance) increases slightly from 0.2%(1979) to 0.3%(1984) of sales.

CAPITAL EXPENDITURES
Level expenditures totaling $2.3 million (1980-84). Assumes OEM purchased line of tractors.

PROFILE (P-1)

		HISTORY		PROJECTION						GROWTH	
		1977	1978	1979	1980	1981	1982	1983	1984	77-79	79-84
A.AMAIL RETAIL MARKET	$MM	727.6	731.8	840.2	888.1	953.7	997.2	1096.9	1160.1	7.5	6.7
B.IM RETAIL MARKET SHARE	%	7.28	7.11	5.49	6.04	5.05	5.77	5.65	5.95		
C.LARGEST COMP SHARE	%	19.0	20.2	22.2	24.3	24.4	24.1	22.8	22.8		
D.COMPETITIVE STRENGTH		L	L	L	L	L	L	L	L		
E.NET SALES	$MM	41.5	42.1	43.1	46.1	35.5	69.6	57.5	66.5	1.9	9.1
F.CONTRIBUTION MARGIN	%	32.80	29.02	24.22	24.21	24.53	23.88	21.93	22.11		
G.TOTAL FIXED COST	$MM	20.1	18.5	16.4	16.2	5.4	5.2	4.8	5.0	(8.8)	(11)
H.OPERATING INCOME	$MM	(6.5)	(6.3)	(6.0)	(5.1)	3.3	11.4	7.8	9.7	3.9	29.3
I.PROFIT BEFORE TAX	$MM	(6.8)	(6.6)	(6.9)	(6.2)	2.6	10.8	7.4	9.3	(.6)	27.3
J.TOTAL AVG ASSETS	$MM	10.1	16.6	21.3	21.6	15.7	17.5	15.5	16.1		
K.TOTAL ENDING ASSETS	$MM	10.1	15.7	22.3	20.3	14.2	15.9	14.1	14.8	48.6	(6.0)
L.CASH FLOW (OPERATIONAL)	$MM	NA	(12.3)	(1.5)	.1	6.2	6.8	4.7	5.3		
M.MARKET GROWTH	%		.57	14.81	5.70	7.39	4.55	10.00	5.76		
N.SALES GROWTH	%		1.55	2.31	7.02	(23.07)	96.24	(17.48)	15.71		
O.PBT GROWTH	%						311.09	(32.01)	26.01		
P.ANNUNAL MKT INFLATION	INDX		1.071	1.061	1.054	1.055	1.055	1.062	1.061		
Q.ANNUAL PR INFL	INDX		1.098	1.076	1.067	1.063	1.086	1.072	1.068		
R.ANNUAL UC INFL	INDX		1.093	1.073	1.066	1.061	1.091	1.068	1.072		
S.ANNUAL FC INFL	INDX		1.095	1.076	1.076	1.049	1.075	1.076	1.073		
T.ROA (AVG ASSETS)	%	(67.1)	(39.7)	(32.3)	(28.5)	16.8	61.8	47.5	57.5		
U.ROS	%	(16.4)	(15.6)	(16.0)	(13.4)	7.4	15.5	12.8	13.9		
V.ASSET(AVG) TURNOVER	X	4.09	2.54	2.02	2.14	2.26	3.98	3.71	4.12		
W.NET PRDTVTY/OP.INC	%	0	0	0	0	216.1	67.4	107.0	91.0		
X.R&D/SALES	%	1.0	.5	.2	.4	.6	.3	.3	.3		
Y.MAE/SALES	%	14.0	14.0	13.5	12.2	14.5	8.3	11.6	11.3		
Z.ESTIMATED CAPITAL EXPENDITURES	$MM	.8	.4	0	.3	.6	.6	.6	.2		
AA. PRODUCT CAPITAL EXPENDITURES	%TOT	25.0	25.0	0	0	16.7	33.3	66.7	50.0		
BA. COST-REDUCTION CAP. EXPEND.	%TOT	0	0	0	0	16.7	16.7	0	0		
CA. OTHER CAPITAL EXPENDITURES	%	75.0	75.0	100.0	100.0	66.7	50.0	33.3	50.0		
DA.R&D EXPENSE	$MM	.4	.2	.1	.2	.2	.2	.2	.2	(32)	14.9
EA. NEW PRODUCT R&D	%R&D	0	0	300.0	100.0	0	0	0	0		
FA.M&AE EXPENSE	$MM	5.8	5.9	5.8	5.6	5.1	5.8	6.7	7.5	0	5.2
GA.NET PLANNED PRODUCTIVITY (CUM)	$MM	0	0	0	0	7.2	7.7	8.3	8.8		
HA.CONTINGENCY	$MM	0	0	0	0	0	0	0	0		
IA.PARTS REVENUE	$MM	2.7	2.9	3.7	3.7	4.1	4.8	3.6	4.1	16.3	2.4
JA.PARTS VARIABLE MARGIN	%	55.40	44.54	49.86	45.94	44.44	45.87	44.77	47.24		

COMPOUND ANNUAL GROWTH RATES

MAJOR CAPITAL PROGRAMS
(Over $5.0 MM)

UNDER 40 H.P. TRACTORS
GLOBAL SEGMENT
($ MM)

Program	APPROPRIATION		EXPENDITURES						
	Year	Amount	-1979	1980	1981	1982	1983	1984	1985+
No major capital programs in excess of $5.0 million identified.									

*Project appropriation approved.

PRODUCT DEVELOPMENT TIMETABLE
(PROJECTS KNOWN OR LOGICALLY ANTICIPATED)

UNDER 40 H.P. TRACTORS
GLOBAL SEGMENT

	1979	1980	1981	1982	1983	1984
International Harvester				OEM purchase of Mitsubishi sourced line of compact Tractors.		

No final sourcing decision has yet been made for this segment, but NAO P-1 data reflects Mitsubishi sourcing beginning in 1982.

Kubota

Deere

CAUSAL FACTOR ANALYSIS
UNDER 40 H. P. TRACTORS
GLOBAL SEGMENT

Sales

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR SALES	$MM		43.09	46.12	35.48	69.63	67.46
B. AVAIL MARKET EFFECT	$MM		.14	.97	(.41)	2.04	(.22)
C. SELLING PRICE EFFECT	$MM		2.9	2.09	5.54	3.84	4.24
D. MARKET SHARE EFFECT	$MM		4.6	(8.95)	6.77	(1.18)	3.29
E. WHL/DLR INV EFFECT	$MM		(3.99)	(4.42)	23.85	(16.37)	2.13
F. REMAINDER SALES	$MM		(.63)	(.34)	(1.6)	(.51)	(.42)
G.NET SALES	$MM	43.09	46.12	35.48	69.63	67.46	66.48

PROFIT BEFORE TAX

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR	$MM		(6.88)	(6.16)	2.63	10.82	7.36
B. AVAIL MARKET EFFECT	$MM		.03	.24	(.1)	.49	(.05)
C. COST INFL EFFECT	$MM		(3.31)	(1.8)	(4.76)	(3.19)	(3.82)
D. SELLING PRICE EFFECT	$MM		2.9	2.09	5.54	3.84	4.24
E. MARKET SHARE EFFECT	$MM		1.11	(2.17)	1.66	(.28)	.72
F. WHOLESALE/DEALER INV EFFECT	$MM		(.96)	(1.07)	5.85	(3.91)	.47
G. VARIABLE COST CHG	$MM		(.04)	.08	(.23)	(1.19)	.28
H. FIXED COST CHG	$MM		1.33	11.11	.53	.7	.16
I. REMAINDER	$MM		(.35)	.32	(.29)	.08	(.09)
J.PROFIT BEFORE TAX	$MM	(6.88)	(6.16)	2.63	10.82	7.36	9.27

Sales Commentary

Annual sales for the Under 40 H.P. Tractor global segment fluctuate over the planning period, but by 1984 are $23 MM greater than 1979 sales. Selling price accounts for about 80% of the increase, market share improvement for about 19%, and market growth for about 11%. Wholesale/dealer inventories decline significantly in 1980 and '81, build up in 1982 to support the new tractor line (purchased from outside source), and stabilize in 1983. Sales decline in '81 is primarily due to market share erosion, and decline in '83 is due to inventory sell-off.

Income Commentary

Income deficits are expected in 1979 and 1980; 1984 income is expected to be about $16 MM greater than the 1979 level. Fixed cost improvement accounts for about 86% of the increase, retention of net inflation benefits accounts for about 10%, and market share gain accounts for about 6%. The purchase of Mitsubishi tractors in 1982 allows the elimination of the fixed cost overhead of production facilities in 1981. The income decline in 1983 is primarily due to inventory sell-off and increased variable costs.

KEY ISSUES

UNDER 40 H.P. TRACTORS
GLOBAL SEGMENT

The proposed "Explore (Aggressive)" categorization of this SBU is appropriate for the following reasons:

- AEG is currently marketing an incomplete and obsolete product line manufactured at Louisville plant. This line will be discontinued in 1979.
- IH dealers have not pushed the IH line -- many carry Japanese products.
- Real market growth is low (1%/year) and characterized by intense price competition from strong, efficient, high volume Japanese producers.

* The key issue is to determine whether the Segment should be continued, and, if so, what is the best source of supply. (1979-A3). Factors complicating this decision include:

- Withdrawal from the business may further strengthen Japanese producers' position within the IH dealer body.
- There appears to be little evidence that AEG can achieve sufficient volume and manufactured content to make an internally sourced product profitable.
- Outsourcing the product with a Japanese producer such as Mitsubishi is complicated by our equity position in KIMCO, and the fact that KIMCO is not cost competitive in this class of product.

The Plan calls for completion of a worldwide marketing study on this SBU. AEG has committed to resolution of the sourcing issue by September 1, 1979. A final decision regarding determination of the worldwide source for the product line will, therefore, be forthcoming. At that time the Segment P-1 should be revised so that it is consistent with the approved strategy. (P-1 currently assumes 1982 OEM sourcing of product line).

* 1979 Key Issue.

40-115 H.P. TRACTORS

GLOBAL SEGMENT

IH SBU STRATEGIC COMMENTARY (P4):

HISTORY - This segment is the largest of the tractor segments, accounting for 62% of the available worldwide tractor market dollar sales, and containing the largest number of strong competitors (many government supported). Continuing trend is to larger tractor horsepowers, as well as increased importance of sophistication, reliability, features, and operator comfort for more productivity; but price remains important.

MARKET AND COMPETITION - Europe, Canada and the U.S. are basically replacement markets (bulk of sales volume).

Industry Unit Sales				Memo: 1978 IH Market Share (Units)		1978 Global Market Share ($)	
Market	1978	1984	% Change	W/O Brazil-Arg.	W/ Brazil-Arg.		
NAO	99,500	102,500	3.0%	14.5%	14.5%	Massey Ferguson	21.9%
EAME*	329,000	387,100	17.7	11.8	11.8	Ford	14.6
PFELA**	38,000	46,400	22.1	12.8	5.5	IH	12.2
	466,500	536,000	14.9%	12.5%	11.3%	Deere	11.1

*Excludes India; **Excludes Brazil (43,800-1978; 80,500-1984) and Argentina (6,200-1978; 19,000-1984).

Massey Ferguson, with most strengths at the lower horsepowers, has been experiencing a financial crisis, but still retains a strong dealer network. Ford Motor Co. has decided to remain in the agricultural tractors business.

IH PRODUCTS AND PRODUCTION

1. Common Market line, introduced in 1965, has rugged features then tailored to European market. Current design trends make them not cost and price competitive in lower horsepowers. Produced in France and Germany.
2. Worldwide (84 Series) line introduced in 1970 (updated 1978) is modern and well accepted. Produced primarily in Britain and Japan, this line is experiencing deteriorating profitability. Doncaster plant has serious 84 Series transmission capacity constraints, and is the single source of worldwide exports.
3. 86 Series line, introduced in 1976, with drive train dating to 1962. Made in the U.S., Mexico, and Australia, this line faces many problems, including high costs vs. competition (only lower horsepower range of this line falls in this Segment) and a horsepower gap in the 80-100 H.P. range.

DISTRIBUTION - Weaknesses include inadequate marketing skills and product knowledge, limited financial capacity and liquidity, poor or lacking representation in some areas, etc.; however, IH network is first or second in many countries.

STRATEGIES - Growth and/or hold for Profitability (where IH market share is strong). - To increase profitability, manufacturing cost reduction and new product development programs are being implemented. The engineering organization must be strengthened concurrently.
To increase market share, an extensive dealer development program will be implemented, weaknesses in the product line will be corrected with the introduction of new tractor models, the IH marketing organization will be strengthened, and IH participation in closed markets such as Brazil will be attempted.

RISKS - A major risk in this SBU segment centers around the TR tractor and C-85 Control Center development programs (timing, cost/ROI targets, features/technology equivalent to current competition only, human resource needs.) Another is the ability of dealer/distributor network to carry the projected volume increases.

July 31, 1979	B. H. Warren	40-115 H.P. Tractors Global Segment	AEG
Date	Manager	SBU	Group

BUSINESS IDENTIFICATION FOR STRATEGIC PLANNING:

STRATEGIC BUSINESS UNIT (SBU) DIRECTORY (P3)

A. SALES $MM IN 1978 1,317

B. NUMBER OF P3 PAGES FOR THIS SBU 1

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		Louisville (USA)	3.5		
TOTAL	100%	TOTAL			100%

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		Massey Ferguson	16
		IH	15
		Ford	11
		Others (Allis-Chalmers, Case, Deutz, Fendt, Fiat, Renault, et al.)	41
TOTAL	100%	TOTAL	100%

July 31, 1979
DATE

Tractors
SBU

AE
GROUP

AGRICULTURAL EQUIPMENT

GROUP

GLOBAL SEGMENT

PLAN SUMMARY

40-115 H.P. TRACTORS

THE PLAN: GROWTH AND PROFITABILITY-Through manufacturing cost reductions, strengthening of dealer organization, and new product introductions.

$ MARKET GROWTH	
Smooth growth 1980-84.	
Real	2.5%
Inflation	6.8
Total	9.3%

NET SALES GROWTH	
Peaks in 1980 and 1984, with low of 3.6% in 1981.	
Real	5.6%
Pricing	6.8
Total	12.4%

COMPETITIVE PRESSURE
Very High-Large number of well entrenched national and government supported competitors and multi-nationals.

$ MARKET SHARE
Smooth increase from 14.2% (1979) to 17.5% (1984).

PROFITABILITY
Inadequate-ROA increases from 11.4% (1979) to 14.6% (1984).
ROS increases from 5.7% (1979) to 7.3% (1984).

CUMULATIVE CASH FLOW (A)
$289 million (1980-84), with $198 MM occurring in 1982-84.

PRICING
Average 6.8% equals market and variable cost inflation rates, but is slightly less than fixed cost inflation rate.

COST INCREASE
Average variable and fixed cost inflation rates of 6.8% and 7.0% respectively.

PRODUCTIVITY %-OPERATING INCOME
Very High - average 119.3% (1980-84), with levels near 150% in 1982-83.

M & A
Increases from 7.6% (1979) to 9.1% (1984), with peak of 9.7% (1983) of sales.

R & D
Increases from 1.8% (1979) to 2.1% (1984) of sales.

CAPITAL EXPENDITURES
Relatively level expenditures totaling $331 million (1980-84), with peak of $95 MM in 1983.

(A) Not adjusted for contingencies.

SEGMENT AEG 40-115 UT GLOBAL (3) 07/24/79 10.45.29

PROFILE (P-1)

		HISTORY		PROJECTION						GROWTH	
		1977	1978	1979	1980	1981	1982	1983	1984	77-79	79-84
A.AVAIL RETAIL MARKET	$MM	5343.2	5069.5	5615.2	5987.5	6503.6	6996.2	7678.2	8746.2	2.5	9.3
B.IH RETAIL MARKET SHARE	%	11.98	12.19	14.20	15.04	15.41	16.08	16.26	17.49		
C.LARGEST COMP SHARE	%	20.90	21.10	20.70	20.70	20.60	20.40	19.80	19.70		
D.COMPETITIVE STRENGTH		L	L	M	M	M	M	M	M		
E.NET SALES	$MM	672.7	700.9	894.2	1040.9	1078.5	1192.0	1334.6	1604.0	15.3	12.4
F.CONTRIBUTION MARGIN	%	26.72	25.09	23.02	20.15	20.47	20.69	20.68	22.80		
G.TOTAL FIXED COST	$MM	120.2	139.3	143.4	161.8	173.3	192.4	208.4	230.2	9.2	9.9
H.OPERATING INCOME	$MM	59.5	36.6	62.4	47.9	47.5	54.4	67.8	135.8	2.4	16.8
I.PROFIT BEFORE TAX	$MM	48.8	23.2	51.3	34.8	33.4	39.7	51.3	116.6	2.5	17.9
J.TOTAL AVG ASSETS	$MM	372.9	402.4	448.8	456.8	506.0	548.4	623.0	800.5		
K.TOTAL ENDING ASSETS	$MM	385.4	430.2	469.6	470.6	530.8	564.4	642.9	832.3	10.4	12.1
L.CASH FLOW (OPERATIONAL)	$MM	NA	34.3	12.9	34.2	56.5	63.7	49.9	84.3		
M.MARKET GROWTH	%		(5.12)	10.76	6.63	8.62	7.57	9.75	13.91		
N.SALES GROWTH	%		4.19	27.58	16.40	3.61	10.52	11.96	20.19		
O.PBT GROWTH	%		(52.59)	121.38	(32.10)	(4.02)	18.86	29.22	127.29		
P.ANNUNAL MKT INFLATION	INDX		1.078	1.071	1.068	1.068	1.067	1.068	1.068		
Q.ANNUAL PR INFL	INDX		1.079	1.071	1.071	1.064	1.069	1.069	1.067		
R.ANNUAL UC INFL	INDX		1.081	1.069	1.073	1.064	1.068	1.069	1.067		
S.ANNUAL FC INFL	INDX		1.072	1.069	1.071	1.068	1.070	1.070	1.069		
T.ROA (AVG ASSETS)	%	13.1	5.8	11.4	7.6	6.6	7.2	8.2	14.6		
U.ROS	%	7.3	3.3	5.7	3.3	3.1	3.3	3.8	7.3		
V.ASSET(AVG) TURNOVER	X	1.80	1.74	1.99	2.28	2.13	2.17	2.14	2.00		
W.NET PRDTUTY/OP.INC	%	0	4.1	44.3	85.3	127.8	148.5	146.0	88.8		
X.R&D/SALES	%	1.8	1.7	1.8	1.7	2.1	2.2	2.2	2.1		
Y.MAE/SALES	%	9.0	10.5	7.6	8.6	8.9	9.4	9.7	9.1		
Z.ESTIMATED CAPITAL EXPENDITURES	$MM	12.3	15.1	30.2	55.6	47.4	63.2	95.3	69.3		
AA. PRODUCT CAPITAL EXPENDITURES	%TOT	25.4	13.1	61.3	39.7	20.0	30.2	55.5	52.0		
BA. COST-REDUCTION CAP. EXPEND.	%TOT	8.5	4.9	3.9	7.3	8.2	5.8	2.2	3.4		
CA. OTHER CAPITAL EXPENDITURES	%	66.2	82.0	34.8	53.1	71.7	64.0	42.2	44.5		
DA.R&D EXPENSE	$MM	11.8	12.1	15.9	17.3	22.9	26.4	29.1	33.2	16.1	15.9
EA. NEW PRODUCT R&D	%R&D	0	57.9	42.1	16.2	3.9	0	0	0		
FA.M&AE EXPENSE	$MM	60.6	73.8	67.5	89.4	96.3	112.4	129.8	146.2	5.6	16.7
GA.NET PLANNED PRODUCTIVITY (CUM)	$MM	0	1.5	27.6	40.9	60.7	80.8	99.0	120.5		
HA.CONTINGENCY	$MM	0	0	0	0	0	0	0	0		
IA.PARTS REVENUE	$MM	109.5	122.2	156.2	172.3	185.3	207.7	227.2	252.0	19.4	10.0
JA.PARTS VARIABLE MARGIN	%	45.42	42.09	45.65	42.08	43.54	45.56	45.28	49.52		

COMPOUND ANNUAL GROWTH RATES

MAJOR CAPITAL PROGRAMS
(Over $5.0 MM)

40-115 H. P. TRACTORS
GLOBAL SEGMENT
($ MM)

Program	APPROPRIATION Year	APPROPRIATION Amount	EXPENDITURES -1979	1980	1981	1982	1983	1984	1985+
*New C-85 Control Center	1979	$28.3	$4.0	$20.4	$ 3.9	$	$	$	$
Increase Plant Capacity - Saltillo	1979	6.5	.5	4.0	2.0				
New Warehouse - Neuss	1979	6.5		3.2	3.3				
Grey Iron Foundry (B)	1979	6.2	5.4	.8					
Tractor Noise Reduction	1980	5.7		2.8	2.0	.9			
Increase WW Tractor Capacity - I	1980	6.0			4.0	2.0			
Increase WW Tractor Capacity - II	1981	23.5				10.0	13.5		
TR-1 Tractor Program	1981	10.0			10.0				
TR-2 Tractor Program	1981	58.4				11.1	44.4	2.9	
Doncaster Foundry Expansion - I	1981	5.6			1.3	3.2	1.1		
Doncaster Foundry Expansion - II	1983	11.0					1.2	4.9	4.9
TR-3 Tractor Program	1983	32.2(A)					9.0	22.1	1.1
TOTAL		$199.9	$9.9	$31.2	$26.5	$27.2	$69.2	$29.9	$ 6.0

(A) Segment Portion of Total TR-3 Program ($41.9MM)
(B) 1979 Reappropriation (Not included in Capital Plan)

*Project appropriation approved.

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PRODUCT DEVELOPMENT TIMETABLE
(PROJECTS KNOWN OR LOGICALLY ANTICIPATED)

40-115 H.P. TRACTORS
GLOBAL SEGMENT

	1979	1980	1981	1982	1983	1984
International Harvester		New Pressure Flow Compensated Hydraulic System - 86 and 88 Series	New C-85 Control Center-84 Series and Common Market Line.		TR-2 Drop Center Trans. w/4WD TR Styling. Enclosed Engine Reverse Air Flow TR-1 Styling with Reverse Air Flow for 84 Series.	
Massey Ferguson				New Styling of front end Enclosed engines		
Ford		New styling of Hood Sheet to match new control center in 1978.	New Synchro transmission Center drive 4 WD	Upgrading of hydraulics to meet European lift requirements.		
Deere		New control center to replace sound gaurd body. Also small tractor control center for Europe.	Upgrade hydraulics Enclosed engine		Sheet metal styling change	

CAUSAL FACTOR ANALYSIS

40 - 115 H. P. TRACTORS

GLOBAL SEGMENT

SALES

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR SALES	$MM		894.22	1040.9	1078.6	1192	1334.55
B. AVAIL MARKET EFFECT	$MM		(1.59)	15.34	7.93	30.79	83.84
C. SELLING PRICE EFFECT	$MM		69.05	64.88	77.29	86.6	100.52
D. MARKET SHARE EFFECT	$MM		47.17	22.5	43.74	12.93	101.22
E. WHL/DLR INV EFFECT	$MM		34.11	(68.64)	(13.45)	13.49	(16.9)
F. REMAINDER SALES	$MM		(2.05)	3.62	(2.01)	(1.15)	1.37
G.NET SALES	$MM	894.22	1040.9	1078.6	1192	1334.55	1604

PROFIT BEFORE TAX

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR	$MM		51.25	34.8	33.4	39.7	51.3
B. AVAIL MARKET EFFECT	$MM		(.37)	3.09	1.62	6.37	17.22
C. COST INFL EFFECT	$MM		(67.29)	(62.87)	(72.72)	(82.2)	(92.33)
D. SELLING PRICE EFFECT	$MM		69.05	64.88	77.29	86.6	100.52
E. MARKET SHARE EFFECT	$MM		10.86	4.53	8.95	2.68	20.93
F. WHOLESALE/DEALER INV EFFECT	$MM		7.85	(13.83)	(2.75)	2.79	(3.5)
G. VARIABLE COST CHG	$MM		(26.44)	3.47	1.35	(.08)	31.76
H. FIXED COST CHG	$MM		(7.72)	(.44)	(6.58)	(2.37)	(7.03)
I. REMAINDER	$MM		(2.39)	(.24)	(.87)	(2.08)	(2.28)
J.PROFIT BEFORE TAX	$MM	51.25	34.8	33.4	39.7	51.3	116.6

Sales Commentary

Annual sales in 1984 are expected to be $710 MM greater than 1979 sales. Selling price accounts for about 56% of the increase, market share growth accounts for about 32%, and market growth for about 19%. The inclusion of Spain in the Common Market in 1983 is a major factor in the large increases of '83 and '84.

Income Commentary

Annual income declines from previous year levels in 1980 and 1981; 1984 income is expected to be about $83 MM greater than the 1981 level. The '80 decline is primarily due to the variable cost increases associated with the planned emphasis on more competitive models in the product mix; the '81 decline is primarily caused by inventory reductions. Of the income rebound from 1981-84, about 40% is caused by variable cost improvement, about 39% by increased market share, 30% by market growth, and 20% by retained net inflation benefits. Manufacturing efficiencies and productivity improvements lead to major variable cost improvements in 1984, and are a primary cause for the dramatic PBT rise that year. Market share gains are expected to be realized through aggressive marketing efforts, product improvements, and the aforementioned product emphasis.

KEY ISSUES

40-115 H. P. TRACTORS
GLOBAL SEGMENT

The 40-115 H.P. Tractor Segment is an important part of the total world tractor market, with IH Sales of $894MM in 1979. The available market is slow growing and highly competitive, with retail sales concentrated in EAME (68%) and NAO (23%). The markets are diverse, and regional characteristics determine demand and competition. In several areas government supported competitors are dominant market forces. The Plan calls for large investments in marketing, R&D, and in capital. Concurrently, profitability is low, and ROS, which ranges from 3% to 4% from 1980 to 1983, only improves to 7.3% in 1984. Planned profitability is largely dependent on market share gains and productivity retention. Parts contribute a substantial portion of operating income through much of the planning period.

Since presence in this large market is undoubtedly necessary despite low profitability, the primary issue is how to prioritize strategies for each specific product/market segment to maximize gains and minimize risk. The magnitude of existing and planned investments and the intensity of competition make it important to concentrate resources on product/market segments with the highest potential:

- Monitor programs closely to insure implementation of programs within planned timetables and expense levels.
- Develop contingency plans and/or strategy adjustments if share gains are not achieved as planned.

The following questions/issues seem to be particularly relevant to the ultimate outcome of the plan:

- Much of the Plan is conditioned on the success of the TR program, yet this program may not result in product superiority over market leaders. What additional R&D is required to move ahead of competition?

- Assuming limited resources, what priority structure should apply to market segments (including geographic areas)? How should these be developed in terms of available potential and competitive resistance?

* Given problems of high inflation rates, difficulties of repatriation of profits, etc., and the scarcity of human and financial resources, should an attempt be made to penetrate the Brazilian market? Has an economic/risk analysis been completed on this issue? (1979-A14).

Given the apparent weakness of Massey-Ferguson, what programs have been developed to take share from Massey-Ferguson in EAME and PFELA and/or to convert good Massey-Ferguson dealers to IH?

- What is the timetable for initiating dealer development programs such as XL and FACTS in EAME and PFELA? Are plans also made for proper training of IH personnel in the usage of these tools?

* 1979 Key Issue

115-200 H.P. TRACTORS

GLOBAL SEGMENT

IH SBU STRATEGIC COMMENTARY (P4):

HISTORY - This segment accounts (1979) for 23% of net sales and 29% of PBT for the total AE Group. These tractors are used in heavy tillage and planting, PTO work, and material handling. Available in 2 wheel drive or 4 wheel drive, rigid frame or articulated, the largest user segment farms 750-1,000 acres. Trend is to larger H.P. units with greater productivity, reliability, and operator comfort. Pricing and parts/service availability are important.

MARKET AND COMPETITION - Largest market is NAO (especially U.S.), but EAME/PFELA offer growth potential.

Industry Unit Sales				Memo: 1978 IH Market Share (Units)	
Market	1978	1984	% Change	W/O Brazil-Arg.	W/Brazil-Arg.
NAO	73,900	83,200	12.6%	26.7%	26.7%
EAME	8,700	14,100	62.1	9.3	9.3
PFELA*	4,500	7,200	60.0	20.1	16.6
Total	87,100	104,500	20.0%	24.6%	24.4%

1978 Global Market Share ($)	
Deere	36.6%
IH	24.3
Case	10.4
Massey Ferguson	8.2

*Excludes Brazil (870-1978; 2,500-1984) and Argentina (90-1978; 750-1984).

Deere is the dominant market force among many strong competitors. Massey Ferguson is currently weak financially but still retains a strong dealer network. Case concentrates on tractors vs. full line of equipment.

IH PRODUCTS AND PRODUCTION

1. 86 Series line, introduced in 1976, but with drive train dating to 1962. Made in the U.S. and Australia, this line faces many problems, including high costs vs. competition, outdated design and features, and a horsepower limited power train that prevents IH participation in the 160-190 PTO H.P. market (growing).
2. 88 Series line, introduced in 1979 and produced in the U.S. This is an innovative, 4 wheel drive articulated row crop ("2+2" design) tractor with 85% of its componentry common to other IH tractor lines.
3. Common Market line D, introduced in 1973, and assembled in France and Germany with purchased power train. Weak in features, reliability, and specifications, it leaves a H.P. gap in range above 120 H.P. (DIN).

DISTRIBUTION - Weaknesses include inadequate marketing skills and product knowledge, limited financial capacity and liquidity, poor or lacking representation in some areas, etc.; however, IH network is first or second in many countries. Attempts are being made to enter Brazilian tractor market.

STRATEGIES - Aggressive Growth - Aggressive marketing of 88 Series and TR-4 Series (1981) is aimed at market leadership by 1985/87. Large future capital expenditures for new products and facilities, and large R&D expenditures will be necessary to achieve this objective.
An extensive worldwide dealer/distributor development program will be implemented including: upgrading or replacing weak dealers/distributors; filling of geographic gaps in coverage; increased financial support; increased product/sales training; increase in number of sales personnel. The IH marketing organization will also be upgraded.

RISKS - One major risk concerns competitive response to the 88 Series (timing of new product, competitive pricing, etc.). Others involve the capability of NAO to supply competitively priced tractors or skid units to EAME/PFELA; the ability of our dealer/distributor network to carry projected volume increases; and, the ability of Components Group to supply engine needs.

BUSINESS IDENTIFICATION FOR STRATEGIC PLANNING:

STRATEGIC BUSINESS UNIT (SBU) DIRECTORY (P3)

A. SALES $MM IN 1978 1,317

B. NUMBER OF P3 PAGES FOR THIS SBU 1

PRODUCTS/MANUFACTURING/ASSEMBLY

(1) PRODUCTS	1978 % OF SBU SALES	(2) MANUFACTURING/ASSEMBLY PLANTS			1978 % OF SBU SALES
TRACTORS		Farmall (USA)	33.8	Bradford (Britain)	2.7
Under 40 h.p. (DIN)	3.2	Neuss (Germany)	19.0	Mooka (Japan)	1.3
40-115 h.p. (DIN)	53.2	Doncaster (Britain)	15.1	Geelong(Australia)	1.2
115-200 h.p. (DIN)	39.3	St. Dizier (France)	13.3	Saltillo (Mexico)	1.1
Over 200 h.p. (DIN)	4.3	Fargo (USA)	3.9	Outside Purchase	5.1
		Louisville (USA)	3.5		
TOTAL	100%	TOTAL			100%

DISTRIBUTION/REGIONS

(3) DISTRIBUTOR/DEALERS, OEMS, DIRECT USERS	1978 % OF SBU SALES	(4) GEOGRAPHIC MARKETS	1978 % OF SBU SALES
Distributor/Dealer	97	North America	54.1
OEM	2	Europe/Africa/Middle East	40.6
Direct User	1	Pacific/Far East/Latin America	5.3
TOTAL	100%	TOTAL	100%

END USE MARKETS/COMPETITORS

(5) END USE MARKETS	1978 % OF SBU SALES	(6) COMPETITORS	1978 % MKT. SHARE
		Deere	17
		Massey Ferguson	16
		IH	15
		Ford	11
		Others (Allis-Chalmers, Case, Deutz, Fendt, Fiat, Renault, et al.)	41
TOTAL	100%	TOTAL	100%

July 31, 1979
DATE

Tractors
SBU

AE
GROUP

AGRICULTURAL EQUIPMENT
GROUP

GLOBAL SEGMENT
PLAN SUMMARY
115-200 H.P. TRACTORS

THE PLAN: AGGRESSIVE GROWTH-Through new products, strengthening of dealer organization, and better product support.

$ MARKET GROWTH	
Smooth growth 1980-1984..	
Real	5.4%
Inflation	6.8
Total	12.2%

NET SALES GROWTH	
Generally smooth, but EAME increases 63% in 1980.	
Real	7.6%
Pricing	6.8
Total	14.4%

COMPETITIVE PRESSURE
Very high from John Deere - expect new products and features. Massey Ferguson may be vulnerable.

$ MARKET SHARE
Smooth increase from 26.8% (1979) to 30.4% (1984).

PROFITABILITY
ROA increases from 29.8% (1979) to 30.5% (1984).
ROS increases from 8.2% (1979) to 11.4% (1984).

CUMULATIVE CASH FLOW (A)
$404 million (1980-84), with $319 MM in 1983-84; negative in 1980.

PRICING
Average 6.8% equals market and variable cost inflation rates, but is less than fixed cost inflation rate.

COST INCREASE
Average variable and fixed cost inflation rates of 6.8% and 7.0% respectively.

PRODUCTIVITY %-OPERATING INCOME
High-average 93.6% (1980-84), with peaks around 110% in 1981-82.

M & A
Increases from 12.0% (1979) to 14.3% (1984) of sales.

R & D
Decreases from 2.8% (1979) to 2.1% (1984) of sales.

CAPITAL EXPENDITURES
$341 million total (1980-84) - $230 million in 1980-81.

(A) Not adjusted for contingencies.

7/31/79

PROFILE (P-1)

		HISTORY		PROJECTION						GROWTH	
		1977	1978	1979	1980	1981	1982	1983	1984	77-79	79-84
A.AVAIL RETAIL MARKET	$MM	1628.8	1931.9	2157.3	2472.7	2791.4	3082.0	3408.1	3830.8	15.1	12.2
B.IH RETAIL MARKET SHARE	%	28.22	24.29	26.76	28.07	27.95	28.89	29.86	30.43		
C.LARGEST COMP SHARE	%	33.21	36.60	34.04	34.47	34.32	34.13	34.06	33.74		
D.COMPETITIVE STRENGTH		M	L	M	M	M	H	H	H		
E.NET SALES	$MM	601.4	517.5	713.6	818.1	901.7	1087.8	1284.9	1399.4	8.9	14.4
F.CONTRIBUTION MARGIN	%	32.31	27.46	28.20	27.61	27.72	28.31	29.68	29.83		
G.TOTAL FIXED COST	$MM	101.6	121.1	133.1	149.8	172.7	194.9	212.1	231.6	14.5	11.7
H.OPERATING INCOME	$MM	92.9	21.0	68.1	76.3	77.4	113.0	169.3	185.8	(13)	22.2
I.PROFIT BEFORE TAX	$MM	85.3	12.0	58.5	62.0	55.9	88.2	143.6	159.4	(15)	22.2
J.TOTAL AVG ASSETS	$MM	153.5	174.8	196.4	279.2	391.4	458.7	496.9	521.8		
K.TOTAL ENDING ASSETS	$MM	158.1	182.1	212.0	335.4	434.9	465.3	503.4	518.7	15.8	19.6
L.CASH FLOW (OPERATIONAL)	$MM	NA	(10.4)	33.9	(34.6)	2.6	116.5	141.6	177.8		
M.MARKET GROWTH	%		18.61	11.66	14.62	12.89	10.41	10.58	12.40		
N.SALES GROWTH	%		(13.95)	37.88	14.65	10.22	20.64	18.12	8.91		
O.PBT GROWTH	%		(85.94)	388.16	5.93	(9.84)	57.78	62.81	11.00		
P.ANNUNAL MKT INFLATION	INDX		1.105	1.078	1.068	1.068	1.068	1.068	1.068		
Q.ANNUAL PR INFL	INDX		1.104	1.078	1.068	1.063	1.072	1.069	1.067		
R.ANNUAL VC INFL	INDX		1.096	1.079	1.068	1.067	1.068	1.069	1.067		
S.ANNUAL FC INFL	INDX		1.097	1.077	1.072	1.069	1.069	1.069	1.070		
T.ROA (AVG ASSETS)	%	55.6	6.9	29.8	22.2	14.3	19.2	28.9	30.5		
U.ROS	%	14.2	2.3	8.2	7.6	6.2	8.1	11.2	11.4		
V.ASSET(AVG) TURNOVER	X	3.92	2.96	3.63	2.93	2.30	2.37	2.59	2.68		
W.NET PRDTVTY/OP.INC	%	0	71.9	53.8	84.0	111.4	107.7	82.1	82.9		
X.R&D/SALES	%	2.4	3.0	2.8	2.9	2.5	2.2	2.1	2.1		
Y.MAE/SALES	%	10.0	13.8	12.0	13.0	13.1	13.4	13.7	14.3		
Z.ESTIMATED CAPITAL EXPENDITURES	$MM	14.1	26.1	30.2	121.9	108.2	42.3	42.4	25.8		
AA. PRODUCT CAPITAL EXPENDITURES	%TOT	79.4	61.7	36.9	79.2	73.4	71.9	70.8	60.7		
BA. COST-REDUCTION CAP. EXPEND.	%TOT	4.8	28.5	37.3	15.9	21.7	11.9	9.6	12.0		
CA. OTHER CAPITAL EXPENDITURES	%	15.8	9.8	25.8	4.9	4.9	16.2	19.6	27.3		
DA.R&D EXPENSE	$MM	14.7	15.7	19.9	23.7	22.5	23.9	26.7	29.2	16.4	8.0
EA. NEW PRODUCT R&D	%R&D	30.6	36.3	18.6	19.0	22.2	20.1	19.1	11.0		
FA.M&AE EXPENSE	$MM	60.2	71.5	85.5	106.1	118.4	146.0	175.5	200.1	19.2	18.5
GA.NET PLANNED PRODUCTIVITY (CUM)	$MM	0	15.1	36.7	64.1	86.2	121.7	139.0	154.0		
HA.CONTINGENCY	$MM	0	0	0	0	0	0	0	0		
IA.PARTS REVENUE	$MM	81.5	95.9	108.4	123.9	141.6	156.1	179.2	211.6	15.3	14.3
JA.PARTS VARIABLE MARGIN	%	42.29	41.78	42.94	42.83	42.79	42.45	42.50	42.38		

COMPOUND ANNUAL GROWTH RATES

MAJOR CAPITAL PROGRAMS
(Over $5.0 MM)

115-200 H.P. TRACTORS
GLOBAL SEGMENT

($ MM)

Program	APPROPRIATION Year	APPROPRIATION Amount	EXPENDITURES -1979	1980	1981	1982	1983	1984	1985+
*TR-4 Tractor Program - I & II	1978	$122.6	$ 7.5	$ 93.4	$ 21.7	$	$	$	$
*Auto Storage and Dist. System - Farmall	1979	25.0	.6	13.8	10.6				
TR-4 Tractor Program III	1980	31.8		6.9	24.9				
Increase Series 88 Capacity	1980	8.5		2.0	3.5	3.0			
3988 Tractor	1981	23.3			10.0	13.3			
Increase Farmall Capacity	1981	10.0			10.0				
Louisville Salt Bath Furnace	1981	5.0			.1	2.5	2.4		
Vari-Range Transmission (TR-4)	1982	36.0				4.0	22.0	10.0	
New Control Center (TR-4)	1982	5.5				3.5	2.0		
TR-3 Tractor Program	1983	9.7(A)					2.3	5.0	2.4
TOTAL		$277.4	$8.1	$116.1	$80.8	$26.3	$28.7	$15.0	$2.4

(A) Segment Portion of Total TR-3 Program ($41.9 MM)

*Project appropriation approved.

PRODUCT DEVELOPMENT TIMETABLE
(PROJECTS KNOWN OR LOGICALLY ANTICIPATED)

115 - 200 H.P. TRACTORS
GLOBAL SEGMENT

	1979	1980	1981	1982	1983	1984
International Harvester			New C-85 Control Center. TR-4 Synchro Transmission. Enclosed engine.	New 3988 "2+2" Tractor.		Hydro-Mechanical Transmission. New Control Center - TR-4 and 88 Series
Deere		New Control Center w/2 doors and more operator comfort. Improved 8-speed transmission w/power shift.	Pressure flow compensated hydraulics. Enclosed engine.		New Hyrdo-Mechanical Transmission & upgrade sheet metal styling	
Case			Upgrade hydraulics New front end styling.	New control Center. Enclosed engine.		New line of light weight high speed diesel engines.
Massey Ferguson				New styling Front end only; enclosed engine.		

CAUSAL FACTOR ANALYSIS

115 - 200 H.P. TRACTORS
GLOBAL SEGMENT

SALES

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR SALES	$MM		713.58	818.1	901.7	1087.8	1284.91
B. AVAIL MARKET EFFECT	$MM		42.15	39.58	26.74	31.62	53.75
C. SELLING PRICE EFFECT	$MM		51.74	53.68	73.45	83.48	88.06
D. MARKET SHARE EFFECT	$MM		30.34	(3.16)	27.27	30.77	20.72
E. WHL/DLR INV EFFECT	$MM		(20.14)	(9.71)	62.44	52.67	(48.52)
F. REMAINDER SALES	$MM		.44	3.21	(3.8)	(1.44)	.61
G.NET SALES	$MM	713.58	818.1	901.7	1087.8	1284.91	1399.43

PROFIT BEFORE TAX

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR	$MM		58.53	62	55.9	88.2	143.6
B. AVAIL MARKET EFFECT	$MM		11.88	10.93	7.41	8.95	15.95
C. COST INFL EFFECT	$MM		(47.82)	(51.99)	(62.32)	(72.09)	(76.76)
D. SELLING PRICE EFFECT	$MM		51.74	53.68	73.45	83.48	88.06
E. MARKET SHARE EFFECT	$MM		8.56	(.87)	7.56	8.71	6.15
F. WHOLESALE/DEALER INV EFFECT	$MM		(5.68)	(2.68)	17.31	14.91	(14.4)
G. VARIABLE COST CHG	$MM		(4.15)	2.94	2.99	16.15	1.86
H. FIXED COST CHG	$MM		(6.64)	(11.68)	(9.67)	(3.49)	(4.41)
I. REMAINDER	$MM		(4.41)	(6.42)	(4.44)	(1.23)	(.65)
J.PROFIT BEFORE TAX	$MM	58.53	62	55.9	88.2	143.6	159.4

Sales Commentary

Annual sales in 1984 are expected to be about $685 MM greater than 1979 sales. Selling price accounts for about 50% of the increase, market growth for about 28%, and market share for about 15%. The largest segment, North America, is basically a replacement market; growth comes from the trend to more sophisticated models, and from expansion of the overseas markets. Market share gain and inventory increases in 1982 are associated with the new 3988 "2+2" tractors.

Income Commentary

Annual income in 1984 is expected to be $100 MM greater than the 1979 level, a 170% increase. Market growth accounts for about 55% of the gain, net inflation effect accounts for about 39%, market share for about 30% and variable cost efficiencies for about 20%. Fixed costs are rising over the period, due to higher depreciation on new capital investments. The income decline in 1981 is caused primarily by start-up costs associated with the High Horsepower Tractor Program. Inventories are declining in 1980 and 1981, as demand for "2+2" tractors outpaces capacity. Rebuilding of inventories occurs in 1982 and 1983, and contributes significantly to the income gains in those years.

KEY ISSUES

115-200 H.P. TRACTORS
GLOBAL SEGMENT

Accounting for almost one-third of total AEG profits, this tractor segment is the backbone of major agricultural equipment companies. Industry capital spending and product development efforts are intense, as are pricing and sales programs. The "Aggressive Growth" strategy is certainly appropriate; the key issues are whether the Plan recognizes sufficient costs for gaining market share, and whether adequate programs are being planned and implemented to achieve forecast results.

Since the Plan may not produce product superiority, and may still provide unfavorable product costs (relative to Deere), the most important program is related to dealer/distributor quantity and quality. What specific programs and human resources requirements are needed to upgrade the distribution network in terms of:

- Replacement or upgrading of weak dealer/distributors, and closing of geographic gaps in coverage?
- Increased financial support to improve dealer/distributor liquidity (IHCC involvement?)?
- Personnel training in product knowledge and sales/marketing techniques?
- Additional salesmen requirements to achieve parity with, or surpass leading competitors - both in quantity and quality of sales personnel?

A basic question is the ability of the Plan to simultaneously increase market share (in a very competitive market), and increase ROA by two-thirds - especially when Deere is planning to spend $400MM per year in capital, twice as much as AEG on R&D, and will merchandise new products through a stronger dealer network. What additional steps can be taken to move ahead in technology, marketing performance, and manufacturing efficiency and capacity in order to take and keep share from Deere?

The Plan assumes pricing with inflation. Given a significant share gain forecast, competitive pressures in most markets, the historical tendency to grant sales concessions or to raise prices at less than inflationary rates -- it may be difficult for AEG to hold to an inflation pricing policy, even with the large contingency allocated to this Segment.

* What is the status of plans to accelerate international marketing of the "2+2" tractors (1979-A4)? Is capacity at Farmall adequate to support this program concurrent with high volume demands associated with a very high NAO market acceptance of the product, especially given the tradeoff between 88 Series and 86 Series production at the plant? Do the marketing research studies of early "2+2" buyers indicate any product dissatisfaction? Are these studies being continued?

Is acquisition of a competitor (e.g., Fendt or Deutz) a viable alternative for gaining product/market dominance in certain high growth EAME product/markets.

*1979 Key Issue.

OVER 200 H.P. TRACTORS

GLOBAL SEGMENT

IH SBU STRATEGIC COMMENTARY (P4):

HISTORY - Tractors in this segment are used primarily in heavy tillage and planting operations. There is a clear customer preference for more features such as wider speed range, power shift transmission, draft control hitch, power take-off, and operator comfort. Because these tractors are so expensive and productivity so significant, the proximity to effective parts and service availability from dealers is a very important buying consideration.

MARKET AND COMPETITION - The largest market for these tractors is in the U.S., where the market - depressed since 1977 - is not expected to recover until 1981.

Industry Unit Sales				Memo: 1978 IH Market Share (Units)		1978 Global Market Share ($)	
Market	1978	1984	% Change	W/O Brazil	With Brazil	Deere	22.6%
						Case	21.7
NAO	11,390	14,820	30.1%	12.4%	12.4%	Versatile	14.1
EAME	190	220	15.8	2.1	2.1	IH	11.7
PFELA*	720	790	9.7	22.2	17.6		
Total	12,300	15,830	28.7%	12.8%	12.6%		

* Excludes Brazil (190-1978; 400-1984).

Market leader Deere attained its position without product entries in the high H.P. end of this segment (current components are H.P. limited). Case aggressively pursues the tractor business but sells a limited line of implements.

IH PRODUCTS AND PRODUCTION - IH tractors (3 models) are built by Steiger Tractor (IH owns 35.6%) with 50% IH componentry. Product costs are excessive and many features are lacking.

STRATEGIES - Explore (Aggressive) - At present, negotiations with Steiger are in progress, and the status of this Segment is uncertain. The P-1 presented assumes that the models currently being merchandised by Steiger will be OEM purchased (without product revisions) and sold as IH replacement models for the current line. This will fill the product/market gap until 1983 (if Steiger negotiations are successful) when a new IH line, based on the 88 Series concept, will be produced in-house.

BUSINESS IDENTIFICATION FOR STRATEGIC PLANNING:
STRATEGIC BUSINESS UNIT (SBU) DIRECTORY (P3)

A. SALES $MM IN 1978 1,317

B. NUMBER OF P3 PAGES FOR THIS SBU 1

PRODUCTS/MANUFACTURING/ASSEMBLY

(1) PRODUCTS	1978 % OF SBU SALES	(2) MANUFACTURING/ASSEMBLY PLANTS			1978 % OF SBU SALES
TRACTORS		Farmall (USA)	33.8	Bradford (Britain)	2.7
Under 40 h.p. (DIN)	3.2	Neuss (Germany)	19.0	Mooka (Japan)	1.3
40-115 h.p. (DIN)	53.2	Doncaster (Britain)	15.1	Geelong(Australia)	1.2
115-200 h.p. (DIN)	39.3	St. Dizier (France)	13.3	Saltillo (Mexico)	1.1
Over 200 h.p. (DIN)	4.3	Fargo (USA)	3.9	Outside Purchase	5.1
		Louisville (USA)	3.5		
TOTAL	100%	TOTAL			100%

DISTRIBUTION/REGIONS

(3) DISTRIBUTOR/DEALERS, OEMS, DIRECT USERS	1978 % OF SBU SALES	(4) GEOGRAPHIC MARKETS	1978 % OF SBU SALES
Distributor/Dealer	97	North America	54.1
OEM	2	Europe/Africa/Middle East	40.6
Direct User	1	Pacific/Far East/Latin America	5.3
TOTAL	100%	TOTAL	100%

END USE MARKETS/COMPETITORS

(5) END USE MARKETS	1978 % OF SBU SALES	(6) COMPETITORS	1978 % MKT. SHARE
		Deere	17
		Massey Ferguson	16
		IH	15
		Ford	11
		Others (Allis-Chalmers, Case, Deutz, Fendt, Fiat, Renault, et al.)	41
TOTAL	100%	TOTAL	100%

July 31, 1979
DATE

Tractors
SBU

AE
GROUP

AGRICULTURAL EQUIPMENT

GROUP

GLOBAL SEGMENT
PLAN SUMMARY

OVER 200 H.P. TRACTORS*

THE PLAN: EXPLORE (AGGRESSIVE) - Major sourcing problem may cause absence from marketplace for two years.

$ MARKET GROWTH	
Relatively smooth growth (1980-84). Real	6.5%
Inflation	7.0
Total	13.5%

NET SALES GROWTH	
Fluctuating-peak of 60.6% (1980); low of 9.0% (1981).	
Real	21.7%
Pricing	7.1
Total	28.8%

COMPETITIVE PRESSURE
High to very high - John Deere, Case, and Versatile.

$ MARKET SHARE
Smooth Growth from 12.8% (1979) to 19.1% (1984).

PROFITABILITY
Turnaround-ROA from negative (9.7)% (1979) to 41.1% (1984).
ROS from negative (4.6)% (1979) to 10.6% (1984).

CUMULATIVE CASH FLOW
$35 million (1980-84), with $16 MM occurring in last year.

PRICING
Average 7.1% annual increase is slightly greater than market and avg. cost inflation rates.

COST INCREASE
Average variable and fixed cost inflation rates of 7.1% and 4.5% respectively-reflects OEM purchased product 1980-82.

PRODUCTIVITY %-OPERATING INCOME
Decreases from 54.7% (1980) to 11.6% (1984)-average of 29.8% (1980-84).

M & A
Decreases from 15.9% (1979) to 9.1% (1984) of sales.

R & D
Increases from 2.2% (1979) to 3.3% (1984).

CAPITAL EXPENDITURES
$17.3 MM total (1980-84) - $16.6 MM occurring 1980-82.

*Assumes that contract negotiations with Steiger were successful and that the tractor models currently being merchandised by that company will be OEM purchased (without product revisions) and sold as IH replacement models for the current line. This would fill the product/market gap until 1983 when a new in-house produced IH line would be available.

PROFILE (P-1)

		HISTORY		PROJECTION						GROWTH	
		1977	1978	1979	1980	1981	1982	1983	1984	77-79	79-84
A.AVAIL RETAIL MARKET	$MM	403.1	472.1	494.8	544.6	626.2	734.8	850.4	933.4	10.8	13.5
B.IH RETAIL MARKET SHARE	%	14.11	11.52	12.77	14.49	16.42	17.28	18.37	19.09		
C.LARGEST COMP SHARE	%	23.75	22.62	21.68	21.35	21.07	21.01	21.00	20.89		
D.COMPETITIVE STRENGTH		L	L	L	L	M	M	M	M		
E.NET SALES	$MM	72.9	56.2	67.6	108.5	118.3	152.4	204.4	239.2	(3.6)	28.8
F.CONTRIBUTION MARGIN	%	15.96	6.84	9.77	11.88	14.20	14.74	20.63	21.02		
G.TOTAL FIXED COST	$MM	9.0	8.8	8.0	8.3	9.9	11.6	17.5	22.2	(5.9)	22.8
H.OPERATING INCOME	$MM	2.6	(5.0)	(1.4)	4.6	6.9	10.9	24.7	28.0	(59)	86.8
I.PROFIT BEFORE TAX	$MM	1.2	(6.2)	(3.1)	3.0	5.4	9.1	22.5	25.4	(114)	59.2
J.TOTAL AVG ASSETS	$MM	28.3	33.2	31.8	30.4	34.0	41.7	57.3	61.7		
K.TOTAL ENDING ASSETS	$MM	28.0	25.7	31.5	33.4	36.5	43.6	57.1	61.8	6.1	14.4
L.CASH FLOW (OPERATIONAL)	$MM	NA	(4.9)	.8	8.4	2.7	3.6	4.6	15.8		
M.MARKET GROWTH	%		17.13	4.80	10.07	14.98	17.33	15.74	9.76		
N.SALES GROWTH	%		(22.98)	20.30	60.58	8.97	28.86	34.14	17.02		
O.PBT GROWTH	%					81.69	70.13	146.65	12.82		
P.ANNUAL MKT INFLATION	INDX		1.105	1.081	1.070	1.070	1.070	1.070	1.069		
Q.ANNUAL PR INFL	INDX		1.098	1.077	1.076	1.068	1.071	1.070	1.069		
R.ANNUAL UC INFL	INDX		1.092	1.075	1.078	1.068	1.070	1.070	1.069		
S.ANNUAL FC INFL	INDX		1.097	1.076	1.030	1.034	1.033	1.058	1.069		
T.ROA (AVG ASSETS)	%	4.2	(18.5)	(9.7)	9.7	15.8	21.8	39.2	41.1		
U.ROS	%	1.6	(11.0)	(4.6)	2.7	4.5	6.0	11.0	10.6		
V.ASSET(AVG) TURNOVER	X	2.58	1.69	2.13	3.57	3.48	3.65	3.57	3.88		
W.NET PRDTVTY/OP.INC	%	0	(2.0)	(164.3)	54.7	40.3	29.1	13.2	11.6		
X.R&D/SALES	%	2.9	3.2	2.2	2.7	2.7	2.1	2.4	3.3		
Y.M&E/SALES	%	12.5	19.7	15.9	9.9	10.5	9.1	8.5	9.1		
Z.ESTIMATED CAPITAL EXPENDITURES	$MM	.2	.2	.6	5.9	5.7	5.0	.4	.3		
AA. PRODUCT CAPITAL EXPENDITURES	%TOT	100.0	100.0	33.3	88.1	87.6	96.0	75.0	33.3		
BA. COST-REDUCTION CAP. EXPEND.	%TOT	0	0	16.7	10.2	8.8	0	0	0		
CA. OTHER CAPITAL EXPENDITURES	%	0	0	50.0	1.7	3.5	4.0	25.0	66.7		
DA.R&D EXPENSE	$MM	2.1	1.8	1.5	2.9	3.2	3.2	5.0	7.8	(13)	39.1
EA. NEW PRODUCT R&D	%R&D	0	0	0	0	0	0	0	0		
FA.M&E EXPENSE	$MM	9.1	11.1	10.7	10.8	12.6	13.8	17.3	21.7	8.5	15.1
GA.NET PLANNED PRODUCTIVITY (CUM)	$MM	0	.1	2.2	2.5	2.8	3.2	3.3	3.3		
HA.CONTINGENCY	$MM	0	0	0	0	0	0	0	0		
IA.PARTS REVENUE	$MM	10.9	13.1	15.8	17.5	22.2	27.1	35.8	48.3	20.6	26.0
JA.PARTS VARIABLE MARGIN	%	37.18	37.97	42.89	42.42	41.95	41.60	41.45	41.46		

COMPOUND ANNUAL GROWTH RATES

MAJOR CAPITAL PROGRAMS
(Over $5.0 MM)

OVER 200 H. P. TRACTORS
GLOBAL SEGMENT
($ MM)

Program	APPROPRIATION Year	APPROPRIATION Amount	EXPENDITURES -1979	1980	1981	1982	1983	1984	1985+
New 4 W D Tractors	1981	$ 10.0	$	$	$ 5.0	$ 5.0	$	$	$
TOTAL		$ 10.0			$ 5.0	$ 5.0			

*Project appropriation approved.

PRODUCT DEVELOPMENT TIMETABLE
(PROJECTS KNOWN OR LOGICALLY ANTICIPATED)

OVER 200 H.P. TRACTORS
GLOBAL SEGMENT

	1979	1980	1981	1982	1983	1984
International Harvester					New line of Operator Rear 4-WD tractors, New engine, Transmission, Styling	New Control Center
Deere		New Control Center two doors, more operator comfort		Large engine, new styling, enclosed engine		Upgrade hydraulics
Case		Pressure Flow Compensated Hydraulic System	New High Speed Engine		New 18-speed Synchro-torque transmission	New Control Center Enclosed Engine
Versatile			New front end styling, enclosed engine		Multi-speed transmission	New Control Center

CAUSAL FACTOR ANALYSIS

OVER 200 H.P. MODELS
GLOBAL SEGMENT

SALES

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR SALES	$MM		67.59	108.53	118.26	152.39	204.42
B. AVAIL MARKET EFFECT	$MM		1.79	5.92	9.92	10.34	4.13
C. SELLING PRICE EFFECT	$MM		7.71	7.53	10.04	13.46	15.39
D. MARKET SHARE EFFECT	$MM		8.76	11.29	5.88	8.64	6.37
E. WHL/DLR INV EFFECT	$MM		23.1	(15.15)	8.34	19.62	8.82
F. REMAINDER SALES	$MM		(.42)	.14	(.05)	(.03)	.1
G.NET SALES	$MM	67.59	108.53	118.26	152.39	204.42	239.22

PROFIT BEFORE TAX

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR	$MM		(3.08)	2.95	5.36	9.12	22.49
B. AVAIL MARKET EFFECT	$MM		.17	.7	1.41	1.52	.85
C. COST INFL EFFECT	$MM		(7.17)	(6.74)	(8.93)	(11.6)	(13.62)
D. SELLING PRICE EFFECT	$MM		7.71	7.53	10.04	13.46	15.39
E. MARKET SHARE EFFECT	$MM		.86	1.34	.83	1.27	1.31
F. WHOLESALE/DEALER INV EFFECT	$MM		2.26	(1.8)	1.18	2.89	1.82
G. VARIABLE COST CHG	$MM		2.26	2.53	.77	11.19	.89
H. FIXED COST CHG	$MM		(.1)	(1.27)	(1.35)	(4.94)	(3.28)
I. REMAINDER	$MM		.06	.12	(.21)	(.44)	(.48)
J.PROFIT BEFORE TAX	$MM	(3.08)	2.95	5.36	9.12	22.49	25.38

Sales Commentary

Annual sales in 1984 are expected to be about $170 MM greater than 1979 sales, a 250% increase. Selling price accounts for about 31% of the increase, wholesale/dealer inventories for about 26%, market share gains for about 24%, and market growth for about 19%. Inventory stocks rise in 1980 in anticipation of the 1981 market recovery; the inventory increase in '83 supports introduction of an in-house produced 4-W.D. line. A six market-share point gain is planned over 1979-1984, to 19%.

Income Commentary

Annual income in 1984 is expected to be about $28 MM greater than 1979 income (a loss year). Variable cost improvement accounts for about 62% of the increase, inventory buildup for about 22%, retained net inflation benefits for 21%, market share gain for 20%, and market growth for 16%. The variable cost improvement reflects outside sourcing in 1979, with in-house production beginning in 1983.

KEY ISSUES

OVER 200 H.P. TRACTORS
GLOBAL SEGMENT

* The failure of negotiations with Steiger have left IH vulnerable in the over 200 H.P. tractor market. The plan indicates that IH will continue to market its current Steiger built tractors, which are not well accepted, until the contract expires in 1980. The TX-190 Series tractor program requires considerable developmental and component sourcing decisions before in-house production can be started. Consequently, even with an accelerated program, sufficient product may not be available for a period of two to three years. (1979-A5).

Given the current distribution/market/production circumstances, it is necessary to determine whether IH should continue in this Segment. The following factors bear on this decision:

- Competition is likely to remain very aggressive: Deere is building an efficient new plant; Case is emphasizing this segment; etc. With our low market share, high product costs, and anticipated absence from the marketplace, it seems unlikely that we can achieve forecasted share increases, and at the same time, price with inflation and reduce M&A expense as a percent of sales.

- The projected ROA turnaround seems unlikely when it requires initiating our own production and retention of productivity concurrent with re-establishing of market position.

- The Group feels that linkage of the over 200 H.P. tractors with other agricultural products may be quite important. However, Deere achieved its position with no high horsepower product offerings, and Case achieved its position with a limited line of implements and a weak dealer network.

- What is the possibility that the existing "2+2", 193 H.P. tractor will take volume away from the over 200 H.P. market due to its exceptional performance/flexibility? What research is being done to determine whether it is a viable option to develop a totally new over 200 H.P. tractor? What would be the substitution effect on the 88 Series if a higher horsepower line is developed, and how would this affect the economics of both programs?

*1979 Key Issue.

GRAIN HARVESTING

GLOBAL SEGMENT

IH SBU STRATEGIC COMMENTARY (P4):

HISTORY - Grain Harvesting equipment's importance in terms of sales volume and profit contribution is second only to tractors. IH is considered one of the leaders in this market, which consists of the worldwide grain, bean, and seed producing countries. The SBU Segment is defined to include self-propelled and pull-behind combines, related crop-gathering attachments, and parts.

MARKET AND COMPETITION - There is a general trend in this market towards increased machine productivity and reduced manpower requirements which translates into demand for increased combine size, greater control sophistication, and increased operator comfort. With the exception of forecasted growth in Brazil and developing countries, the outlook for combines in traditional markets is generally considered stable or declining on a unit basis, but growing on a real dollar basis due to the trend towards larger and more sophisticated machines.

Industry Unit Sales*

Market	1979	1984	% Change
NAO	36,200	36,300	0.3%
EAME	23,800	26,400	10.9
PFELA	7,400	16,700	125.7
	67,400	79,400	17.8%

*Self-propelled combines

Memo: 1978 IH Market Share (Units)*

U.S.	21.1%
Canada	11.8
Australia	26.0
France	9.8
Germany	8.5

1978 Global Market Share (Units)*

John Deere	29%
Massey Ferguson	18
IH	14
New Holland/Clayson	13
Claas	8

John Deere has a broad range of well accepted combines, modern low-cost production facilities, and an excellent image and reputation in the marketplace. Massey-Ferguson enjoys strong worldwide distribution of its very complete line of combines, but market share has been deteriorating and they do have reliability problems. New Holland produces a twin rotor combine, but it has not experienced the same acceptance as the IH axial flow line.

IH PRODUCTS AND PRODUCTION - In NAO, beginning in 1980 all S.P. combines will be of the axial flow type (manufactured in the U.S. - currently capacity constrained), which has obtained excellent acceptance and provided IH with a significant competitive advantage. EAME produces conventional combines, but the present range is incomplete and lacks modern features on some models with profitability generally low. Axial flow combines have been successfully tested in Europe and will be sold in limited quantities starting in 1979. In Australia, our present product offering consists of pull-behind and self-propelled models - both well accepted. These will be supplemented by imported axial flow combines. In Brazil, our joint venture with IDEAL is marketing their new model C-1175, which is a modern machine, larger than any other on the Brazilian market.

STRATEGIES - Aggressive Growth - This SBU segment is considered one of AE Group's most attractive in terms of potential profit contribution and growth. Current acceptance and demand for the IH Axial Flow combines in NAO (exceeding our ability to supply) provide a competitive edge which AE Group is dedicated to exploiting globally. Continued product development efforts will aim at maintaining product superiority in the marketplace. Aggressive marketing programs are being implemented to achieve forecasted share gains in NAO and to accelerate international marketing of the axial flow concept machine.

July 31, 1979	B. H. Warren	Grain Harvesting Global Segment	AEG
Date	Manager	SBU	Group

BUSINESS IDENTIFICATION FOR STRATEGIC PLANNING:
STRATEGIC BUSINESS UNIT (SBU) DIRECTORY (P3)

A. SALES $MM IN 1978 459

B. NUMBER OF P3 PAGES FOR THIS SBU 1

PRODUCTS/MANUFACTURING/ASSEMBLY

(1) PRODUCTS	1978 % OF SBU SALES	(2) MANUFACTURING/ASSEMBLY PLANTS	1978 % OF SBU SALES
CROP HARVESTING		East Moline (USA)	71.4
Grain Harvesting (combines, corn heads, platforms, attachments, and parts)	90.1	Croix (France)	11.5
		Memphis (USA)	7.6
Cotton Harvesting (pickers, harvesters, and parts)	9.9	Geelong (Australia)	3.2
		Outside Purchase	6.3
TOTAL	100%	TOTAL	100%

DISTRIBUTION/REGIONS

(3) DISTRIBUTOR/DEALERS, OEMS, DIRECT USERS	1978 % OF SBU SALES	(4) GEOGRAPHIC MARKETS	1978 % OF SBU SALES
Distributor/Dealer	99	North America	86.6
		Europe/Africa/Middle East	8.9
Direct User	1	Pacific/Far East/Latin America	4.5
TOTAL	100%	TOTAL	100%

END USE MARKETS/COMPETITORS

(5) END USE MARKETS	1978 % OF SBU SALES	(6) COMPETITORS	1978 % MKT. SHARE
		Deere	29
		IH	17
		Massey Ferguson	15
		Others (Allis-Chalmers, Claas, New Holland, et al.)	39
TOTAL	100%	TOTAL	100%

July 31, 1979
DATE

Crop Harvesting
SBU

AE
GROUP

P3

AGRICULTURAL EQUIPMENT

GROUP
GLOBAL SEGMENT
PLAN SUMMARY
GRAIN HARVESTING

THE PLAN: AGGRESSIVE GROWTH-Through new concept products, strengthening of dealer organization, and product support.

$ MARKET GROWTH
Smooth growth 1980-84. Real 5.1% Inflation 6.2 Total 11.3%

NET SALES GROWTH
Rate peaks in 1981 and 1983 due to new product introductions. Real 8.1% Pricing 6.8 Total 14.9%

COMPETITIVE PRESSURE
High-especially as other rotary combines are introduced by competition. Deere is very strong; Massey Ferguson may be vulnerable.

$ MARKET SHARE
Smooth increase from 18.6% (1979) to 25.4% (1984).

PROFITABILITY
Very High - ROA increases from 45.3% (1979) to 58.0% (1984); ROS increases from 16.5% (1979) to 19.5% (1984).

CUMULATIVE CASH FLOW (A)
$498 million (1980-84), with $299MM occurring in 1983-84.

PRICING
Average 6.8% annual increase is greater than market inflation rate, but less than cost inflation rates.

COST INCREASE
Average variable and fixed cost inflation rates are both 7.0%.

PRODUCTIVITY %-OPERATING INCOME
Average 30.9% (1980-84).

M & A
Increases from 11.6% (1979) to 13.1%(1984) of sales.

R & D
Decreases from 2.0% (1979) to 1.9% (1984) of sales.

CAPITAL EXPENDITURES
Level expenditures totaling $161 million (1980-84).

(A) Not adjusted for contingencies.

PROFILE (P-1)

		HISTORY		PROJECTION						GROWTH	
		1977	1978	1979	1980	1981	1982	1983	1984	77-79	79-84
A.AVAIL RETAIL MARKET	$MM	1947.6	2216.4	2607.1	2841.6	3166.8	3519.1	4024.5	4450.8	15.7	11.3
B.IN RETAIL MARKET SHARE	%	12.45	15.75	18.55	20.54	21.57	22.61	24.97	25.41		
C.LARGEST COMP SHARE	%	27.00	27.90	25.70	23.20	21.80	22.40	22.60	22.50		
D.COMPETITIVE STRENGTH		L	L	L	M	M	M	M	M		
E.NET SALES	$MM	328.1	413.5	569.3	641.2	757.8	843.9	1044.9	1139.4	31.7	14.9
F.CONTRIBUTION MARGIN	%	36.02	33.87	35.03	35.00	35.30	35.30	35.18	36.07		
G.TOTAL FIXED COST	$MM	60.5	76.6	96.0	108.4	123.1	141.1	158.0	171.2	26.0	12.3
H.OPERATING INCOME	$MM	57.7	63.4	103.4	115.9	144.7	156.9	209.5	239.9	33.9	18.3
I.PROFIT BEFORE TAX	$MM	49.5	54.8	94.1	104.8	131.3	142.0	193.0	222.3	37.9	18.7
J.TOTAL AVG ASSETS	$MM	147.4	179.3	207.9	234.2	282.5	319.3	360.5	383.3		
K.TOTAL ENDING ASSETS	$MM	163.6	190.3	214.6	241.1	290.5	325.8	365.5	386.3	14.5	12.5
L.CASH FLOW (OPERATIONAL)	$MM	NA	19.3	33.8	47.6	62.1	89.1	138.3	160.7		
M.MARKET GROWTH	%		13.80	17.63	8.99	11.44	11.12	14.36	10.59		
N.SALES GROWTH	%		26.01	37.69	12.64	18.18	11.36	23.82	9.04		
O.PBT GROWTH	%		10.73	71.74	11.31	25.29	8.15	35.92	15.18		
P.ANNUNAL MKT INFLATION	INDX		1.096	1.070	1.061	1.061	1.062	1.063	1.061		
Q.ANNUAL PR INFL	INDX		1.107	1.078	1.068	1.064	1.071	1.071	1.067		
R.ANNUAL VC INFL	INDX		1.098	1.075	1.070	1.067	1.069	1.072	1.070		
S.ANNUAL FC INFL	INDX		1.095	1.074	1.073	1.067	1.069	1.071	1.070		
T.ROA (AVG ASSETS)	%	33.6	30.6	45.3	44.7	46.5	44.5	53.5	58.0		
U.ROS	%	15.1	13.3	16.5	16.3	17.3	16.8	18.5	19.5		
V.ASSET(AVG) TURNOVER	X	2.23	2.31	2.74	2.74	2.68	2.64	2.90	2.97		
W.NET PRDTVTY/OP.INC	%	0	11.0	22.0	29.0	30.4	34.5	30.0	30.8		
X.R&D/SALES	%	2.5	2.2	2.0	2.2	2.1	2.1	1.9	1.9		
Y.MAE/SALES	%	12.9	11.1	11.6	12.1	12.5	12.8	13.1	13.1		
Z.ESTIMATED CAPITAL EXPENDITURES	$MM	18.8	12.9	19.4	32.0	34.6	34.2	29.8	30.0		
AA. PRODUCT CAPITAL EXPENDITURES	%TOT	62.9	30.6	64.7	59.7	43.6	69.4	59.5	64.8		
BA. COST-REDUCTION CAP. EXPEND.	%TOT	.6	3.9	18.8	8.9	2.9	.9	2.3	4.0		
CA. OTHER CAPITAL EXPENDITURES	%	36.5	65.5	16.5	31.4	53.5	29.7	38.2	31.2		
DA.R&D EXPENSE	$MM	8.3	9.3	11.2	14.0	15.9	17.9	20.1	21.9	16.2	14.4
EA. NEW PRODUCT R&D	%R&D	66.3	64.5	15.2	17.9	16.4	16.2	14.4	16.4		
FA.M&AE EXPENSE	$MM	42.4	45.9	66.1	77.4	94.5	108.1	136.6	149.3	24.8	17.7
GA.NET PLANNED PRODUCTIVITY (CUM)	$MM	0	7.0	22.8	33.6	44.0	54.2	62.9	73.8		
HA.CONTINGENCY	$MM	0	0	0	0	0	0	0	0		
IA.PARTS REVENUE	$MM	52.8	64.4	74.9	80.8	97.4	111.1	138.8	158.4	19.1	16.2
JA.PARTS VARIABLE MARGIN	%	48.11	43.45	44.41	43.99	43.52	43.20	43.23	42.86		

COMPOUND ANNUAL GROWTH RATES

MAJOR CAPITAL PROGRAMS
(Over $5.0 MM)

GRAIN HARVESTING
GLOBAL SEGMENT
($ MM)

Program	APPROPRIATION Year	APPROPRIATION Amount	EXPENDITURES -1979	1980	1981	1982	1983	1984	1985+
*1420 Axial Flow Combine (III)	1979	$ 6.9	$ 1.6	$ 3.3	$ 2.0	$	$	$	$
*Combine Capacity - NAO	1979	12.3	1.2	8.2	2.9				
Upgrade 1400 Series Combines	1981	9.0			2.2	4.5	2.3		
1495 Axial Flow Combine (VII)	1983	9.5					5.7	3.8	
European Axial Flow Combines	1984	10.7						1.5	9.2
E. Moline Shipping Facility	1984	5.1							5.1
2100 Series Combines	1984(A)	41.5							41.5
TOTAL		$95.0	$ 2.8	$11.5	$ 7.1	$ 4.5	$ 8.0	5.3	$55.8

(A) Not in capital Plan: contained in P1/P4.

*Project appropriation approved.

PRODUCT DEVELOPMENT TIMETABLE
(PROJECTS KNOWN OR LOGICALLY ANTICIPATED)

GRAIN HARVESTING
GLOBAL SEGMENT

	1979	1980	1981	1982	1983	1984
International Harvester	1480 combine	1482 combine. Hillside 1460. Croix models 923, 933 & 943. 1420 combine (Class III).	New line of corn heads		New small pull-type combine. New grain headers. Facelift 1440 combine, 1460 combine, 1480 combine	Class VII combine. Axial Flow European manufacture.
Deere	New models of conventional combines				Model(s) of non-conventional combines	
Sperry Rand/ New Holland	Announce TR-70 in Europe			Additional models of non-conventional combines.		
Massey Ferguson						Model(s) non-conventional combines.

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CAUSAL FACTOR ANALYSIS

GRAIN HARVESTING
GLOBAL SEGMENT

SALES

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR SALES	$MM		569.26	641.2	757.8	843.91	1044.92
B. AVAIL MARKET EFFECT	$MM		13.22	29.12	31.74	60.43	43.01
C. SELLING PRICE EFFECT	$MM		40.86	45.67	56.22	69.17	71.19
D. MARKET SHARE EFFECT	$MM		53.31	30.54	34.59	89.36	18.5
E. WHL/DLR INV EFFECT	$MM		(31.78)	12.88	(29.81)	(10.9)	(32.09)
F. REMAINDER SALES	$MM		(3.66)	(1.61)	(6.63)	(7.05)	(6.11)
G.NET SALES	$MM	569.26	641.2	757.8	843.91	1044.92	1139.43

PROFIT BEFORE TAX

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR	$MM		94.15	104.8	131.3	142	193
B. AVAIL MARKET EFFECT	$MM		4.63	10.19	11.2	21.33	15.13
C. COST INFL EFFECT	$MM		(34.72)	(38.45)	(44.28)	(56.16)	(58.99)
D. SELLING PRICE EFFECT	$MM		40.86	45.67	56.22	69.17	71.19
E. MARKET SHARE EFFECT	$MM		18.67	16.65	12.21	31.54	6.51
F. WHOLESALE/DEALER INV EFFECT	$MM		(11.13)	4.51	(10.52)	(3.85)	(11.29)
G. VARIABLE COST CHG	$MM		.61	3.23	(1.25)	(.26)	11.74
H. FIXED COST CHG	$MM		(5.07)	(6.89)	(3.85)	(6.54)	(1.9)
I. REMAINDER	$MM		(3.2)	(2.49)	(4.03)	(4.24)	(3.09)
J.PROFIT BEFORE TAX	$MM	94.15	104.8	131.3	142	193	222.3

Sales Commentary

Annual sales in 1984 are expected to be about $570 MM greater than 1979 sales, a 100% increase. Selling price accounts for about 50% of the increase, market share gains account for about 40% of the increase, and market growth accounts for about 31%. Market growth reflects a cyclical recovery in this market, and the trend to larger, more sophisticated models. Market share gains in '80 and '83 stem from new product introductions and product improvements.

Income Commentary

Annual income in 1984 is expected to be about $128 MM greater than 1979 income, a 136% increase. Market share gains account for about 62% of the increase, market growth for about 49%, and retained net inflation benefits for 39%. The variable cost improvement in '84 is due to the favorable impact of productivity programs and manufacturing efficiencies associated with new products.

KEY ISSUES

GRAIN HARVESTING
GLOBAL SEGMENT

Given the apparent product superiority, market acceptance, and profitability of the new axial flow combine, the "Aggressive Growth" strategy for this SBU is appropriate:

- What additional steps should be taken to solidify IH's pre-emption of this market?
- What contingency plans are in place, or being developed, to provide swift and decisive action should competition react to IH gains in this area?
- * Is the R&D budget of 1.9% of sales adequate to maintain product superiority with the current axial flow combine and concurrently produce the next generation of machines (1979-A8)? Current product/market dominance allows this segment to generate 46% of total Group PBT on 19% of net sales in 1979.

The projected share gains for the axial flow combine appear reasonable as long as competition does not radically change relative pricing and promotion levels. However, the profit impact of planned pricing actions may be overstated if competition reacts aggressively:

- What steps are being taken to monitor both pricing and promotional activities of competition to determine at the earliest possible date whether responses are required to maintain projected volumes and/or margins?
- * If competition fails to react during the "capacity limited" stage, can price increases beyond the planned 6.8% be taken which would produce incremental income? (1979-A7).

This SBU, perhaps more than any other, represents a potential area of sizeable and profitable gain against competition. However, despite the fact that axial flow machines have been successfully tested in EAME, the plan calls for development and introduction of new conventional machines (900 Series - capital expense of $2.4 million plus related start-up and introduction costs) prior to production of axial flow machines in the European market:

- What are the cost/benefits of introducing the new 900 Series conventional combine models as an interim program vis-a-vis acceleration of the axial flow introduction into EAME, or spending similar amounts on M&A to maintain share?

* 1979 Key Issue.

COTTON HARVESTING

GLOBAL SEGMENT

IH SBU STRATEGIC COMMENTARY (P4):

HISTORY AND MARKET - IH was first to introduce a mechanical harvester for cotton and is a recognized leader in the production and distribution of cotton harvesting equipment. Cotton is grown in many parts of the world, with a regional breakdown of cotton production (1977-78 Avg.) as follows: North America - 20%; Eastern Bloc - 36%; and Other Foreign - 44%. Because of the many economic, political, and climatic factors that influence this market, the cotton industry is characterized by widely fluctuating swings in production and price. In the long term, cotton will remain a primary staple resource for the textile industry and can be expected to become more competitive vis-a-vis synthetics as a result of escalating costs in the petrochemical industry.

Although the majority of cotton is grown outside the U.S.A., over 90% of the Western World's cotton harvesting equipment is sold in the U.S.A. Most cotton growing countries are of the "developing" kind where increased mechanization of cotton harvesting is contingent upon economic-political considerations, plus the availability of a suitable infrastructure to support this level of technology.

COMPETITION - IH and Deere are the only major producers of cotton pickers in the Western world. Allis-Chalmers and Hesston participate with less than 10% of the Segment's volume through cotton harvester offerings. The NAO market for cotton pickers has ranged from less than 1,000 to 3,000 units per year, with IH market share recently declining from 50% in 1976 to 40% in 1978. Cotton harvester sales are estimated at 1,000 - 1,200 units per year, with IH currently holding a 17% market share. Deere offers an extensive line of cotton harvesting equipment and has established itself as the leader in this business.

IH PRODUCTS AND PRODUCTION - Product offerings include self-propelled cotton pickers, self-propelled cotton harvesters, and related parts. Parts account for approximately 25% of the segment's volume and currently are providing a major offset to machine losses. IH products are well established and accepted in the marketplace, but there is a need to improve and update current models. All IH cotton equipment is produced at Memphis Plant, which in 1979 will only be utilizing 40% of its PPC for this line.

STRATEGIES - Maintain and enhance for improved profitability and Group balance. - Because of production linkages with grain harvesting equipment, dependence of 170 dealers on cotton equipment sales for their viability, and reluctance to give Deere a virtual monopoly in this business, AEG does not intend to divest itself of this Segment. Profitability will be achieved through new product introductions. Emphasis will be increased on developing export sales markets.

RISKS - One high risk is that Deere will maintain a product development lead. Another is that our historical product cost problems will continue to plague this line.

July 31, 1979	B. H. Warren	Cotton Harvesting Global Segment	AEG
Date	Manager	SBU	Group

BUSINESS IDENTIFICATION FOR STRATEGIC PLANNING:
STRATEGIC BUSINESS UNIT (SBU) DIRECTORY (P3)

A. SALES $MM IN 1978 459

B. NUMBER OF P3 PAGES FOR THIS SBU 1

PRODUCTS/MANUFACTURING/ASSEMBLY

(1) PRODUCTS	1978 % OF SBU SALES	(2) MANUFACTURING/ASSEMBLY PLANTS	1978 % OF SBU SALES
CROP HARVESTING		East Moline (USA)	71.4
Grain Harvesting (combines, corn heads, platforms, attachments, and parts)	90.1	Croix (France)	11.5
		Memphis (USA)	7.6
Cotton Harvesting (pickers, harvesters, and parts)	9.9	Geelong (Australia)	3.2
		Outside Purchase	6.3
TOTAL	100%	TOTAL	100%

DISTRIBUTION/REGIONS

(3) DISTRIBUTOR/DEALERS, OEMS, DIRECT USERS	1978 % OF SBU SALES	(4) GEOGRAPHIC MARKETS	198 % OF SBU SALES
Distributor/Dealer	99	North America	86.6
		Europe/Africa/Middle East	8.9
Direct User	1	Pacific/Far East/Latin America	4.5
TOTAL	100%	TOTAL	100%

END USE MARKETS/COMPETITORS

(5) END USE MARKETS	1978 % OF SBU SALES	(6) COMPETITORS	198 % MKT. SHARE
		Deere	29
		IH	17
		Massey Ferguson	15
		Others (Allis-Chalmers, Claas, New Holland, et al.)	39
TOTAL	100%	TOTAL	100%

July 31, 1979
DATE

Crop Harvesting
SBU

AE
GROUP

P-3

AGRICULTURAL EQUIPMENT
GROUP
GLOBAL SEGMENT
PLAN SUMMARY
COTTON HARVESTING

THE PLAN: PROFITABILITY - Through new product introductions.

$ MARKET GROWTH
Fluctuating market growth with decline in 1981. Real 3.5% Inflation 6.7 Total 10.2%

NET SALES GROWTH
Widely fluctuating - decline with market in 1981. Real 11.8% Pricing 7.0 Total 18.8%

COMPETITIVE PRESSURE
High - Deere is the only other major competitor.

$ MARKET SHARE
Increases from 35.9% (1979) to 40.6% (1984). Peak of 42.9% in 1982.

PROFITABILITY
Turnaround-ROA from negative (13.4)% (1979) to 38.9% (1984).
ROS from negative (6.6)% (1979) to 19.5% (1984).

CUMULATIVE CASH FLOW
$29 MM (1980-84) - negative in 1980, with $30MM occurring in 1983-84.

PRICING
Avg. 7.0% annual increase is greater than market inflation factor, but equal to variable and fixed cost inflation rates.

COST INCREASE
Average variable and fixed cost inflation rates are both 7.0%.

PRODUCTIVITY %-OPERATING INCOME
Distorted by low margins in 1980-81, average is 43% for 1982-84.

M & A
Decreases from 14.2% (1979) to 10.6%(1984) of sales.

R & D
High - decreases from 6.6% (1979) to 5.0%(1984) of sales.

CAPITAL EXPENDITURES
$31 million total (1980-84); $18 million in 1980-81.

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PROFILE (P-1)

		HISTORY		PROJECTION						GROWTH	
		1977	1978	1979	1980	1981	1982	1983	1984	77-79	79-84
A.AVAIL RETAIL MARKET	$MM	135.3	120.4	124.5	140.7	134.4	165.3	190.0	202.0	(3.9)	10.2
B.IH RETAIL MARKET SHARE	%	49.93	39.05	35.85	34.43	38.48	42.89	38.44	40.58		
C.LARGEST COMP SHARE	%	45.50	54.36	56.71	57.74	55.16	48.94	53.37	51.55		
D.COMPETITIVE STRENGTH		M	M	L	L	L	M	M	M		
E.NET SALES	$MM	59.1	45.4	46.7	65.5	63.3	105.4	109.8	110.5	(10)	18.8
F.CONTRIBUTION MARGIN	%	34.06	33.35	35.18	38.09	41.11	46.85	49.56	50.34		
G.TOTAL FIXED COST	$MM	16.5	13.0	17.7	20.4	22.4	26.1	28.5	30.7	7.0	11.7
H.OPERATING INCOME	$MM	4.7	2.1	(1.3)	4.6	3.6	23.3	25.9	24.9	(51)	84.6
I.PROFIT BEFORE TAX	$MM	2.1	0	(3.1)	2.6	1.2	20.4	22.5	21.5	(87)	55.1
J.TOTAL AVG ASSETS	$MM	16.5	22.1	22.9	33.5	37.8	52.3	55.7	55.2		
K.TOTAL ENDING ASSETS	$MM	37.5	30.9	25.7	38.1	40.6	54.3	55.7	56.0	(15)	16.9
L.CASH FLOW (OPERATIONAL)	$MM	NA	1.5	(.4)	(6.3)	1.4	3.9	14.7	15.0		
M.MARKET GROWTH	%		(11.02)	3.43	12.95	(4.42)	22.94	14.97	6.29		
N.SALES GROWTH	%		(23.21)	2.82	40.43	(3.37)	66.46	4.21	.59		
O.PBT GROWTH	%					(51.94)	1542.74	10.41	(4.45)		
P.ANNUAL MKT INFLATION	INDX		1.105	1.079	1.068	1.063	1.070	1.068	1.065		
Q.ANNUAL PR INFL	INDX		1.102	1.083	1.069	1.070	1.070	1.070	1.069		
R.ANNUAL VC INFL	INDX		1.095	1.081	1.075	1.063	1.071	1.070	1.070		
S.ANNUAL FC INFL	INDX		1.100	1.080	1.074	1.067	1.068	1.070	1.071		
T.ROA (AVG ASSETS)	%	12.5	0	(13.4)	7.7	3.3	38.9	40.3	38.9		
U.ROS	%	3.5	0	(6.6)	3.9	2.0	19.3	20.5	19.5		
V.ASSET(AVG) TURNOVER	X	3.58	2.05	2.04	1.95	1.67	2.01	1.97	2.00		
W.NET PRDTVTY/OP.INC	%	0	76.6	(320.3)	109.2	219.8	40.8	42.1	47.4		
X.R&D/SALES	%	5.4	8.8	6.6	5.3	6.3	4.2	4.4	5.0		
Y.MAE/SALES	%	11.4	10.2	14.2	11.3	10.9	10.8	10.4	10.6		
Z.ESTIMATED CAPITAL EXPENDITURES	$MM	2.7	3.2	4.1	9.9	7.7	6.2	2.8	4.5		
AA. PRODUCT CAPITAL EXPENDITURES	%TOT	88.9	0	12.2	28.3	57.1	59.7	57.1	37.8		
BA. COST-REDUCTION CAP. EXPEND.	%TOT	0	0	58.5	40.4	22.1	8.1	0	13.3		
CA. OTHER CAPITAL EXPENDITURES	%	11.1	100.0	29.3	31.3	20.8	32.3	42.9	48.9		
DA.R&D EXPENSE	$MM	3.2	4.0	3.1	3.5	4.0	4.4	4.8	5.5	(1.6)	12.2
EA. NEW PRODUCT R&D	%R&D	46.9	80.0	3.2	20.0	27.5	34.1	37.5	43.6		
FA.M&AE EXPENSE	$MM	6.7	4.6	6.6	7.4	6.9	11.4	11.4	11.7	(.8)	12.1
GA.NET PLANNED PRODUCTIVITY (CUM)	$MM	0	1.6	4.1	5.0	8.0	9.5	10.9	11.8		
HA.CONTINGENCY	$MM	0	0	0	0	0	0	0	0		
IA.PARTS REVENUE	$MM	9.9	11.4	13.4	18.2	17.3	29.2	30.7	32.5	16.3	19.5
JA.PARTS VARIABLE MARGIN	%	42.96	37.83	42.32	41.46	40.82	40.98	41.28	40.82		

COMPOUND ANNUAL GROWTH RATES

MAJOR CAPITAL PROGRAMS
(Over $5.0 MM)

COTTON HARVESTING
GLOBAL SEGMENT
($ MM)

Program	APPROPRIATION Year	APPROPRIATION Amount	EXPENDITURES -1979	1980	1981	1982	1983	1984	1985+
New 2/4 Row Cotton Pickers	1980	$ 8.1	$	$ 4.1	$ 1.8	$ 2.2	$	$	$
TOTAL		$ 8.1		$ 4.1	$ 1.8	$ 2.2			

*Project appropriation approved.

7/31/79

PRODUCT DEVELOPMENT TIMETABLE
(PROJECTS KNOWN OR LOGICALLY ANTICIPATED)

COTTON HARVESTING
GLOBAL SEGMENT

	1979	1980	1981	1982	1983	1984
International Harvester	782 cotton picker	1400 cotton harvester		New two and four-row low inertia drum cotton picker		
Deere			New two and four-row cotton pickers and cotton harvesters			

7/31/79

CAUSAL FACTOR ANALYSIS

COTTON HARVESTING
GLOBAL SEGMENT

SALES

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR SALES	$MM		46.65	65.51	63.3	105.37	109.81
B. AVAIL MARKET EFFECT	$MM		2.57	(4.89)	7.31	5.4	(.17)
C. SELLING PRICE EFFECT	$MM		4.21	4.13	6.93	7.2	7.11
D. MARKET SHARE EFFECT	$MM		(1.88)	2.59	9.91	(7.92)	4.05
E. WHL/DLR INV EFFECT	$MM		13.98	(3.75)	17.96	(.12)	(10.09)
F. REMAINDER SALES	$MM		(.03)	(.29)	(.03)	(.12)	(.25)
G.NET SALES	$MM	46.65	65.51	63.3	105.37	109.81	110.46

PROFIT BEFORE TAX

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR	$MM		(3.08)	2.58	1.24	20.37	22.49
B. AVAIL MARKET EFFECT	$MM		.91	(1.86)	3	2.53	(.08)
C. COST INFL EFFECT	$MM		(4.22)	(3.62)	(5.38)	(5.6)	(5.64)
D. SELLING PRICE EFFECT	$MM		4.21	4.13	6.93	7.2	7.11
E. MARKET SHARE EFFECT	$MM		(.66)	.99	4.07	(3.71)	2.01
F. WHOLESALE/DEALER INV EFFECT	$MM		4.92	(1.43)	7.38	(.06)	(5)
G. VARIABLE COST CHG	$MM		2	1.58	5.69	2.76	.88
H. FIXED COST CHG	$MM		(1.28)	(.61)	(2.05)	(.55)	(.15)
I. REMAINDER	$MM		(.21)	(.61)	(.61)	(.55)	(.12)
J.PROFIT BEFORE TAX	$MM	(3.08)	2.58	1.24	20.37	22.49	21.49

Sales Commentary

Annual sales in 1984 are expected to be about $64 MM greater than 1979 sales, a 138% increase. Selling price accounts for about 45% of the gain, wholesale/dealer inventory effect for about 23%, market growth for for about 16%, and market share improvement for about 11%. Inventory buildups in '80 and '82 are in support of new product introductions (1400 harvester and low-inertia drum cotton picker). Market growth reflects the cyclical nature of the market, and the trend toward larger models.

Income Commentary

Annual income in 1984 is expected to be about $25 MM greater than 1979 income (a loss year). Variable cost improvement accounts for about 53% of the increase, wholesale/dealer inventory effect for about 24%, net inflation gain for about 21%, and market growth for about 18%. The major income improvement in '82 comes from inventory, variable cost, and market share effects associated with the new cotton picker.

KEY ISSUES

COTTON HARVESTING
GLOBAL SEGMENT

The "Manage for Profitability" categorization seems contradictory to the 17-18% planned growth in sales and assets. However, accomplishment of the Plan goals may be difficult given the low (3.5%) real market growth, Deere's high competitive strength, and the Segment's low historical profitability.

- Equipment sales (excluding parts) are first projected to become profitable in 1982, with a 6.4% market share gain (and a related 5.7 percentage point gain in contribution margin) caused by the introduction of a new cotton picker. What steps are being taken to monitor competitive activities to determine whether Deere will react competitively?

- What contingency plans are being developed if the projected 1982 product profitability turnaround does not materialize? If only 1981 levels of contribution margins are maintained in 1982, then marginal revenue would be reduced $6.0 million (13%) that year alone on sales of $105.4 million.

An "Aggressive Explore" strategy should be examined in view of the small contribution of this segment to the total Group (i.e., a PBT loss of $3.1 MM on 1.5 percent of AEG net sales in 1979), and the risk in the projected turnaround in 1982.

- What alternative IH business opportunities and plans in the cotton equipment business have been reviewed? Would an in-depth and comprehensive study on worldwide cotton production, harvesting, and utilization techniques and practices, such as that completed on Hay Handling Equipment, produce strategic alternatives not now being considered for this SBU? Completion of such a study may also indicate manufacturing options which would result in more favorable costs than those currently achieved and/or planned using the Memphis facility.

PLANTERS

GLOBAL SEGMENT

IH SBU STRATEGIC COMMENTARY (P4):

HISTORY AND MARKET - On a unit basis, it is estimated that IH obtains 25% of the fixed frame planter market in NAO and 38% of the grain drill market. IH obtains about 20% of the market in Australia. From 1971 through 1973, IH enjoyed substantial market acceptance in NAO and in exports, achieving market shares in excess of 40%. IH market share on the fixed frame planters has since eroded, as well as grain drills, due to the introduction by Deere of innovative products and the age of our planter line.

COMPETITION - The planter market remains somewhat of a local market on a worldwide basis:

Fixed Frame Planters - U.S. and Canada

Competitor	% Market Share	Market Position
John Deere	55%	1
IH	25	2
Allis-Chalmers	9	3
Massey Ferguson	5	5

Grain Drills - U.S. and Canada

Competitor	% Market Share	Market Position
IH	38%	1
John Deere	31	2
Tye	8	3
Massey Ferguson	5	6

John Deere is the strongest, most rapidly growing competitor, with the newest line of product and usage of effective manufacturing and marketing techniques. Deere appears to spend more on engineering than their competitors with the expected results in product innovation, performance, and manufacturing cost reduction.

IH PRODUCTS AND PRODUCTION - Current IH line is inadequate, primarily in the area of positive seed placement and depth control in planters and in the lack of an air-seeding system in grain drills.

Plate planters are produced in Canton Plant and the Cyclo-Planters in East Moline; current capacity utilization is only 59% and 19%. Plans call for the consolidation of production at East Moline. There exists a serious under-utilization problem throughout the Crop Production SBU.

STRATEGIES - Growth - Regain market acceptance and profitability, particularly in the fixed-frame planter business. Regain market share lost to Deere through the introduction of a complete new line of planters combining Cyclo Air feature with depth placement capability. Also, the entire grain drill line is being updated with both Canada and Australia recognizing the need for an air seeder unit. Undertake major modernization and standardization of grain drills and other products in the long term.

RISKS - Projected market share recovery and profit improvements will not occur as projected.

July 31, 1979	B. H. Warren	Planters Global Segment	AEG
Date	Manager	Group SBU	Group

BUSINESS IDENTIFICATION FOR STRATEGIC PLANNING:
STRATEGIC BUSINESS UNIT (SBU) DIRECTORY (P3)

A. SALES $MM IN 1978 288

B. NUMBER OF P3 PAGES FOR THIS SBU 1

PRODUCTS/MANUFACTURING/ASSEMBLY

(1) PRODUCTS	1978 % OF SBU SALES	(2) MANUFACTURING/ASSEMBLY PLANTS			1978 % OF SBU SALES
CROP PRODUCTION		Canton (USA)	30.3	Croix (France)	6.2
Planters (Row-crop planters, grain drills, row-crop cultivators and parts)	38.3	Hamilton (Canada)	26.0	Geelong (Australia)	4.5
		Memphis (USA)	11.2	Saltillo (Mexico)	0.7
Tillage (Moldboard plows, disk harrows, chisel plows, field cultivators and parts)	61.7	East Moline (USA)	10.2	Outside Purchase	10.9
TOTAL	100%	TOTAL			100%

DISTRIBUTION/REGIONS

(3) DISTRIBUTOR/DEALERS, OEMS, DIRECT USERS	1978 % OF SBU SALES	(4) GEOGRAPHIC MARKETS	1978 % OF SBU SALES
Distributor/Dealer	99	North America	88.1
Direct User	1	Europe/Africa/Middle East	6.6
		Pacific/Far East/Latin America	5.3
TOTAL	100%	TOTAL	100%

END USE MARKETS/COMPETITORS

(5) END USE MARKETS	1978 % OF SBU SALES	(6) COMPETITORS	1978 % MKT. SHARE
		North America Only:	
		Deere	25
		IH	22
		Others (Allis-Chalmers, Krause, Massey-Ferguson, Noble, Tye, et al.)	53
TOTAL	100%	TOTAL	100%

July 31, 1979
DATE

Crop Production
SBU

AE
GROUP

P-3

AGRICULTURAL EQUIPMENT
GROUP
GLOBAL SEGMENT
PLAN SUMMARY
PLANTERS

THE PLAN: GROWTH - Through new product introductions.

$ MARKET GROWTH	
Fluctuating market growth with peak in 1982. Real	7.6%
Inflation	7.0
Total	14.6%

NET SALES GROWTH	
Dramatic increases in 1981-1982. Real	17.5%
Pricing	6.9
Total	24.4%

COMPETITIVE PRESSURE

Rises from medium to very high by 1984.

$ MARKET SHARE

Increases from 28.6%(1979) to 34.4% (1984) with 4.6% share gain in 1982.

PROFITABILITY

Turnaround-ROA from negative (5.8)% (1979) to 44.6% (1984). ROS from negative (3.3)% (1979) to 17.3% (1984).

CUMULATIVE CASH FLOW (A)

$118MM (1980-84) - negative in 1980, with $96MM occurring in 1983-1984.

PRICING

Avg. 6.9% annual increase is slightly less than market inflation rate, but equal to cost inflation rates.

COST INCREASE

Average variable and fixed cost inflation rates are both 6.9%.

PRODUCTIVITY %-OPERATING INCOME

Distorted by loss position in 1980, avg. is 47% for 1981-84.

M & A

Very high - decreases slightly from 18.5%(1979) to 18.1%(1984) of sales.

R & D

Decreases from 4.1% (1979) to 3.6% (1984) of sales.

CAPITAL EXPENDITURES

$38 million total(1980-84) - $20 million in 1980-1981.

(A) Not adjusted for contingencies.

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PROFILE (P-1)

		HISTORY		PROJECTION						GROWTH	
		1977	1978	1979	1980	1981	1982	1983	1984	77-79	79-84
A.AVAIL RETAIL MARKET	$MM	400.1	423.4	381.6	405.0	492.9	625.7	701.7	764.2	(2.3)	14.6
B.IH RETAIL MARKET SHARE	%	30.56	83.88	28.64	25.90	27.80	32.40	34.24	34.38		
C.LARGEST COMP SHARE	%	33.74	37.95	41.83	44.31	42.10	38.89	35.30	30.84		
D.COMPETITIVE STRENGTH		M	L	L	L	M	H	H	H		
E.NET SALES	$MM	135.1	110.5	113.8	114.8	188.9	291.7	333.6	339.6	(7.6)	24.4
F.CONTRIBUTION MARGIN	%	62.96	39.55	38.35	37.72	40.22	37.04	40.61	41.45		
G.TOTAL FIXED COST	$MM	46.2	48.2	43.4	47.6	53.4	62.3	69.3	74.7	(2.9)	11.5
H.OPERATING INCOME	$MM	38.9	(4.5)	.2	(4.3)	22.6	45.8	66.2	66.1	(41)	215.9
I.PROFIT BEFORE TAX	$MM	33.7	(6.8)	(3.7)	(8.5)	17.5	39.5	59.1	58.7	(45)	77.8
J.TOTAL AVG ASSETS	$MM	80.9	73.4	64.7	71.2	90.9	120.7	130.6	131.6		
K.TOTAL ENDING ASSETS	$MM	79.9	74.1	66.1	75.5	94.0	122.2	131.6	132.5	(8.3)	14.9
L.CASH FLOW (OPERATIONAL)	$MM	NA	6.3	3.2	(6.4)	8.3	20.6	45.2	50.3		
M.MARKET GROWTH	%		5.84	(9.86)	6.12	21.71	26.94	12.15	7.48		
N.SALES GROWTH	%		(18.21)	2.98	.89	64.55	54.42	14.36	1.80		
O.PBT GROWTH	%						125.71	49.62	(.68)		
P.ANNUAL MKT INFLATION	INDX		1.109	1.081	1.070	1.070	1.070	1.070	1.069		
Q.ANNUAL PR INFL	INDX		1.105	1.083	1.070	1.070	1.068	1.070	1.069		
R.ANNUAL VC INFL	INDX		1.092	1.085	1.075	1.062	1.069	1.069	1.069		
S.ANNUAL FC INFL	INDX		1.096	1.083	1.073	1.065	1.068	1.069	1.070		
T.ROA (AVG ASSETS)	%	41.6	(9.3)	(5.8)	(11.9)	19.3	32.7	45.3	44.6		
U.ROS	%	24.9	(6.2)	(3.3)	(7.4)	9.3	13.5	17.7	17.3		
V.ASSET(AVG) TURNOVER	X	1.67	1.51	1.76	1.61	2.08	2.42	2.55	2.58		
W.NET PRDTVTY/OP.INC	%	0	(63.9)	3009.5	(250.7)	73.5	43.2	35.0	37.6		
X.R&D/SALES	%	3.0	4.8	4.1	5.1	3.8	3.0	3.2	3.6		
Y.MAE/SALES	%	13.4	16.9	18.5	19.1	16.4	18.5	18.1	18.1		
Z.ESTIMATED CAPITAL EXPENDITURES	$MM	3.2	4.3	4.3	10.8	9.5	5.7	5.5	6.2		
AA. PRODUCT CAPITAL EXPENDITURES	%TOT	30.7	33.2	22.7	67.9	44.4	46.1	63.1	41.0		
BA. COST-REDUCTION CAP. EXPEND.	%TOT	16.6	15.7	36.9	12.6	21.1	14.3	8.0	17.7		
CA. OTHER CAPITAL EXPENDITURES	%	52.7	51.2	41.4	19.4	34.6	39.6	28.9	41.3		
DA.R&D EXPENSE	$MM	4.1	5.3	4.7	5.8	7.1	8.7	10.7	12.2	7.1	21.0
EA. NEW PRODUCT R&D	%R&D	0	0	74.5	75.9	69.0	77.0	82.2	82.8		
FA.MAE EXPENSE	$MM	18.1	18.7	21.1	21.9	31.0	53.9	60.4	61.3	7.9	23.8
GA.NET PLANNED PRODUCTIVITY (CUM)	$MM	0	2.9	6.3	10.8	16.6	19.8	23.8	24.9		
HA.CONTINGENCY	$MM	0	0	0	0	0	0	0	0		
IA.PARTS REVENUE	$MM	20.2	24.3	26.3	24.8	45.8	66.6	75.6	79.8	11.8	25.9
JA.PARTS VARIABLE MARGIN	%	46.17	44.58	42.83	42.32	41.68	41.35	41.13	40.68		

COMPOUND ANNUAL GROWTH RATES

MAJOR CAPITAL PROGRAMS
(Over $5.0 MM)

PLANTERS

GLOBAL SEGMENT
($ MM)

Program	APPROPRIATION Year	APPROPRIATION Amount	EXPENDITURES -1979	1980	1981	1982	1983	1984	1985+
New Air Planter Models	1979	$ 10.8	$ 0.7	$ 7.2	$ 2.2	$ 0.6	$ 0.1		
Air Seeder	1982	5.9				0.5	2.1	2.7	.6
TOTAL		$ 16.7	$ 0.7	$ 7.2	$ 2.2	$ 1.1	$ 2.2	2.7	.6

*Project appropriation approved.

7/31/79

PRODUCT DEVELOPMENT TIMETABLE
(PROJECTS KNOWN OR LOGICALLY ANTICIPATED)

PLANTERS
GLOBAL SEGMENT

	1979	1980	1981	1982	1983	1984
International Harvester	New end-wheel drills	New model shovel drill	New models air planters New row crop cultivators	Additional models of new planters	Additional models of planters. Air delivery system for seeders	
Deere	New model folding cultivator		Air metering system for planters Air delivery system for seeders		New planter line with air metering and air delivery.	
Allis-Chalmers	New models of planters					

7/31/79

CAUSAL FACTOR ANALYSIS

PLANTERS
GLOBAL SEGMENT

SALES

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR SALES	$MM		113.79	114.8	188.9	291.7	333.6
B. AVAIL MARKET EFFECT	$MM		(.88)	14.46	25.46	9.76	1.28
C. SELLING PRICE EFFECT	$MM		7.55	12.3	18.65	21.77	21.88
D. MARKET SHARE EFFECT	$MM		(10.36)	8.74	26.87	12.09	.93
E. UHL/DLR INV EFFECT	$MM		4.76	38.59	31.43	(1.76)	(18.16)
F. REMAINDER SALES	$MM		(.06)	0	.39	.04	.06
G.NET SALES	$MM	113.79	114.8	188.9	291.7	333.6	339.6

PROFIT BEFORE TAX

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR	$MM		(3.72)	(8.5)	17.5	39.5	59.1
B. AVAIL MARKET EFFECT	$MM		(.34)	5.45	10.24	3.61	.52
C. COST INFL EFFECT	$MM		(8.21)	(9.9)	(15.79)	(17.34)	(17.69)
D. SELLING PRICE EFFECT	$MM		7.55	12.3	18.65	21.77	21.88
E. MARKET SHARE EFFECT	$MM		(3.97)	3.3	10.81	4.48	.38
F. WHOLESALE/DEALER INV EFFECT	$MM		1.83	14.56	12.64	(.65)	(7.37)
G. VARIABLE COST CHG	$MM		(.39)	3.69	(8.61)	11.66	2.64
H. FIXED COST CHG	$MM		(.92)	(2.52)	(4.92)	(2.56)	(.47)
I. REMAINDER	$MM		(.33)	(.88)	(1.01)	(.76)	(.29)
J.PROFIT BEFORE TAX	$MM	(3.72)	(8.5)	17.5	39.5	.59.1	58.7

Sales Commentary

Annual sales in 1984 are expected to increase by about $225 MM over 1979 sales, a threefold increase. Selling price accounts for about 36% of the increase, inventory effects for about 24%, market growth for about 22%, and market share improvement for about 17%. Inventory increases support the new product introductions, which are slated to occur each year from 1980 to 1983. Stocks are scaled down toward the end of the period. The market undergoes a cyclical peaking in 1982, when real growth reaches 20%. An aggressive growth program is designed to achieve market share gain of six points by 1984.

Income Commentary

Annual income declines $5 MM in 1980, primarily because of a market share loss. Income increases dramatically over the remainder of the period, and 1984 income is expected to be about $67 MM greater than 1980 income. Of that $67 MM increase, about 29% is due to market growth, about 29% to inventory increases, about 28% to market share gains, about 21% to retention of net inflation benefits, and about 13% to variable cost improvement. A market share gain of 4.6 points is expected in 1982. The unfavorable variable cost impact in 1982 is due to higher M&A expenses; the improvement in '83 is due to the impact of productivity programs, and reduction of warranty expenses and sales allowances because of the updated and more competitive product line. The income decline in 1984 is primarily due to inventory reduction.

KEY ISSUES

PLANTERS
GLOBAL SEGMENT

The "Manage for Growth" categorization is justified by planned significant product improvements and a forecast of dramatic turnaround in profitability and cash flow provided through quantum increases in sales, M&A expense, and market share -- starting in 1981-1982. The forecasted economic benefits occur primarily in the last two years of the plan, with $118MM of the five year total profits before tax of $166MM, and $96MM of the five year cumulative cash flow of $118MM occurring in 1983-84.

- What marketing research has been done, or is planned, to ascertain whether farmers perceive a competitive benefit of the new IH planter vis-a-vis Deere?

* • What marketing programs are being planned to produce the relative share gain against Deere? (1979-A16).

- The primary investment risks (other than accomplishing the forecasted economic benefits) appear to be large increases in M&A expenditures and possible competitive (pricing) reactions from Deere when their large projected share loss (11 points) actually starts to take place.

* • What alternatives exist to deal with facility under-utilization problems if a "Growth" plan is not successful? (1979-A11).

* 1979 Key Issue.

TILLAGE

GLOBAL SEGMENT

IH SBU STRATEGIC COMMENTARY (P4):

HISTORY AND MARKET - Equipment of this nature is used to prepare the soil for planting, prevent erosion, and for light cultivation throughout the world. IH participation is limited primarily to the U.S., Canada, Mexico, Australia, France and South Africa. In North America, IH provides a full line of major tillage equipment with a good dealer organization and a product reputation for performance and reliability. Current French design of plows is dated by five to ten years and is incomplete in range.

COMPETITION - In the North American market, the position of IH is very strong with products accepted as leader of the four major tillage categories. Deere will increase capacity and is considered the leading, aggressive NAO competitor.

	IH	J. Deere	Krause	Massey Ferguson
Moldboard Plows	32%	22%	-%	12%
Disk Harrows	16	10	13	3
Chisel Plows	18	15	3	2
Field Cultivators	23	12	1	1

In general, local manufacturers dominate the tillage businesses outside of North America. These short-line manufacturers are flexible and knowledgeable about local farming practices, and have excellent reputations.

PRODUCTION - Capacity utilization at Canton is: moldboard plows - 53.6%; disk harrow - 40.1%. At Memphis, disk harrows are at 60% of PPC. In Australia, production plans of 1,500 units in 1980 fall at 53.3% of practical plant capacity of 2,800 units. In Croix, plow production is less than 50% of practical plant capacity, resulting in a high manufacturing overhead rate. Tillage production in NAO is under review for facilities rationalization.

STRATEGIES - Hold in NAO. Grow in overseas markets, in particular, EAME.- In Europe, IH France will introduce a completely new line of roll-over plows aimed at increasing market share in France to at least 20% by 1983, and obtaining more significant share of markets such as Germany and the U.K. In NAO, the primary emphasis will be on modernization and standardization of the current, well-accepted product line. EAME will strive to obtain and hold a reasonable market share, provide a full line for dealers, and generate cash/improve profits.

RISKS - Pressure for conservation tillage and increased fuel prices could change product/market requirements.

July 31, 1979	B. H. Warren	Tillage Global Segment	AEG
Date	Manager	SBU	Group

BUSINESS IDENTIFICATION FOR STRATEGIC PLANNING:
STRATEGIC BUSINESS UNIT (SBU) DIRECTORY (P3)

A. SALES $MM IN 1978 288

B. NUMBER OF P3 PAGES FOR THIS SBU 1

PRODUCTS/MANUFACTURING/ASSEMBLY

(1) PRODUCTS	1978 % OF SBU SALES
CROP PRODUCTION	
Planters (Row-crop planters, grain drills, row-crop cultivators and parts)	38.3
Tillage (Moldboard plows, disk harrows, chisel plows, field cultivators and parts)	61.7
TOTAL	100%

(2) MANUFACTURING/ASSEMBLY PLANTS			1978 % OF SBU SALES
Canton (USA)	30.3	Croix (France)	6.2
Hamilton (Canada)	26.0	Geelong (Australia)	4.5
Memphis (USA)	11.2	Saltillo (Mexico)	0.7
East Moline (USA)	10.2	Outside Purchase	10.9
TOTAL			100%

DISTRIBUTION/REGIONS

(3) DISTRIBUTOR/DEALERS, OEMS, DIRECT USERS	1978 % OF SBU SALES
Distributor/Dealer	99
Direct User	1
TOTAL	100%

(4) GEOGRAPHIC MARKETS	1978 % OF SBU SALES
North America	88.1
Europe/Africa/Middle East	6.6
Pacific/Far East/Latin America	5.3
TOTAL	100%

END USE MARKETS/COMPETITORS

(5) END USE MARKETS	1978 % OF SBU SALES
TOTAL	100%

(6) COMPETITORS	1978 % MKT. SHARE
North America Only:	
Deere	25
IH	22
Others (Allis-Chalmers, Krause, Massey-Ferguson, Noble, Tye, et al.)	53
TOTAL	100%

July 31, 1979
DATE

Crop Production
SBU

AE
GROUP

P 3

AGRICULTURAL EQUIPMENT
GROUP
GLOBAL SEGMENT
PLAN SUMMARY
TILLAGE

THE PLAN: PROFITABILITY - NAO; GROWTH - EAME, PFELA: Through maintenance of modern product line.

$ MARKET GROWTH
Smooth growth 1980-1984. Real 1.4% Inflation 7.1 Total 8.5%

NET SALES GROWTH
Annual growth declines from 14.2% (1980) to 9.2%(1984). Real 4.1% Pricing 6.9 Total 11.0%

COMPETITIVE PRESSURE
High: Deere in U.S., and short-liners overseas.

$ MARKET SHARE
Increases from 16.3%(1979) to 18.1% (1984).

PROFITABILITY
ROA increases from 17.4% (1979) to 40.3% (1984).
ROS increases from 6.5% (1979) to 18.5% (1984).

CUMULATIVE CASH FLOW
$109 million (1980-84)-negative in 1980, with $80 million occurring in 1983-1984.

PRICING
Average 6.9% annual increase is slightly less than market and cost inflation rates.

COST INCREASE
Average variable and fixed cost inflation rates are both 7.0%.

PRODUCTIVITY %-OPERATING INCOME
High: Average is 74% for 1980-1984.

M & A
Decreases from 13.7% (1979) to 12.0%(1984) of sales.

R & D
Increases from 1.6%(1979) to 2.5%(1984) of sales.

CAPITAL EXPENDITURES
$79 million total (1980-1984)- $48 million in 1980-1981.

7/31/79

PROFILE (P-1)

		HISTORY		PROJECTION						GROWTH	
		1977	1978	1979	1980	1981	1982	1983	1984	77-79	79-84
A.AVAIL RETAIL MARKET	$MM	932.6	905.6	1077.2	1158.3	1306.2	1409.8	1524.6	1620.2	7.5	8.5
B.IM RETAIL MARKET SHARE	%	15.08	15.38	16.27	16.46	16.56	17.41	17.69	18.02		
C.LARGEST COMP SHARE	%	11.43	11.03	11.13	11.62	11.64	11.80	11.79	11.81		
D.COMPETITIVE STRENGTH		M	M	H	H	H	H	H	H		
E.NET SALES	$MM	189.6	177.7	237.7	271.4	305.5	338.5	366.1	399.7	12.0	11.0
F.CONTRIBUTION MARGIN	%	36.32	39.06	33.52	35.71	38.47	42.77	44.29	44.03		
G.TOTAL FIXED COST	$MM	56.2	54.8	60.0	66.0	74.7	80.0	84.8	92.8	3.3	9.1
H.OPERATING INCOME	$MM	12.7	14.6	19.7	30.9	42.9	64.8	77.3	83.3	24.9	33.4
I.PROFIT BEFORE TAX	$MM	6.9	10.8	15.4	25.2	35.2	56.1	68.2	73.9	62.0	36.9
J.TOTAL AVG ASSETS	$MM	112.8	98.8	88.1	116.3	142.5	163.2	171.5	183.2		
K.TOTAL ENDING ASSETS	$MM	116.7	88.2	92.6	126.6	148.8	164.6	173.2	186.2	(9.8)	15.0
L.CASH FLOW (OPERATIONAL)	$MM	NA	28.3	12.2	(13.3)	9.6	32.8	40.1	39.6		
M.MARKET GROWTH	%		(2.90)	18.96	7.52	12.77	7.93	8.15	6.27		
N.SALES GROWTH	%		(6.26)	33.77	14.18	12.56	10.78	8.16	9.19		
O.PBT GROWTH	%		84.10	42.53	64.36	39.36	59.56	21.64	8.22		
P.ANNUAL MKT INFLATION	INDX		1.094	1.121	1.072	1.070	1.071	1.070	1.072		
Q.ANNUAL PR INFL	INDX		1.097	1.118	1.066	1.063	1.077	1.070	1.070		
R.ANNUAL VC INFL	INDX		1.092	1.081	1.072	1.066	1.070	1.070	1.070		
S.ANNUAL FC INFL	INDX		1.090	1.080	1.073	1.067	1.070	1.070	1.070		
T.ROA (AVG ASSETS)	%	5.2	10.9	17.4	21.7	24.7	34.4	39.8	40.3		
U.ROS	%	3.1	6.1	6.5	9.3	11.5	16.6	18.6	18.5		
V.ASSET(AVG) TURNOVER	X	1.68	1.80	2.70	2.33	2.14	2.07	2.13	2.18		
W.NET PRDTVTY/OP.INC	%	0	33.6	57.6	66.1	94.9	71.7	66.1	69.0		
X.R&D/SALES	%	1.9	1.7	1.6	1.7	1.8	1.9	2.2	2.5		
Y.MAE/SALES	%	14.7	15.6	13.7	12.6	12.4	12.1	12.1	12.0		
Z.ESTIMATED CAPITAL EXPENDITURES	$MM	4.4	7.4	7.9	25.3	23.0	8.6	8.3	13.3		
AA. PRODUCT CAPITAL EXPENDITURES	%TOT	53.6	47.6	5.6	11.7	14.4	45.9	49.7	53.8		
BA. COST-REDUCTION CAP. EXPEND.	%TOT	10.2	7.7	14.2	29.4	30.0	7.6	9.3	19.5		
CA. OTHER CAPITAL EXPENDITURES	%	36.1	44.7	80.2	58.9	55.6	46.5	41.0	26.7		
DA.R&D EXPENSE	$MM	3.6	3.0	3.8	4.7	5.5	6.5	8.0	9.9	2.7	21.1
EA. NEW PRODUCT R&D	%R&D	0	0	0	0	0	0	0	0		
FA.M&AE EXPENSE	$MM	27.9	27.8	32.6	34.1	37.8	41.1	44.4	47.8	8.1	7.9
GA.NET PLANNED PRODUCTIVITY (CUM)	$MM	0	4.9	11.4	20.4	40.7	46.5	51.1	57.5		
HA.CONTINGENCY	$MM	0	0	0	0	0	0	0	0		
IA.PARTS REVENUE	$MM	43.3	50.6	53.8	61.8	70.3	75.9	81.5	93.6	11.6	11.7
JA.PARTS VARIABLE MARGIN	%	46.77	42.29	41.40	40.48	39.99	39.78	39.54	39.25		

COMPOUND ANNUAL GROWTH RATES

MAJOR CAPITAL PROGRAMS
(Over $5.0 MM)

TILLAGE

GLOBAL SEGMENT
($ MM)

Program	APPROPRIATION Year	APPROPRIATION Amount	EXPENDITURES -1979	1980	1981	1982	1983	1984	1985+
Disc Blade Mfg.	1979	$ 10.3	$	$ 5.6	$ 4.7	$	$	$	$
New Plow Line	1983	5.0						2.5	2.5
TOTAL		$ 15.3		$ 5.6	$ 4.7			$ 2.5	$ 2.5

*Project appropriation approved.

PRODUCT DEVELOPMENT TIMETABLE
(PROJECTS KNOWN OR LOGICALLY ANTICIPATED)

TILLAGE
GLOBAL SEGMENT

	1979	1980	1981	1982	1983	1984
International Harvester	596 Disk Harrow. 730 Semi-Mount Plow. Toggle trip beam sub-soiler.	496 Disk Harrow. 21.5' Field Cultivator. 41.5' Field Cultivator. New Process Plow Bottoms.	790 Offset Disk Harrow. 64' Field Cultivator. 45' Chisel Plow.	Uni-Frame Disk Harrow. 498 Disk Harrow.		New low-draft moldboard plows.
Deere	New models of disk harrows		Facelift plows, Disk Harrows, Cultivators.			
Krause		Expand line of chisel plows	Facelift of disk harrow line			
Massey Ferguson	Massey Ferguson is making plans to buy in-stead of manu-facture tillage line.					

7/31/79

CAUSAL FACTOR ANALYSIS

TILLAGE
GLOBAL SEGMENT

SALES

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR SALES	$MM		237.72	271.44	305.54	338.48	366.09
B. AVAIL MARKET EFFECT	$MM		.59	10.34	1.59	2.67	(2.23)
C. SELLING PRICE EFFECT	$MM		16.76	17.99	24.24	23.82	25.98
D. MARKET SHARE EFFECT	$MM		1.94	1.41	11.1	4.12	4.88
E. WHL/DLR INV EFFECT	$MM		13.45	3.02	(2.78)	(3.05)	4.47
F. REMAINDER SALES	$MM		.97	1.35	(1.22)	.05	.52
G.NET SALES	$MM	237.72	271.44	305.54	338.48	366.09	399.72

PROFIT BEFORE TAX

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR	$MM		15.35	25.23	35.16	56.1	68.24
B. AVAIL MARKET EFFECT	$MM		.2	3.69	.61	1.14	(.99)
C. COST INFL EFFECT	$MM		(16.16)	(16.29)	(17.88)	(18.89)	(20.62)
D. SELLING PRICE EFFECT	$MM		16.76	17.99	24.24	23.82	25.98
E. MARKET SHARE EFFECT	$MM		.65	.5	4.27	1.76	2.16
F. WHOLESALE/DEALER INV EFFECT	$MM		4.51	1.08	(1.07)	(1.3)	1.98
G. VARIABLE COST CHG	$MM		6.49	8.45	12.3	5.25	(.96)
H. FIXED COST CHG	$MM		(1.58)	(3.97)	(.07)	.74	(1.87)
I. REMAINDER	$MM		(.98)	(1.53)	(1.47)	(.38)	(.08)
J.PROFIT BEFORE TAX	$MM	15.35	25.23	35.16	56.1	68.24	73.85

Sales Commentary

Annual sales in 1984 are expected to be about $162 MM greater than 1979 sales. Selling price is responsible for about 67% of the increase, market share gain for about 14%, wholesale/dealer inventory effect for about 9%. The available market trend shows a gradual decline in unit sales, but a shift to more sophisticated and expensive models. An extensive product development program is scheduled over the period.

Income Commentary

Annual income in 1984 is expected to be about $58 MM greater than 1979 income. Variable cost improvement accounts for about 54% of the increase (major facilities rationalization program to utilize excess capacity), retained net inflation benefits account for about 32% of the increase, and market share improvements account for about 16%.

KEY ISSUES

TILLAGE
GLOBAL SEGMENT

The "Manage for Profitability" categorization of this Segment seems appropriate, given the present leading market position (in an essentially flat market), and the strong product reputation for performance and reliability.

* Given the serious capacity under-utilization of this Segment worldwide, the key issue is one of facilities rationalization. (1979-All).

- Would a more aggressive pricing strategy (in the early years of this Plan) yield more favorable economic results?
- Could OEM sales to competitors have favorable economic benefits?
- What programs are being implemented and/or emphasized to achieve a more aggressive participation in the Canadian tillage market?

Is there a technological threat to the Tillage Segment by virtue of changing trends in farming practices in various geographic areas--"minimum" or "no" tillage--in view of the fact that these approaches would substantially reduce agricultural consumption of diesel fuel?

*1979 Key Issue.

HAY, FORAGE, & MATERIAL HANDLING

GLOBAL SBU

IH SBU STRATEGIC COMMENTARY (P4):

SBU SEGMENTATION - 1979 Fiscal Year Projections

Sales by Operation	
NAO	62%
EAME	36%
PFELA	2%

Sales, Profit Before Tax, Cash Flow and ROA by Segment

Segment	Net Sales	PBT	Cash Flow	ROA
Hay	68%	$ 0.1 MM	$(6.3)MM	0.1%
Forage	10%	$(1.1)MM	$(0.1)MM	(12.9%)
Mat'l Handling	22%	$ 7.8 MM	$ 5.9 MM	68.4%*

*Material Handling Segment primarily consists of OEM purchased products merchandised as IH product.

SBU STRATEGY - Profitability

HAY SEGMENT - IH market shares generally are around 10% or less—an undesirable position which has contributed to the poor profit performance of this segment. Contributing to poor market performance have been an incomplete product line, poor marketing/sales support, and consequent encroachment into IH dealerships by competitive lines. In general, the IH product line is incomplete; past insufficient R & D and marketing commitment have resulted in an obsolete and expensive product line. The strongest offering in NAO is the mower conditioner. Other major product line categories are rectangular and roll balers, mowers, and windrowers.

HAY STRATEGY - Hold near term. Growth longer term.

FORAGE SEGMENT - The NAO market for forage equipment is considered mature, with forecasted unit declines being largely offset by the trend towards larger pull-behind and toward self-propelled machines. In the past, EAME forage harvesters were imported from North America, though not very successfully. Presently, only outside purchased machines are sold by IH - Pottinger in France and PZ in Germany. In NAO, IH expects to obtain 12% of the forage harvester market in 1979.

FORAGE STRATEGY - Growth in NAO; explore in EAME.

MATERIAL HANDLING SEGMENT - The material handling product line is diverse and includes attachment and powered equipment to mix, move and disperse feed, silage, their by-products and other materials. Projections for this product segment indicate that a gradual unit industry decline will occur between 1979 and 1984. Because of the wide variety of products offered in the market place, in calculating market size and market share, comparisons are only made against comparable products. On this basis, we estimate that our market share is around 8%. In general, the IH line in each of these categories is limited when compared to some of the full line offerings made by the specialized manufacturers active in this segment, such as Melroe in compact loaders. In the products offered, IH is very competitive in features, terms, and price. IH asset requirements are lower in this SBU segment because most of the products are outside sourced. Pre-tax ROA on machines only is in excess of 60%. Gradual growth in products handled by outside vendors is aimed at increased sales and profits, as well as providing a full selection of product offerings.

MATERIAL HANDLING STRATEGY - Hold, or moderate growth, for continued profitability.

BUSINESS IDENTIFICATION FOR STRATEGIC PLANNING: STRATEGIC BUSINESS UNIT (SBU) DIRECTORY (P3)

A. SALES $MM IN 1978 164

B. NUMBER OF P3 PAGES FOR THIS SBU 1

PRODUCTS/MANUFACTURING/ASSEMBLY

(1) PRODUCTS	1978 % OF SBU SALES	(2) MANUFACTURING/ASSEMBLY PLANTS	1978 % OF SBU SALES
HAY, FORAGE, MATERIAL HANDLING		Hamilton (Canada)	27.3
Hay Harvesting (Balers, windrowers, mower conditioners, mowers, swathers, tedders, rakes, and parts	63.7	Memphis (USA)	21.1
		Doncaster (Britain)	14.9
		Croix (France)	8.5
Forage Harvesting (Forage harvesters, flail choppers, forage boxes, blowers and parts)	11.6	Outside Purchase	28.2
Material Handling (Compact loaders, tractor-mounted loaders, manure spreaders, wagons, grinder-mixers, shredders, snow blowers, miscellaneous and parts.	24.7		
TOTAL	100%	TOTAL	100%

DISTRIBUTION/REGIONS

(3) DISTRIBUTOR/DEALERS, OEMS, DIRECT USERS	1978 % OF SBU SALES	(4) GEOGRAPHIC MARKETS	1978 % OF SBU SALES
Distributor/Dealer	98	North America	63.3
		Europe/Africa/Middle East	35.5
Direct User	2	Pacific/Far East/Latin America	1.2
TOTAL	100%	TOTAL	100%

END USE MARKETS/COMPETITORS

(5) END USE MARKETS	1978 % OF SBU SALES	(6) COMPETITORS	1978 % MKT. SHARE
		Hay Harvesting Only:	
		New Holland (Sperry Rand)	33
		Deere	22
		Massey Ferguson	8
		IH	7
		Others (Claas, Hesston, Vermeer, Welger, etc.)	30
TOTAL	100%	TOTAL	100%

July 31, 1979
DATE

Hay, Forage, Material Handling
SBU

AE
GROUP

AGRICULTURAL EQUIPMENT

GROUP

GLOBAL SBU

PLAN SUMMARY

HAY, FORAGE, & MATERIAL HANDLING

THE PLAN: PROFITABILITY- Through new product introductions designed to support the "full line supplier" concept.

$ MARKET GROWTH		
Slight decline in real market size 1980-84.	Real	(0.6)%
	Inflation	7.0
	Total	6.4%

NET SALES GROWTH		
Generally smooth, but 18.6% peak in 1980.	Real	5.0%
	Pricing	6.9
	Total	11.9%

COMPETITIVE PRESSURE
Competitive resistance is high due to multitude of specialty producers.

$ MARKET SHARE
Smooth increase from 9.8% (1979) to 12.8% (1984).

PROFITABILITY
ROA increases from 6.7% (1979) to 22.2% (1984).
ROS increases from 3.2% (1979) to 9.9% (1984).

CUMULATIVE CASH FLOW (A)
$89 million (1980-84), with $70 MM occurring in 1982-84.

PRICING
Average 6.9% annual increase is slightly less than market and avg. cost inflation rates.

COST INCREASE
Avg. variable and fixed cost inflation rates of 7.0% and 6.9% respectively.

PRODUCTIVITY %-OPERATING INCOME
High: Average is 71% for 1980-84.

M & A
Increases from 13.4% (1979) to 14.5% (1984) of sales.

R & D
Increases slightly from 2.6% (1979) to 2.7% (1984) of sales, with interim years at lower rates.

CAPITAL EXPENDITURES
Level expenditures totaling $50 million (1980-84).

(A) Not adjusted for contingencies.

PROFILE (P-1)

		HISTORY		PROJECTION						GROWTH	
		1977	1978	1979	1980	1981	1982	1983	1984	77-79	79-84
A.AVAIL RETAIL MARKET	$MM	1203.8	1324.6	1468.4	1580.8	1702.0	1778.3	1907.1	2006.8	10.4	6.4
B.IH RETAIL MARKET SHARE	%	8.20	7.63	9.75	10.18	10.68	11.52	12.40	12.84		
C.LARGEST COMP SHARE	%	27.41	27.98	26.56	26.20	25.70	25.26	24.57	24.20		
D.COMPETITIVE STRENGTH		L	L	L	L	L	L	L	L		
E.NET SALES	$MM	161.0	164.1	214.1	253.9	275.0	305.4	344.4	376.2	15.3	11.9
F.CONTRIBUTION MARGIN	%	32.57	31.76	30.23	30.50	32.34	32.18	33.42	33.67		
G.TOTAL FIXED COST	$MM	42.0	45.0	53.6	57.5	62.6	68.6	74.9	82.6	12.9	9.1
H.OPERATING INCOME	$MM	10.5	7.1	11.2	19.9	26.4	29.7	40.2	44.1	3.4	31.6
I.PROFIT BEFORE TAX	$MM	6.4	3.6	6.8	14.8	20.7	23.6	33.7	37.3	3.3	40.7
J.TOTAL AVG ASSETS	$MM	83.2	88.0	101.7	114.2	125.0	140.2	155.5	167.8		
K.TOTAL ENDING ASSETS	$MM	85.0	89.9	105.4	118.9	129.7	143.3	158.2	171.7	11.4	10.2
L.CASH FLOW (OPERATIONAL)	$MM	NA	1.7	(.5)	4.0	14.7	16.2	25.8	28.3		
M.MARKET GROWTH	%		10.03	10.86	7.66	7.66	4.49	7.24	5.22		
N.SALES GROWTH	%		1.93	30.46	18.56	8.31	11.05	12.77	9.23		
O.PBT GROWTH	%		(43.31)	88.07	118.42	39.86	14.01	42.79	10.68		
P.ANNUAL MKT INFLATION	INDX		1.104	1.080	1.071	1.070	1.070	1.070	1.070		
Q.ANNUAL PR INFL	INDX		1.087	1.082	1.072	1.069	1.070	1.069	1.067		
R.ANNUAL VC INFL	INDX		1.083	1.082	1.072	1.069	1.070	1.068	1.069		
S.ANNUAL FC INFL	INDX		1.091	1.081	1.070	1.070	1.070	1.067	1.069		
T.ROA (AVG ASSETS)	%	7.6	4.1	6.7	13.0	16.6	16.8	21.7	22.2		
U.ROS	%	3.9	2.2	3.2	5.8	7.5	7.7	9.8	9.9		
V.ASSET(AVG) TURNOVER	X	1.93	1.87	2.11	2.22	2.20	2.18	2.21	2.24		
W.NET PRDTVTY/OP.INC	%	0	40.8	86.5	58.7	69.4	78.1	74.7	73.5		
X.R&D/SALES	%	2.8	2.8	2.6	2.4	2.4	2.5	2.5	2.7		
Y.MAE/SALES	%	12.1	14.5	13.4	13.3	13.4	14.7	14.5	14.5		
Z.ESTIMATED CAPITAL EXPENDITURES	$MM	2.3	4.8	5.2	9.9	12.9	11.6	5.9	9.2		
AA. PRODUCT CAPITAL EXPENDITURES	%TOT	47.8	25.0	17.3	24.2	31.8	50.0	44.1	78.3		
BA. COST-REDUCTION CAP. EXPEND.	%TOT	4.3	33.3	44.2	31.3	25.6	13.8	11.9	1.1		
CA. OTHER CAPITAL EXPENDITURES	%	47.8	41.7	38.5	44.4	42.6	36.2	44.1	20.7		
DA.R&D EXPENSE	$MM	4.5	4.6	5.6	6.0	6.6	7.5	8.7	10.1	10.6	12.9
EA. NEW PRODUCT R&D	%R&D	8.9	26.1	41.8	83.3	18.2	12.0	8.0	4.0		
FA.M&AE EXPENSE	$MM	19.6	23.9	28.7	33.8	36.9	44.9	50.0	54.6	21.0	13.8
GA.NET PLANNED PRODUCTIVITY (CUM)	$MM	0	2.9	9.7	11.7	18.3	23.2	30.0	38.4		
HA.CONTINGENCY	$MM	0	0	0	0	0	0	0	0		
IA.PARTS REVENUE	$MM	32.7	36.5	42.7	47.2	51.4	57.0	63.6	71.1	14.3	10.7
JA.PARTS VARIABLE MARGIN	%	37.81	37.79	36.37	38.05	38.06	38.56	38.78	39.09		

COMPOUND ANNUAL GROWTH RATES

MAJOR CAPITAL PROGRAMS
(Over $5.0 MM)

HAY, FORAGE, & MATERIAL HANDLING
GLOBAL SBU

($ MM)

Program	APPROPRIATION Year	APPROPRIATION Amount	EXPENDITURES -1979	1980	1981	1982	1983	1984	1985+
New WBX Rectangular Balers	1981	$ 5.9	$	$	$ 1.6	$ 4.3	$	$	$
TOTAL		$ 5.9			$ 1.6	$ 4.3			

*Project appropriation approved.

PRODUCT DEVELOPMENT TIMETABLE
(PROJECTS KNOWN OR LOGICALLY ANTICIPATED)

HAY, FORAGE & MATERIAL HANDLING
GLOBAL SBU

	1979	1980	1981	1982	1983	1984
International Harvester	3100 Series mower. 425-baler test market. New twine knotters. Ear corn hopper.	425 baler-full production. 16' dual auger self-propelled windrower. Corn snapper attachment. New forage blower.	New mower-conditioner. New round baler. Direct throw forage harvester.	32' canvas platform for windrower. One row forage harvester.	New WBX line of hay balers.	New round baler in small size. New line of windrowers. Replacement models for 720/830 forage harvesters.
Sperry Rand/ New Holland	New models of forage harvesters.		New models of balers-rectangular and round.			New line of forage harvesters.
Deere	New models of forage harvesters.		Update of current models.			Haylage-in-a-day equipment.
Welger		Update of current models.				

CAUSAL FACTOR ANALYSIS

HAY, FORAGE, & MATERIAL HANDLING
GLOBAL SBU

SALES

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR SALES	$MM		214.15	253.9	275	305.4	344.4
B. AVAIL MARKET EFFECT	$MM		.79	1.02	(4.23)	.56	(3.85)
C. SELLING PRICE EFFECT	$MM		17.1	17.85	19.98	22.24	23.77
D. MARKET SHARE EFFECT	$MM		6.41	7.96	13.95	15.61	8.24
E. WHL/DLR INV EFFECT	$MM		15.67	(5.8)	.75	.49	3.14
F. REMAINDER SALES	$MM		(.22)	.07	(.04)	.1	.49
G.NET SALES	$MM	214.15	253.9	275	305.4	344.4	376.2

PROFIT BEFORE TAX

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR	$MM		6.78	14.8	20.7	23.6	33.7
B. AVAIL MARKET EFFECT	$MM		.24	.31	(1.37)	.18	(1.29)
C. COST INFL EFFECT	$MM		(15.68)	(16.17)	(17.95)	(19.23)	(21.49)
D. SELLING PRICE EFFECT	$MM		17.1	17.85	19.98	22.24	23.77
E. MARKET SHARE EFFECT	$MM		1.94	2.43	4.51	5.03	2.75
F. WHOLESALE/DEALER INV EFFECT	$MM		4.74	(1.77)	.24	.16	1.05
G. VARIABLE COST CHG	$MM		.67	4.7	(.52)	3.65	1.23
H. FIXED COST CHG	$MM		(.17)	(.95)	(1.57)	(1.53)	(2.38)
I. REMAINDER	$MM		(.8)	(.5)	(.42)	(.43)	(.05)
J.PROFIT BEFORE TAX	$MM	6.78	14.8	20.7	23.6	33.7	37.3

Sales Commentary

Annual sales in 1984 are expected to be about $162 MM greater than 1979 sales. Selling price accounts for 62% of the increase, market share increases for 32%, and wholesale/dealer inventory increases for 9%. Market share grows steadily over the period, as product line gaps are filled and current offerings are updated. There is a substantial inventory buildup in 1980, in support of new product developments and aggressive marketing efforts. A real decline in the available market takes place over the period.

Income Commentary

Annual income in 1984 is expected to be about $30 MM greater than 1979 income. Market share gains account for about 55% of the increase, retained net inflation benefits for about 34%, variable cost improvement for about 32%, and wholesale/dealer inventory increases for about 14%. The variable cost improvement in 1983 is due to the more profitable line of hay balers introduced that year.

KEY ISSUES

HAY, FORAGE & MATERIAL HANDLING GLOBAL SBU

This SBU is positioned in a very diverse and fragmented product market with industry real dollar sales declining, high competitive resistance (due to a multitude of capable specialty producers), and low IH competitive strength (less than 10% market share and 0.37x largest competitor). Portions of IH's product line are considered inadequate or obsolete, and IH dealers sell competitive equipment in key market areas.

- If market share remains at 9.75% (which is high on a historical basis), then 1984 ROS and ROA will not achieve acceptable levels. A follow-up control system could reveal critical control points where program support might be curtailed.

- Is it possible to source some products outside, thereby improving profits and diverting assets to other SBU's; conversely, have opportunities been investigated to private label some IH products in order to achieve more economical volumes?

- The economics of the WBX square baler program are currently being reviewed, with a decision imminent on whether it is economically desirable to source NAO requirements from a central IHGB source. What contingency plan exists if the proposed sourcing situation is not viable?

What is the current status of the more revolutionary technological approaches being considered in these businesses?

- With "Hay-in-a-Day" only at a "concept" level of development, is the possibly energy intensive nature of the process going to seriously affect the economic viability of the program?

* - What contingency steps have been taken relative to a possible breakthrough with the forage combine concept? How will it affect other related Segments, e.g., Grain Harvesting? (1979-A10).

*1979 Key Issue.

OUTDOOR POWER

GLOBAL SBU

IH SBU STRATEGIC COMMENTARY (P4):

HISTORY AND MARKET - This SBU encompasses garden tractors, lawn tractors, riding mowers, and outside purchased home maintenance products such as push mowers and snow blowers. These products have worldwide application, but are primarily used in the economically developed countries. Although U.S. manufacturers are not specifically excluded from foreign markets, economic considerations and strong local producers have inhibited them from actively participating. IH sales have been limited primarily to the U.S. and Canada, therefore, the P1 data and P4 refer only to these markets. Outdoor Power products, with a broad base of end-users classified as residential homeowners or commercial operations, in contrast with other AE Group manufactured products, fall mostly into consumer goods categorization and, hence, are subject to image and positioning criteria more like consumer goods than agricultural equipment.

COMPETITION -

U. S. Industry and IH Market Share (Units)

	Garden Tractor		Lawn Tractor		Riding Mower	
1976	210,000	13.3%	360,000	1.5%	324,000	2.2%
1977	200,000	12.8	350,000	1.3	320,000	1.8
1978	200,000	13.9	363,000	1.6	314,000	2.0

1978 North American Market Shares ($)

Mass Merchants		Specialty Producers		Ag. Eq Mfgs.		Other
Sears	18.0%	Wheel Horse	9.0%	Deere	7.0%	20.0%
MTD	11.5	A-C/Simplicity	8.5	IH	7.0	
Murray Ohio	10.7	Ariens	8.3			
	40.2%		25.8%		14.0%	20.0%

Due to the broad base of potential customers, the distribution methods employed by the various manufacturers have a direct bearing on their ability to penetrate the market place. There are three channels of distribution for this SBU: mass merchants, specialty merchandisers, and agricultural equipment dealers.

IH PRODUCTS AND PRODUCTION - IH products have a reputation for quality, excellent performance, reliability, and good parts/service support. The major weaknesses of the product line in comparison to the mass merchandisers are high costs due to fewer economies of scale and narrower range of product offerings. Production of all lawn and garden equipment is now at the Louisville Plant after removal of riding mowers from the Memphis Plant. Distribution is through the Louisville Distribution Center, including home maintenance items purchased from outside sources.

STRATEGIES - Growth - IH is not attempting to meet the mass merchandisers head-on in the lawn and garden business. Rather, IH is seeking to sell to the segment of market that wants a premium, high quality product with good parts and service availability. To increase market share, the dealer network will be upgraded and expanded, new products will be developed, and the scope of the product line offered will be expanded by more outside sourced products.

July 31, 1979	B.H. Warren	Outdoor Power Global SBU	AEG
Date	Manager		Group

BUSINESS IDENTIFICATION FOR STRATEGIC PLANNING:
STRATEGIC BUSINESS UNIT (SBU) DIRECTORY (P3)

A. SALES $MM IN 1978 66

B. NUMBER OF P3 PAGES FOR THIS SBU 1

PRODUCTS/MANUFACTURING/ASSEMBLY

(1) PRODUCTS	1978 % OF SBU SALES	(2) MANUFACTURING/ASSEMBLY PLANTS	1978 % OF SBU SALES
OUTDOOR POWER - Lawn & Garden (Garden tractors, lawn tractors, riding mowers, related equipment, and parts)	100	Louisville (USA) Memphis (USA) Outside Purchase	59.6 27.8 12.6
TOTAL	100%	TOTAL	100%

DISTRIBUTION/REGIONS

(3) DISTRIBUTOR/DEALERS, OEMS, DIRECT USERS	1978 % OF SBU SALES	(4) GEOGRAPHIC MARKETS	198 % OF SBU SALES
Distributor/Dealer Direct Users	99 1	North America	100
TOTAL	100%	TOTAL	100%

END USE MARKETS/COMPETITORS

(5) END USE MARKETS	1978 % OF SBU SALES	(6) COMPETITORS			198 % MKT. SHARE
Homeowner	90	Roper (Sears)	18	Ariens	8
		MTD	12	Deere	7
Commercial	10	Murray-Ohio	11	IH	7
		Wheel Horse	9	Others	19
		Allis-Chalmers/Simplicity	9		
TOTAL	100%	TOTAL			100%

July 31, 1979
DATE

Outdoor Power
SBU

AE
GROUP

AGRICULTURAL EQUIPMENT

GROUP
GLOBAL SBU

PLAN SUMMARY
OUTDOOR POWER

THE PLAN: GROWTH-Through distribution network upgrading and expansion, and increased product offerings.

$ MARKET GROWTH	NET SALES GROWTH	COMPETITIVE PRESSURE
Growth rate increases to high of 10.5% in 1982, then begins declining. Real 0.7%, Inflation 7.0, Total 7.7%	Erratic growth-peaks in 1980 and 1982 with decline in 1981. Real 5.6%, Pricing 7.0, Total 12.6%	High to moderate-mass merchandisers such as Sears provide most resistance due to pricing policies and distribution systems.

$ MARKET SHARE	PROFITABILITY	CUMULATIVE CASH FLOW
Small increase from 7.9% (1979) to 9.0% (1984), with slight decline in 1981.	ROA increases from 17.4% (1979) to 33.4% (1984). ROS increases from 4.9% (1979) to 11.0% (1984).	$31 million (1980-84). Inflows distributed relatively uniformly year by year.

PRICING	COST INCREASE	PRODUCTIVITY %-OPERATING INCOME
Average 7.0% annual increase equals market inflation rate and cost inflation rates.	Avg. variable and fixed cost inflation rates are both 7.0%.	Average 41% (1980-84); relatively level each year.

M & A	R & D	CAPITAL EXPENDITURES
Increases from 18.4% (1979) to 18.9% (1984) of sales, with lower levels in interim years (17.3% average).	Increases slightly from 1.8% (1979) to 2.0% (1984) of sales.	$21 million total (1980-84). General increase year by year.

PROFILE (P-1)

		HISTORY		PROJECTION						GROWTH	
		1977	1978	1979	1980	1981	1982	1983	1984	77-79	79-84
A.AVAIL RETAIL MARKET	$MM	1001.3	1007.1	1002.5	1048.4	1128.2	1246.1	1359.4	1449.3	.1	7.7
B.IH RETAIL MARKET SHARE	%	7.50	7.00	7.90	8.40	8.20	8.70	8.89	9.00		
C.LARGEST COMP SHARE	%	18.00	18.00	18.00	18.00	18.00	18.00	18.00	18.00		
D.COMPETITIVE STRENGTH		M	M	M	M	M	M	M	M		
E.NET SALES	$MM	60.9	65.3	78.3	103.9	102.8	123.5	131.8	142.0	13.4	12.6
F.CONTRIBUTION MARGIN	%	31.20	29.40	34.23	40.04	39.69	40.73	41.12	39.93		
G.TOTAL FIXED COST	$MM	21.3	15.2	21.9	24.0	27.7	31.0	34.4	38.5	1.4	11.9
H.OPERATING INCOME	$MM	(2.3)	4.0	4.9	17.6	13.1	19.3	19.8	18.2	103.2	30.0
I.PROFIT BEFORE TAX	$MM	(4.2)	2.8	3.8	16.3	11.6	17.4	17.5	15.6	70.4	32.6
J.TOTAL AVG ASSETS	$MM	27.2	21.9	21.8	27.6	29.4	37.5	42.1	46.7		
K.TOTAL ENDING ASSETS	$MM	27.5	22.2	23.7	28.2	30.7	39.4	43.4	48.3	(6.7)	15.3
L.CASH FLOW (OPERATIONAL)	$MM		1.9	4.6	7.1	4.9	4.2	8.7	5.7		
M.MARKET GROWTH	%		.58	(.46)	4.58	7.61	10.45	9.09	6.61		
N.SALES GROWTH	%		7.22	19.91	32.69	(1.06)	20.14	6.72	7.74		
O.PBT GROWTH	%			35.71	328.95	(28.83)	50.00	.57	(10.86)		
P.ANNUAL MKT INFLATION	INDX		1.000	1.000	1.070	1.070	1.070	1.070	1.070		
Q.ANNUAL PR INFL	INDX		1.000	1.000	1.070	1.068	1.072	1.070	1.070		
R.ANNUAL VC INFL	INDX		1.000	1.000	1.070	1.068	1.072	1.070	1.070		
S.ANNUAL FC INFL	INDX		1.000	1.000	1.070	1.069	1.071	1.070	1.070		
T.ROA (AVG ASSETS)	%	(15.4)	12.8	17.4	59.1	39.5	46.4	41.6	33.4		
U.ROS	%	(6.9)	4.3	4.9	15.7	11.3	14.1	13.3	11.0		
V.ASSET(AVG) TURNOVER	X	2.24	2.98	3.59	3.76	3.50	3.29	3.13	3.04		
W.NET PRDTVTY/OP.INC	%	0	0	71.4	29.0	45.0	38.3	41.9	48.9		
X.R&D/SALES	%	3.6	2.9	1.8	2.0	2.2	2.0	2.0	2.0		
Y.MAE/SALES	%	13.3	7.4	18.4	15.2	17.6	16.9	18.1	18.9		
Z.ESTIMATED CAPITAL EXPENDITURES	$MM	1.8	2.6	2.8	1.8	3.7	5.0	3.9	6.7		
AA. PRODUCT CAPITAL EXPENDITURES	%TOT	11.1	11.5	39.3	61.1	56.8	12.0	0	100.0		
BA. COST-REDUCTION CAP. EXPEND.	%TOT	0	0	21.4	0	0	10.0	23.1	0		
CA. OTHER CAPITAL EXPENDITURES	%	88.9	88.5	39.3	38.9	43.2	78.0	76.9	0		
DA.R&D EXPENSE	$MM	2.2	1.9	1.4	2.1	2.3	2.5	2.7	2.9	(17)	15.7
EA. NEW PRODUCT R&D	%R&D	0	0	21.4	33.3	30.4	20.0	18.5	20.7		
FA.M&AE EXPENSE	$MM	8.1	4.8	14.4	15.8	18.1	20.9	23.9	26.8	33.3	13.2
GA.NET PLANNED PRODUCTIVITY (CUM)	$MM	0	0	3.5	5.1	5.9	7.4	8.3	8.9		
HA.CONTINGENCY	$MM	0	0	0	0	0	0	0	0		
IA.PARTS REVENUE	$MM	4.1	4.9	5.3	6.6	7.3	8.3	9.4	11.1	13.7	16.9
JA.PARTS VARIABLE MARGIN	%	46.34	46.94	45.28	45.45	43.84	44.58	42.55	41.44		

COMPOUND ANNUAL GROWTH RATES

MAJOR CAPITAL PROGRAMS
(Over $5.0 MM)

OUTDOOR POWER

GLOBAL SBU
($ MM)

Program	APPROPRIATION Year	APPROPRIATION Amount	EXPENDITURES -1979	1980	1981	1982	1983	1984	1985+
New O. P. Paint System	1982	$ 5.0	$	$	$	$ 2.5	$ 2.5	$	$
TOTAL		$ 5.0				$ 2.5	$ 2.5		

*Project appropriation approved.

PRODUCT DEVELOPMENT TIMETABLE
(PROJECTS KNOWN OR LOGICALLY ANTICIPATED)

OUTDOOR POWER
GLOBAL SBU

	1979	1980	1981	1982	1983	1984
International Harvester		New line Cadet Tractors		New line Cadet Riding Tractors		
Deere		New line garden tractors				
Wheel Horse						
Allis-Chalmers/ Simplicity						

All competition will advance product offerings to stay competitive and meet market requirements.

CAUSAL FACTOR ANALYSIS

OUTDOOR POWER
GLOBAL SBU

SALES

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR SALES	$MM		78.3	103.9	102.8	123.5	131.8
B. AVAIL MARKET EFFECT	$MM		(1.79)	.5	2.99	2.1	(.46)
C. SELLING PRICE EFFECT	$MM		6.8	6.55	8.29	8.65	9.31
D. MARKET SHARE EFFECT	$MM		4.93	(2.15)	5.83	2.47	1.5
E. WHL/DLR INV EFFECT	$MM		15.67	(6.16)	3.79	(4.92)	(.15)
F. REMAINDER SALES	$MM		0	.17	(.19)	0	0
G.NET SALES	$MM	78.3	103.9	102.8	123.5	131.8	142

PROFIT BEFORE TAX

		PROJECTION					
		1979	1980	1981	1982	1983	1984
A.PREVIOUS YEAR	$MM		3.8	16.3	11.6	17.4	17.5
B. AVAIL MARKET EFFECT	$MM		(.61)	.2	1.19	.85	(.19)
C. COST INFL EFFECT	$MM		(5.65)	(5.73)	(6.98)	(7.35)	(8.12)
D. SELLING PRICE EFFECT	$MM		6.8	6.55	8.29	8.65	9.31
E. MARKET SHARE EFFECT	$MM		1.69	(.85)	2.31	1.01	.62
F. WHOLESALE/DEALER INV EFFECT	$MM		5.36	(2.46)	1.5	(2)	(.26)
G. VARIABLE COST CHG	$MM		5.64	(.34)	1.2	.49	(1.58)
H. FIXED COST CHG	$MM		(.53)	(1.92)	(1.24)	(1.14)	(1.58)
I. REMAINDER	$MM		(.2)	(.13)	(.48)	(.4)	(.3)
J.PROFIT BEFORE TAX	$MM	3.8	16.3	11.6	17.4	17.5	15.6

Sales Commentary

Annual sales in 1984 are expected to be about $64 MM greater than 1979 sales. Selling price accounts for about 62% of the increase, market share growth for about 20%, and inventory effect for about 13%. The large sales increases in '80 and '82 are due to inventory and market share effects associated with new Cadet product introductions. Market growth is expected to be flat.

Income Commentary

Annual income is expected to increase by $12.5 MM in 1980, a fourfold increase over 1979, primarily due to inventory buildup and variable cost improvements. Income in 1984 is expected to be about $12 MM greater than 1979 income. About 49% of the profit improvement is due to retained net inflation benefits, 46% to variable cost improvement, and 41% to market share gains. Variable cost improvements are due to a more favorable "high quality" product offering, and associated manufacturing efficiencies. The income decline in 1981 is primarily due to inventory reduction and higher depreciation charges on new capital investment; in '84 the decline is caused by higher depreciation and variable cost charges.

KEY ISSUES

OUTDOOR POWER
GLOBAL SBU

* The lawn and garden equipment market offers minimal real growth and places IH in direct competition with major agricultural equipment manufacturers, strong mass merchandisers, and specialty producers. The SBU strategy is to pursue the difficult-to-define "premium market niche" where the IH product's reputation for quality, excellent performance, reliability, and good parts/service support can be used to greatest advantage. (1979-A12).

The dramatic improvement in 1980 in sales and profits appears to be based on the introduction of a new line of Cadets with lower manufacturing costs and higher(apparently) customer appeal.

- What marketing programs are planned to produce the 33% increase in sales?

* • What research has been done to determine whether potential customers will perceive a meaningful benefit and competitive advantage of the new Cadets? (1979-A12).

Although the investment has been made for the new line, IH is still at a disadvantage in terms of distribution and widely available service support.

- What steps will be taken to monitor the launch of the new line and determine whether the projected volumes will be forthcoming at planned expense levels?

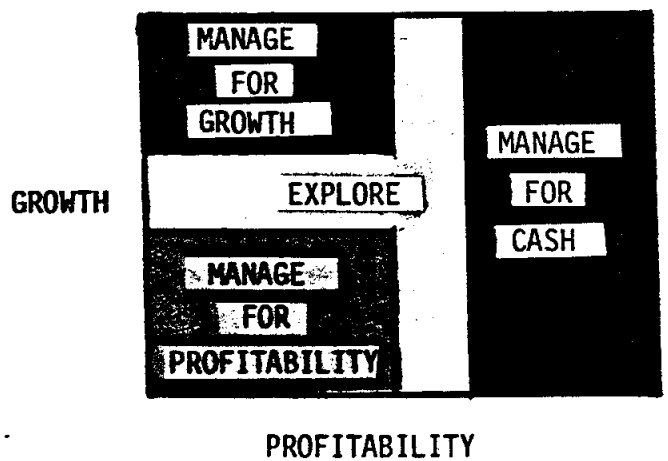
Should the improvements in volume and profit fail to materialize as planned, a "Cash" strategy may be warranted in view of the high priority needs of AEG in core business areas.

*1979 Key Issue

APPENDIX

IH CATEGORIZATION FRAMEWORK

IH
CATEGORIZATION
COLOR CODES



AGRICULTURAL EQUIPMENT GROUP

MANAGE FOR GROWTH: A SBU/segment in this category should spend as needed to increase its share of the end use market, subject to approval of the soundness of individual programs. Profitability objectives for such SBU/segments are understood to be long-term rather than short-term in nature. If the long-run prospects are attractive enough, cash flow in such SBU/segments may be below corporate objectives in the short-term.
MANAGE FOR PROFITABILITY: A SBU/segment in this category is expected to participate in whatever growth is to be found in the end use market, but the principal focus is on profitability in the form of cost reduction, maintenance of full levels of capacity (even, in times, risking short-term loss of share).
EXPLORE: A SBU/segment in this category is treated as a Manage for Growth SBU/segment on a short-term basis, but considered uncertain as to long-term resource commitments. A SBU/segment in this category should spend whatever is necessary to determine the attractiveness of the SBU/segment's market potential and its ability to compete. By its nature, an Explore categorization can be used as a two-tiered classification. For example, a SBU/segment may be run at the operation level as a Manage for Growth SBU/segment, but at the group and corporate level be considered an exploration.
MANAGE FOR CASH: A SBU/segment in this category should seek to develop cash flow by avoiding new capital expenditures (except for fast-pay-back cost-reduction programs), reducing working capital, tightening discretionary spending, and pricing for the highest possible margin. The focus should be on the highest possible cash flow, even at the expense, if necessary, of market share and/or percentage profitability.

AGRICULTURAL EQUIPMENT GROUP
"LINKAGE" MATRIX

SBU SEGMENTS	AGRICULTURAL EQUIPMENT										COMPONENTS		PAYLINE		TRUCK		FINANCIAL SERVICES		SOLAR	
	TRACTORS		CROP HARVESTING		CROP PRODUCTION		HAY, FORAGE, MAT. HAND.		OUTDOOR POWER											
	P	D	P	D	P	D	P	D	P	D	P	D	P	D	P	D	P	D	P	D
TRACTORS																				
Under 40 h.p.	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	-	●	-	-
40-115 h.p.	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	-	●	-	-
115-200 h.p.	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	-	●	-	-
Over 200 h.p.	●	●	●	●	●	●	●	●	●	●	●	●	-	●	-	●	-	●	-	-
CROP HARVESTING																				
Grain	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	-	●	-	-
Cotton	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	-	●	-	-
CROP PRODUCTION																				
Planters	●	●	●	●	●	●	●	●	●	●	●	●	-	●	-	●	-	●	-	-
Tillage	●	●	●	●	●	●	●	●	●	●	●	●	●	●	-	●	-	●	-	-
HAY, FORAGE, MATERIAL HANDLING																				
Hay	●	●	●	●	●	●	●	●	●	●	●	●	●	●	-	●	-	●	-	-
Forage	●	●	●	●	●	●	●	●	●	●	●	●	-	●	-	●	-	●	-	-
Material Handling	●	●	●	●	●	●	●	●	●	●	●	●	-	●	-	●	-	●	-	-
OUTDOOR POWER	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	-	●	-	-

CODE:
P = Production Aspects
D = Distribution Aspects

Strong Linkage = Red = ●
Medium Linkage = Yellow = ●
Weak Linkage = Green = ●

COMMENTS:
Linkages in the marketing and distribution of agricultural equipment are considered particularly strong with a full product line, providing the Company and the dealer a very real competitive advantage. Medium to strong linkages also exist between current product lines and manufacturing facilities.

AGRICULTURAL EQUIPMENT GROUP
IMPACT PROFILE

SGU Segments	Weather	Farming Practices	Net Farm Income	Technology Food/Crop Usage	Technology Agricultural Equipment	World Agricultural Trade	General Economy	Government Policy
Tractors (DIN)								
Under 40 hp	●	●●	●●	●●	●	●	●	●●
40-115 h.p.	●	●●	●	●●	●	●	●	●
115-200 h.p.	●	●●	●	●●	●	●	●	●
Over 200 h.p.	●	●●	●	●●	●	●	●	●
Crop Harvesting								
Grain	●	●	●	●●	●●	●	●	●
Cotton	●	●	●	●	●●	●	●	●
Crop Production								
Planters	●	●	●	●	●	●	●	●●
Tillage	●	●	●	●	●	●	●	●●
Hay Forage & Material Handl.								
Hay	●	●	●	●	●	●	●	●
Forage	●	●	●	●	●	●	●	●
Mat. Handl.	●	●	●	●	●	●	●●	●
Outdoor Power	●	●	●	●	●	●	●	●

Level of Risk to IH:

High = Red = ●
Medium = Yellow = ●
Low = Green = ●

COMMENTS:

By its very nature, the selection of impact factors relative to the SBU/Segments leads to those elements having generally a high potential impact on IH and the industry. Factors with the most significant impact are weather, net farm income, agricultural equipment technology, and government policy.

AGRICULTURAL EQUIPMENT GROUP

STATUS REPORT

1979 STRATEGIC PLAN REVIEW ASSIGNMENTS

	ISSUE	STATUS/ACTION REQUIRED
A1	Redefine SBU segmentation for Agricultural Tractors - W.W. to show product/market SBU's.	COMPLETE.
A2	Develop data on end-use market segments for tractors within horsepower classes.	* COMPLETE. Follow-up on implementation needed.
A3	Analyze sourcing alternatives for 20-40 H.P. tractors.	*INCOMPLETE. Warren/Musgjerd letter of 5/24/79 indicates that issue will be resolved by 9/1/79.
A4	Feasibility and cost/benefit of accelerating the TX Tractor program.	*COMPLETE - INCORPORATE UPDATE TO APPROVED RESPONSE IN STRATEGIC PLAN SUBMISSION.
A5	Resolve long-range strategy for Steiger relationship.	*INCOMPLETE - CONTINUED VERBAL BRIEFINGS REQUESTED.
A6	High task profitability goals for 40-200 H.P. tractors.	COMPLETE - MONTHLY REPORTS IN MARKET SHARE MEETINGS.
A7	Cost/benefit of increasing axial flow production capacity.	COMPLETE. Appropriation approved.
A8	Strategy to maintain IH leadership versus Deere rotary combine.	*COMPLETE - SET UP SPECIFIC ACTIVITIES AND CHECK POINTS OF PROGRESS AFTER STRATEGIC PLAN REVIEW. Deere expected to achieve product parity by 1982. Advanced concepts under study.
A9	Cost/benefit of accelerating the planter development project.	COMPLETE. Advance funds approved.
A10	R&D proposal, to determine potential of the innovative hay/ forage combine.	*AEG RESPONSE COMPLETE - NEW ASSIGNMENT MADE TO R.J. POTTER (DUE BEFORE 9/15/79). AEG is proceeding with "hay in a day" project - timetable due 10/31/79. Mr. Potter's group is proceeding with forage combine concept-resource requirements presentation in August, 1979.

*Addressed in 1980 Strategic Plan Key Issues.

7/31/79

AGRICULTURAL EQUIPMENT GROUP

STATUS REPORT (cont'd)

1979 STRATEGIC PLAN REVIEW ASSIGNMENTS

	ISSUE	STATUS/ACTION REQUIRED
A11	Alternatives to use excess plant capacity.	*BRIEF DISCUSSION OF ISSUE AT STRATEGIC PLAN REVIEW - COMPLETE PRESENTATION AND PLAN UPDATE LATER (ON WORLD-WIDE BASIS).
A12	Study "premium" market niche for lawn and garden tractors.	*INCOMPLETE. However, current programs do pursue this strategy.
A13	Group and Corporate cost/benefit of standard seasonal waiver of finance and other marketing tactics used by competitors.	*INCOMPLETE. Task force at Group is developing implementation plans.
A14	Strategic options available for Brazil.	*INCOMPLETE - PROGRESS REPORT AT STRATEGIC PLAN REVIEW ON ENTRY INTO BRAZILIAN TRACTOR MARKET AND NEED FOR NEW ISSUE ASSIGNMENT.
A15	Human resource requirements to achieve the plan.	*INCOMPLETE - ISSUE SHOULD BE SPECIFICALLY ADDRESSED IN STRATEGIC PLAN.
A16	Strategies for "quantum" increases in worldwide market share.	*INCOMPLETE. 1980-84 Strategic Plan shows large market share increases in many segments. New assignment required to define additional breakthrough approaches.
A17	Report on R&D projects leading to the development of innovative products and/or new businesses.	*INCOMPLETE - CATEGORIZED R&D ACTIVITIES WITHIN STRATEGIC PLAN SUBMISSION REQUIRE FINAL REVIEW.
A18	Aggressive programs to increase U.S. exports.	COMPLETE - FOLLOW-UP VIA SPECIAL EXPORTS MEETING ISSUE ASSIGNMENTS.

*Addressed in 1980 Strategic Plan Key Issues.